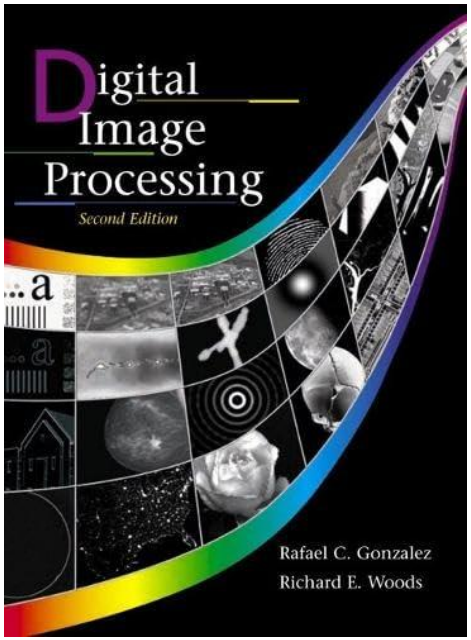
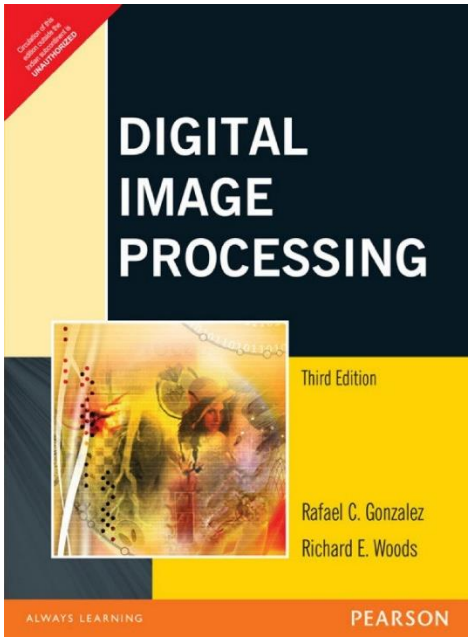
Digital Image Processing

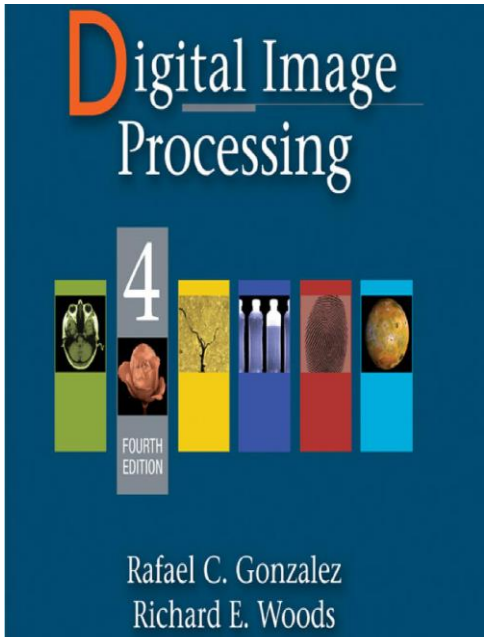
Course Teacher: Prof. Dr. Rameswar Debnath

Ref. Book: Rafael. C. Gonzalez and Richard E. Woods, Digital Image Processing, 2nd/3rd ed., Pearson Education (Singapore) Pte. Ltd.

Recommended Book







1 Introduction

- Digital image processing focuses on two major tasks:
  - Improvement of pictorial information for human interpretation.
  - Processing of image data for storage, transmission, and representation for autonomous machine perception.

Introduction

- Images play the single most important role in human perception.
- One picture is worth more than ten thousand words.
- Computer vision is a field of artificial intelligence that trains computers to interpret and understand the visual world.
- Objective of computer vision is to emulate human intelligence.

Image Overview

- An image may be defined as a two-dimensional function, f(x, y), where x and are *spatial* (plane) coordinates, and the amplitude of f at any pair of coordinates (x, y) is called the *intensity* or *gray level* of the image at that point.
- When *x*, *y*, and the amplitude values of *f* are all finite, discrete quantities, we call the image a *digital image*.

Digital Image Processing

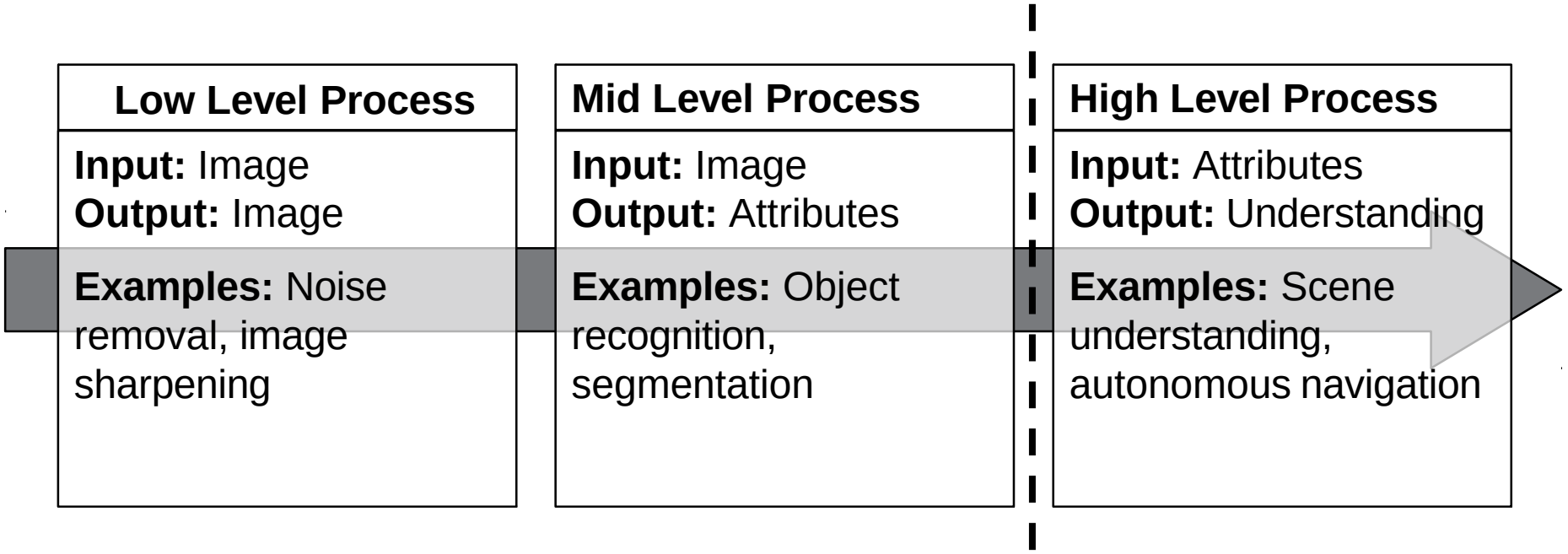
- Digital Image Processing (DIP) is a software which is used to manipulate the digital images by the use of computer system.
- It is also used to enhance the images, to get some important information from it.
- For example: Adobe Photoshop, MATLAB, etc.
- Digital image processing is one of the fastest growing industry which affects everyone's life.

Types of Image Processing

- Three types of computerized processes in digital image:
  - Low-level process
  - Mid-level process
  - High-level process
- Low-level processes involve primitive operations such as image preprocessing to reduce noise, contrast enhancement, and image sharpening.
- A low-level process is characterized by the fact that both its inputs and outputs are images.

Types of Image Processing

- Mid-level processing on images involves tasks such as segmentation (partitioning an image into regions or objects), description of those objects to reduce them to a form suitable for computer processing, and classification (recognition) of individual objects.
- A mid-level process is characterized by the fact that its inputs generally are images, but its outputs are attributes extracted from those images (e.g., edges, contours, and the identity of individual objects).
- Higher-level processing involves “making sense” of an ensemble of recognized objects and performing the cognitive functions normally associated with vision.


Low Level Process
Input: Image
Output: Image
Examples: Noise removal, image sharpening
Mid Level Process
Input: Image
Output: Attributes
Examples: Object recognition, segmentation
High Level Process
Input: Attributes
Output: Understanding
Examples: Scene understanding, autonomous navigation

Applications of Digital Image Processing

- Image sharpening and restoration
- Medical field
  - Gamma ray imaging
  - PET scan
  - X Ray Imaging
  - Medical CT
  - UV imaging
  - Ultrasound imaging

Applications of Digital Image Processing

- Remote sensing
- Transmission and encoding
- Machine/Robot vision
- Color processing
- Pattern recognition
- Video processing
- Microscopic Imaging
- Others

Fundamental Steps in DIP

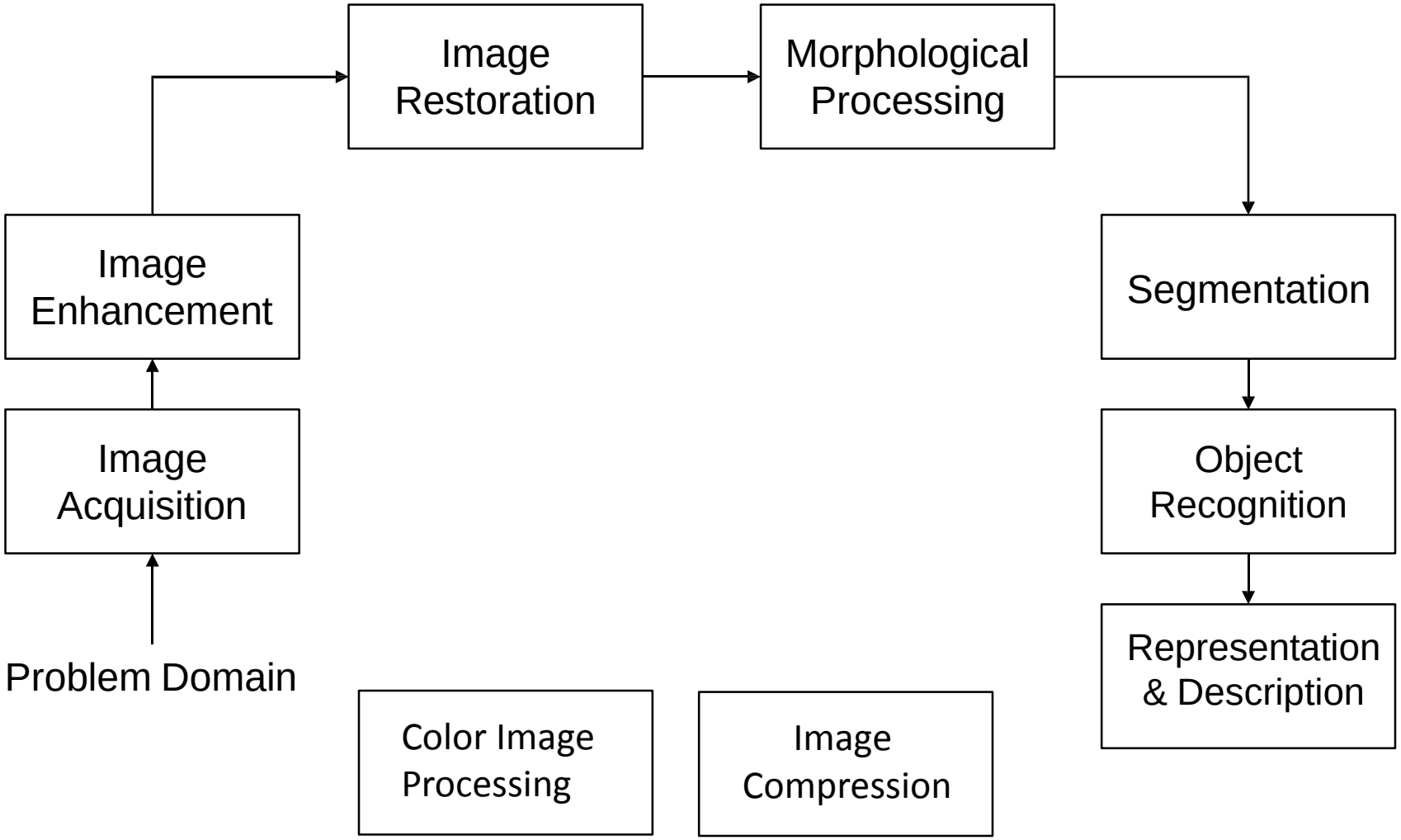


Image Acquisition

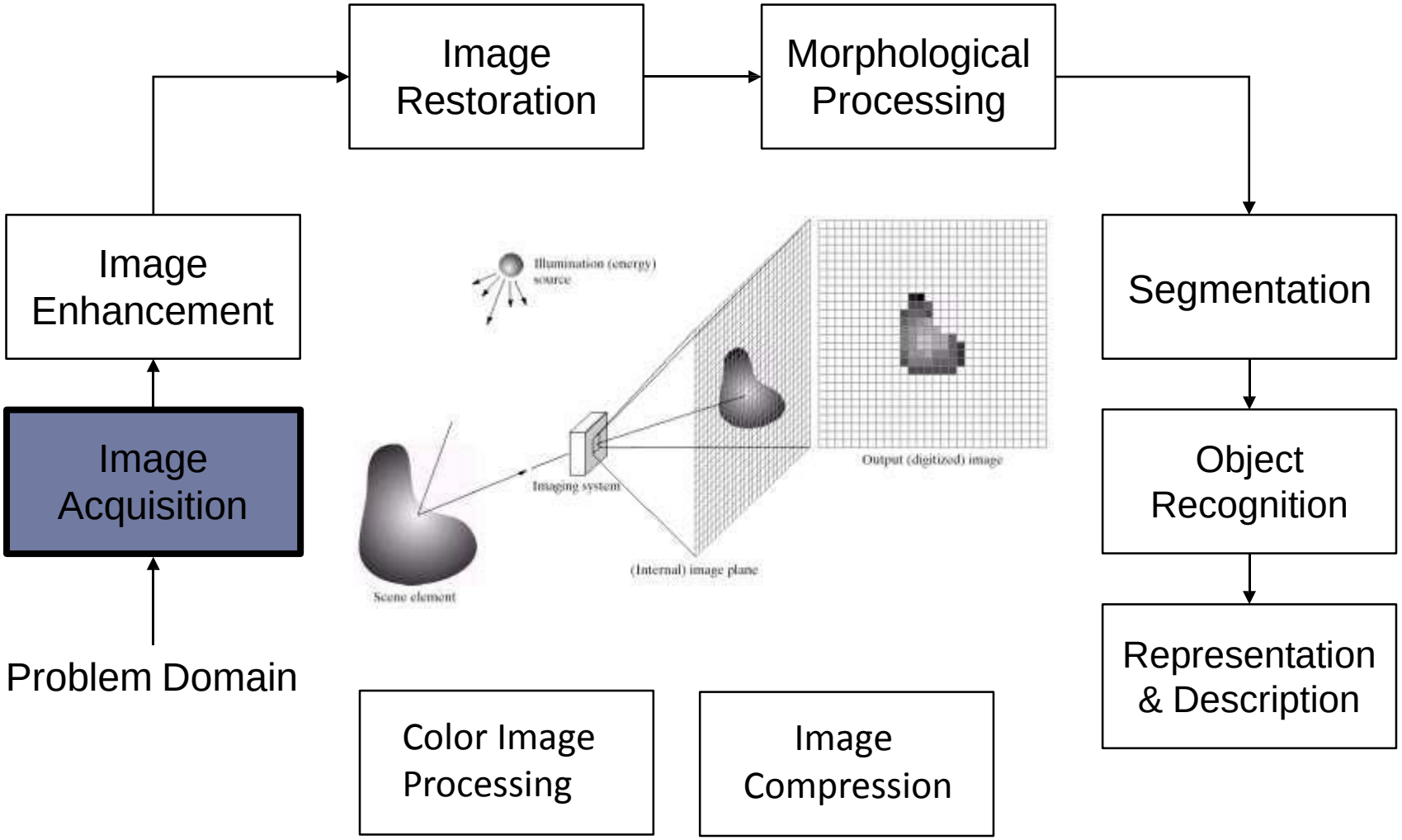


Image Enhancement

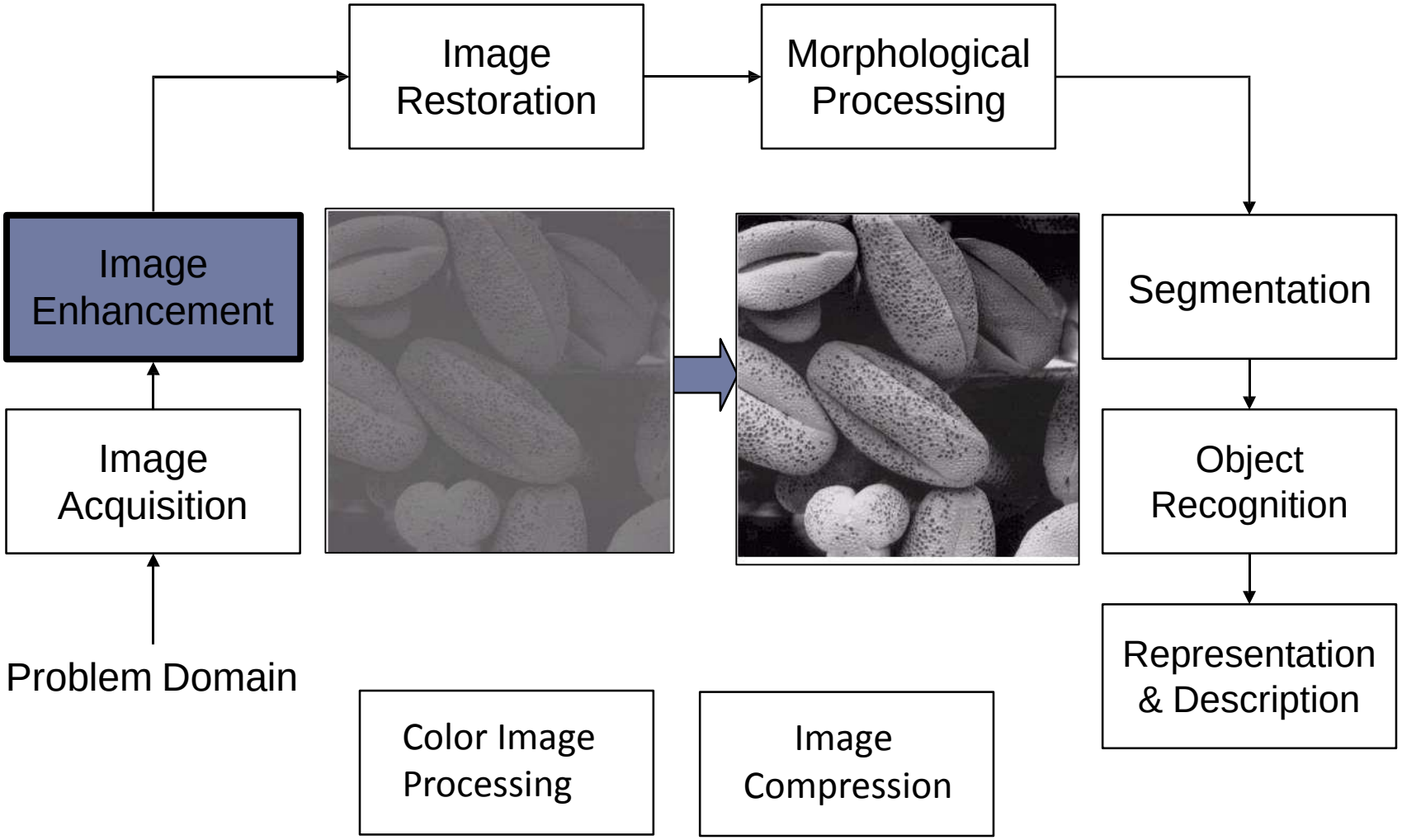
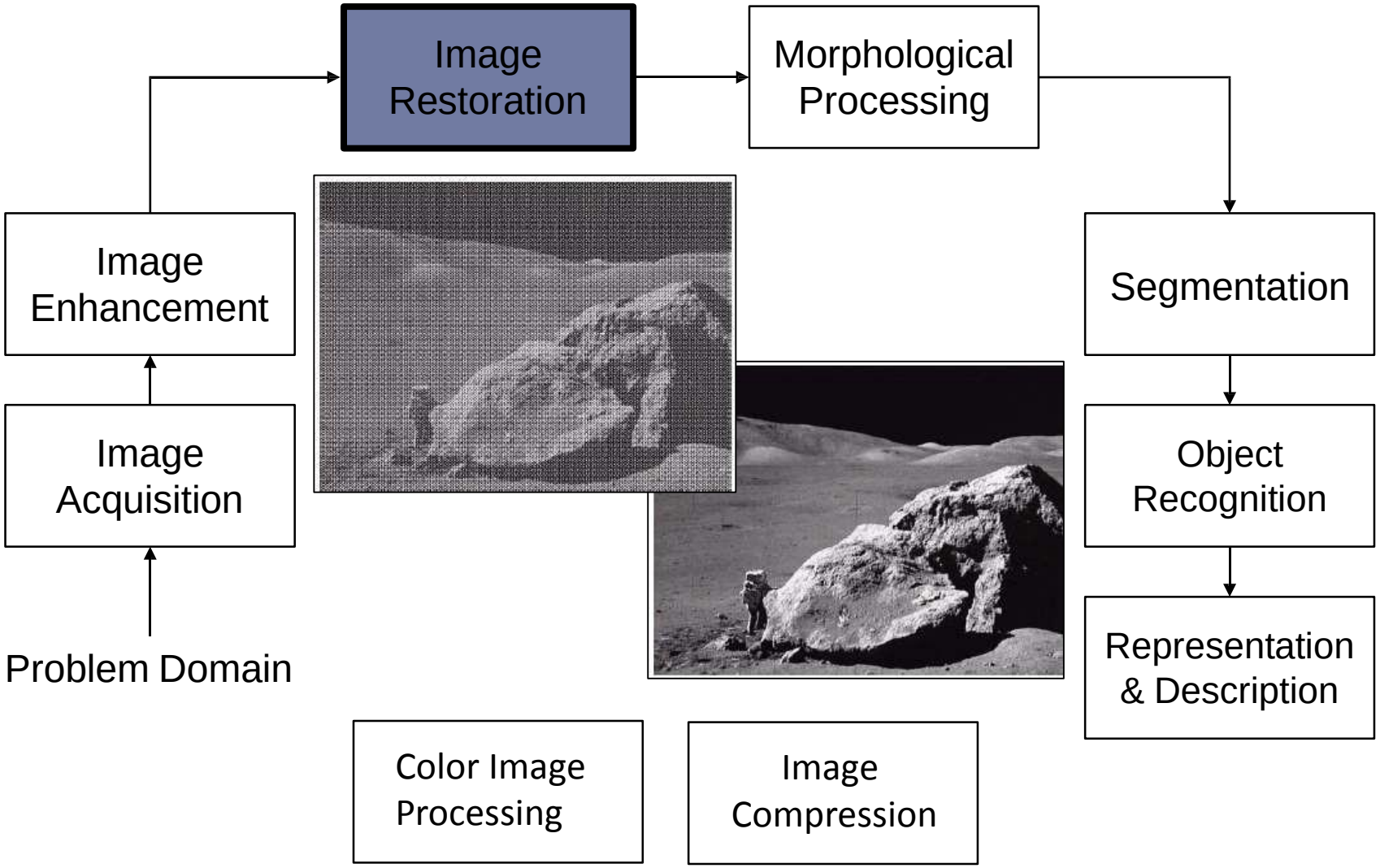
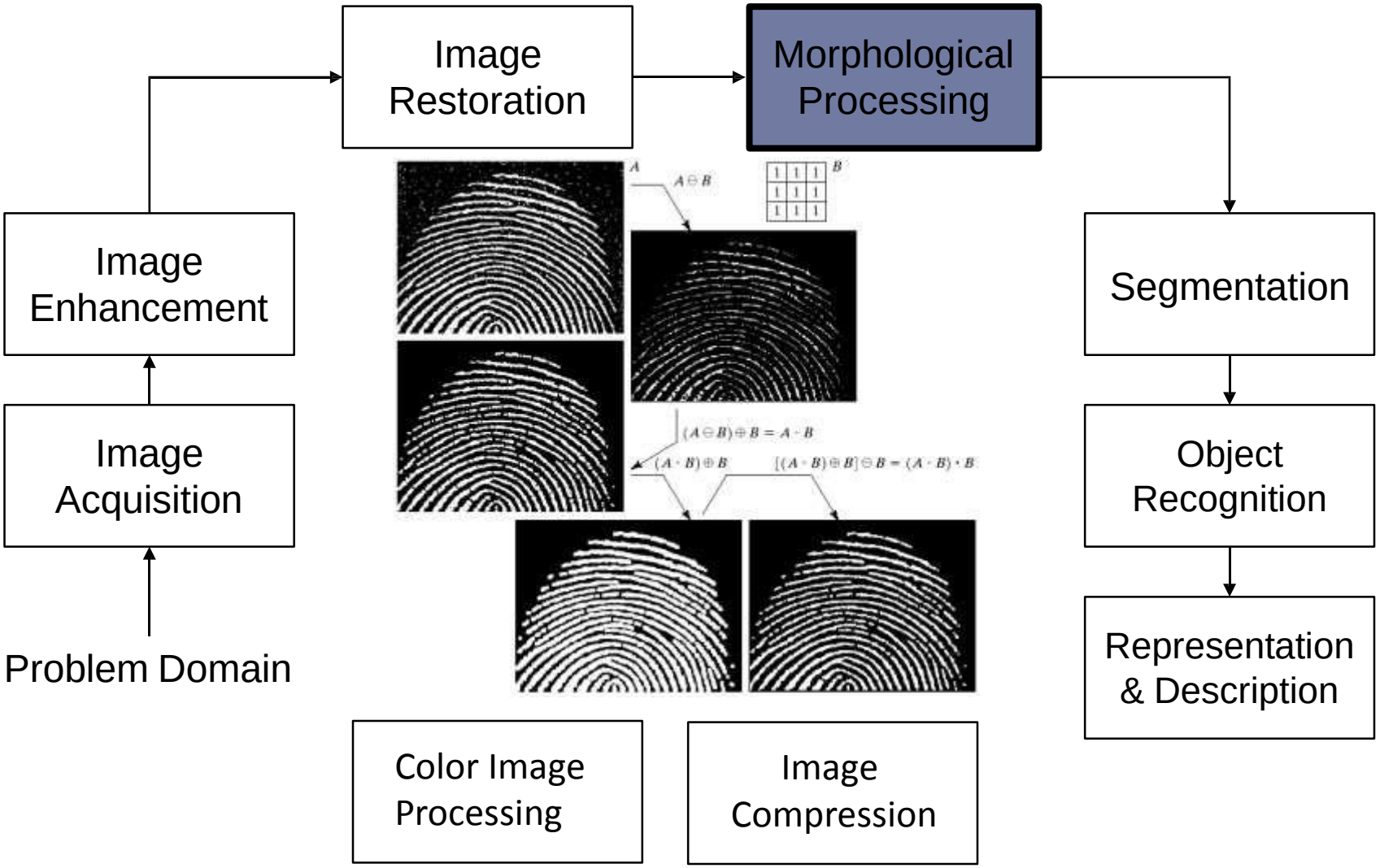


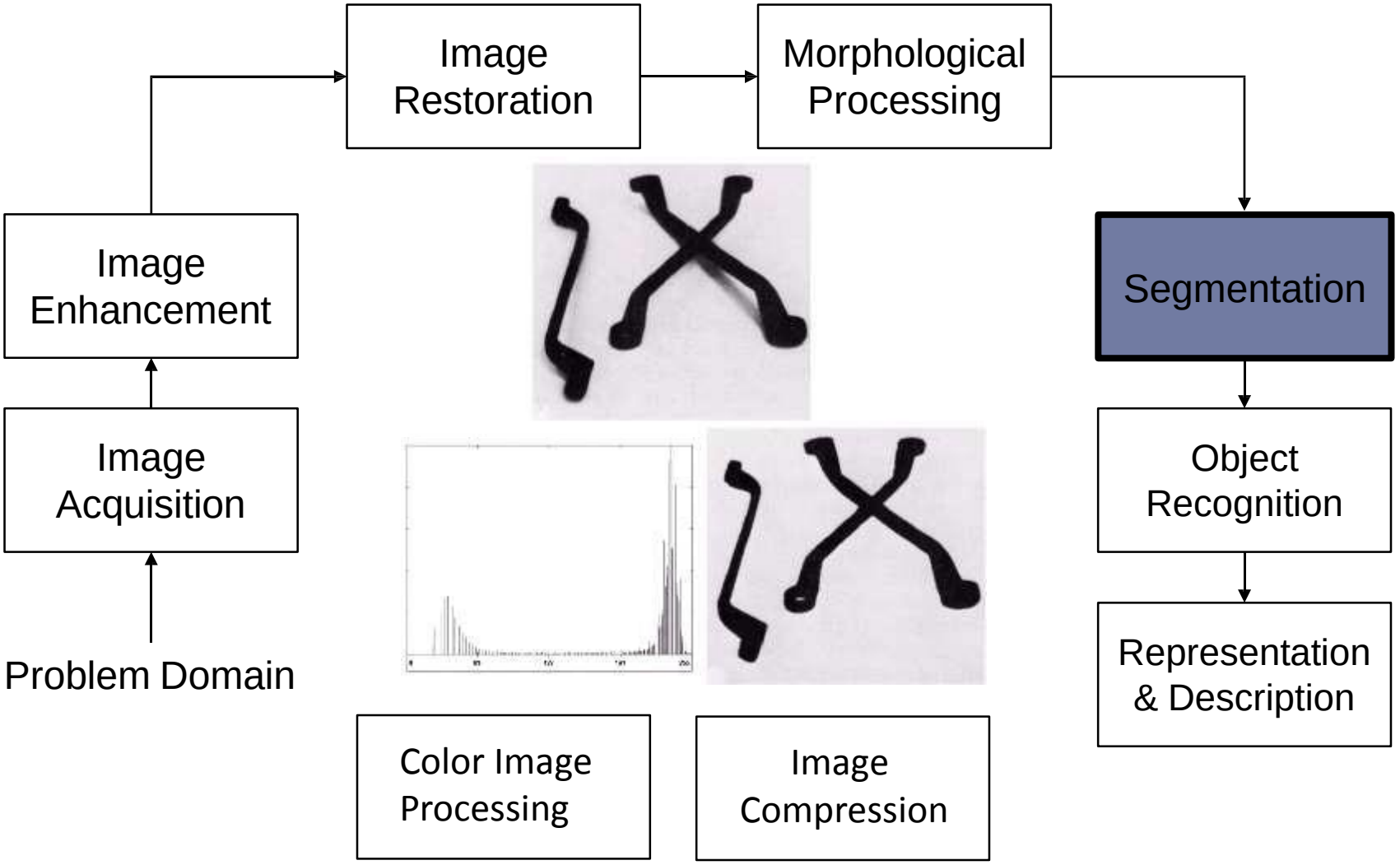
Image Restoration



Morphological Processing



Segmentation



Object Recognition

Image Restoration

Morphological Processing

Image Enhancement

Segmentation

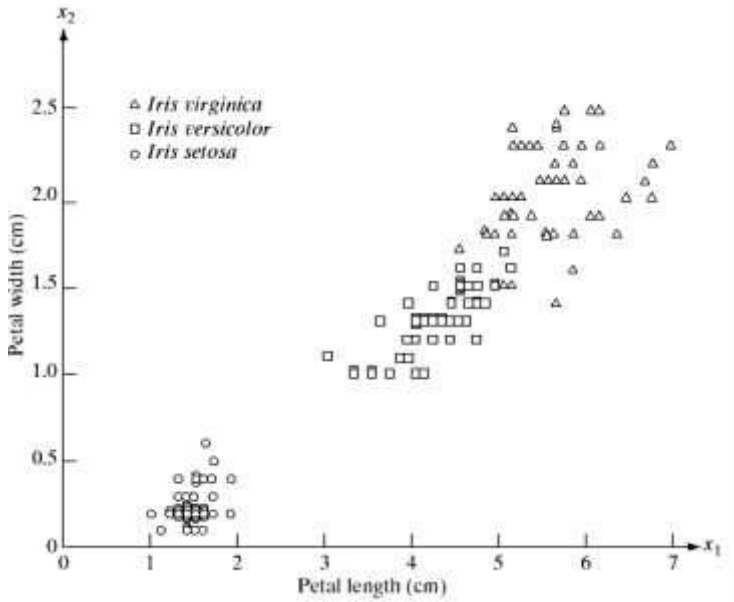


Image Acquisition

Object Recognition

Problem Domain

Representation & Description

Color Image Processing

Image Compression

Representation and Description

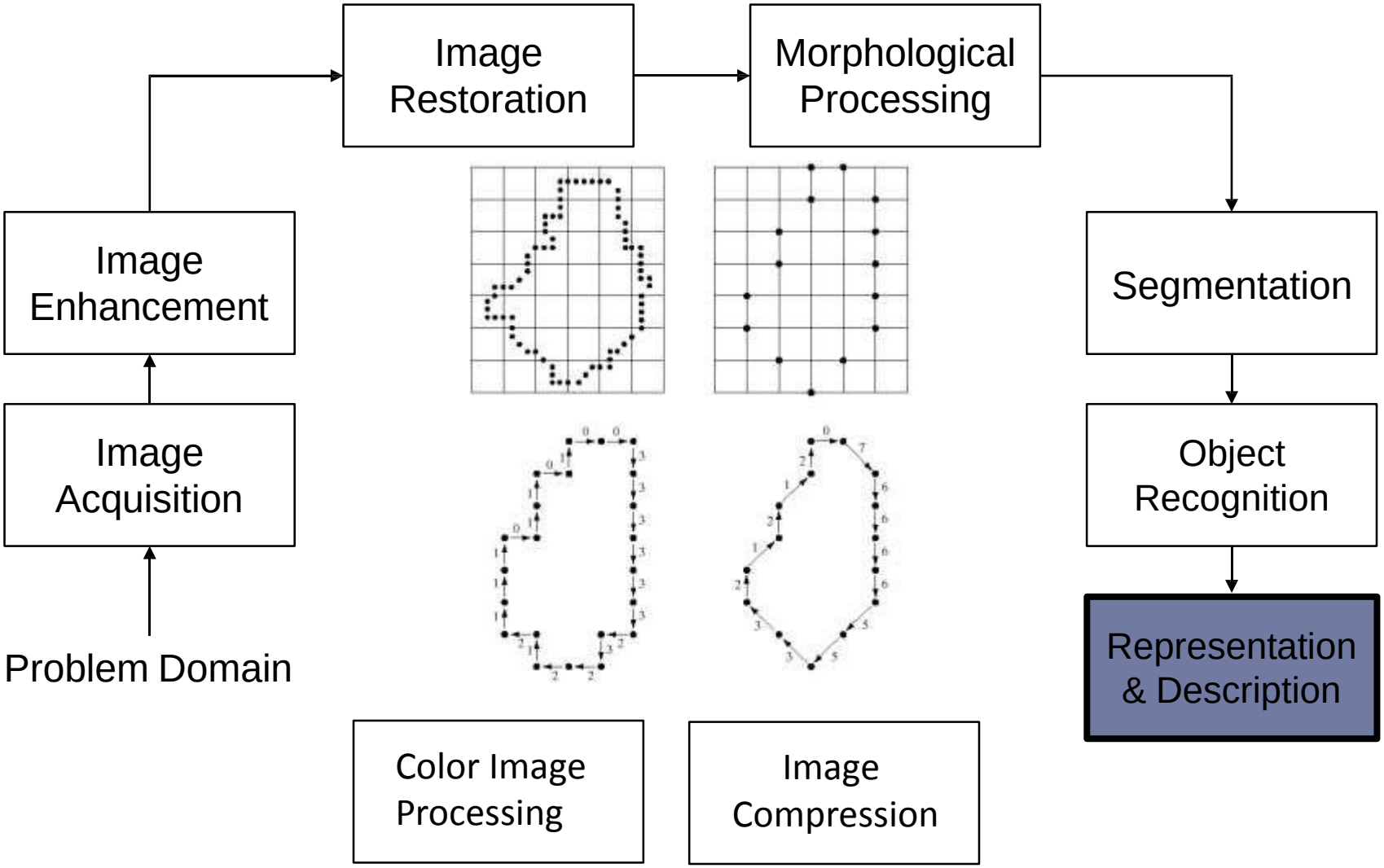
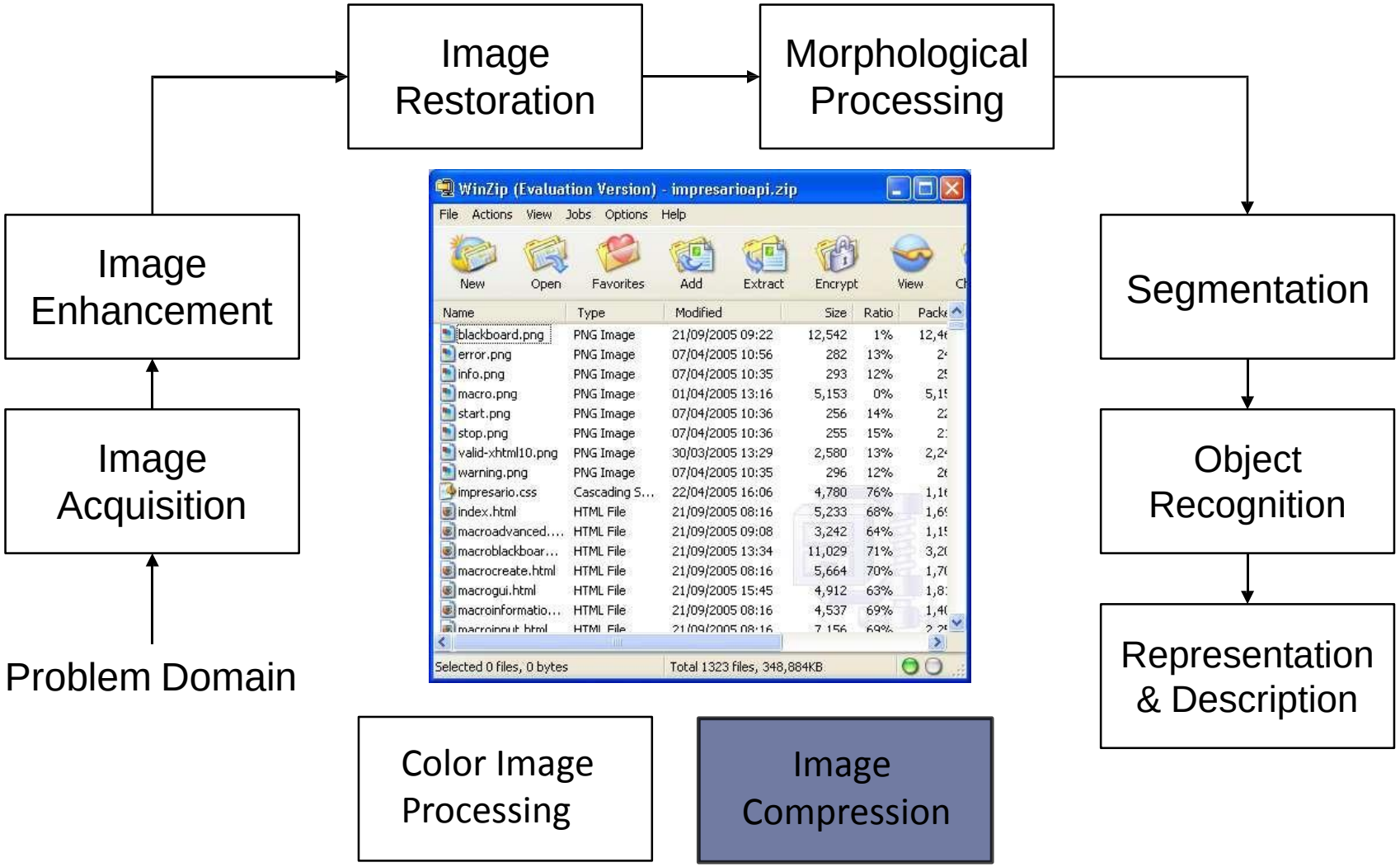
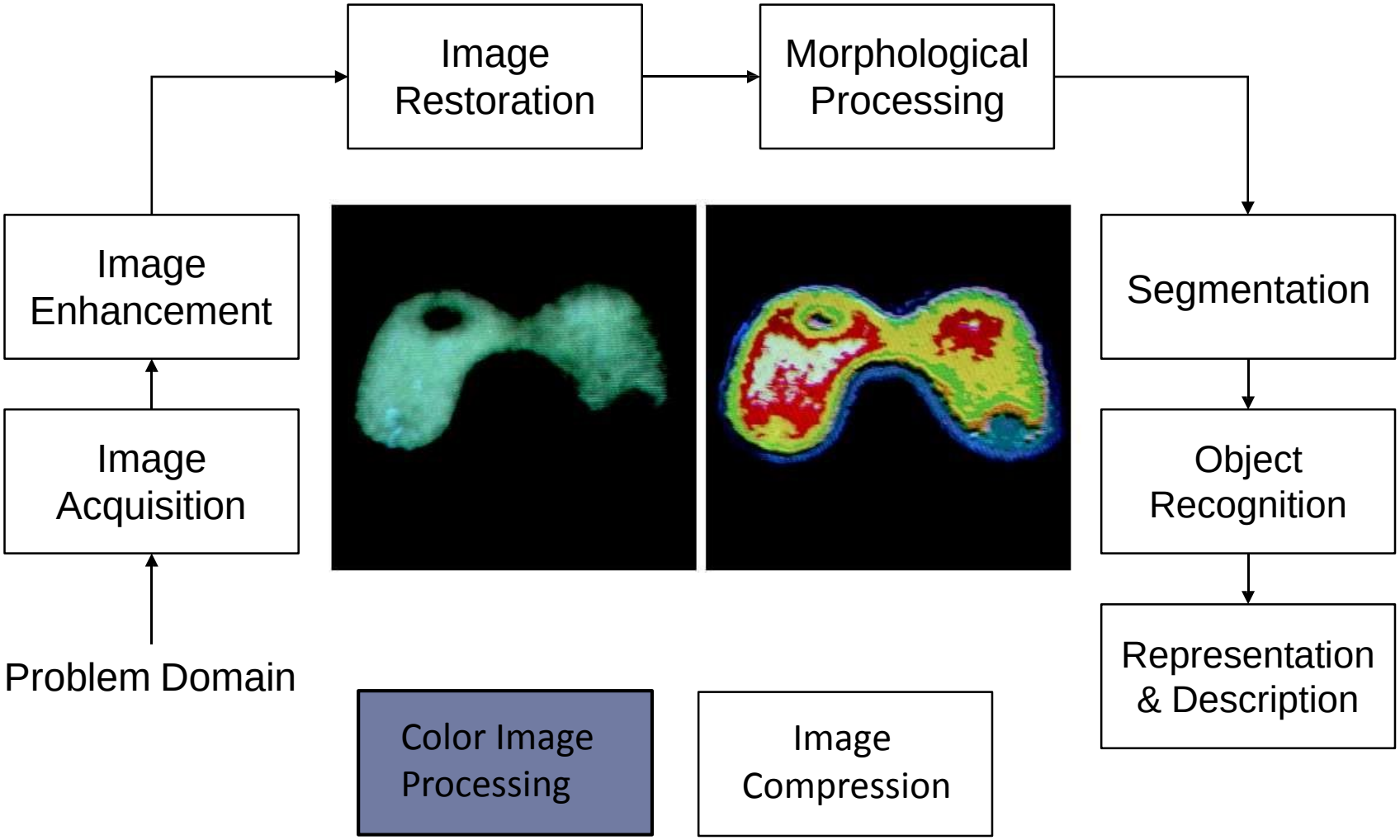


Image Compression



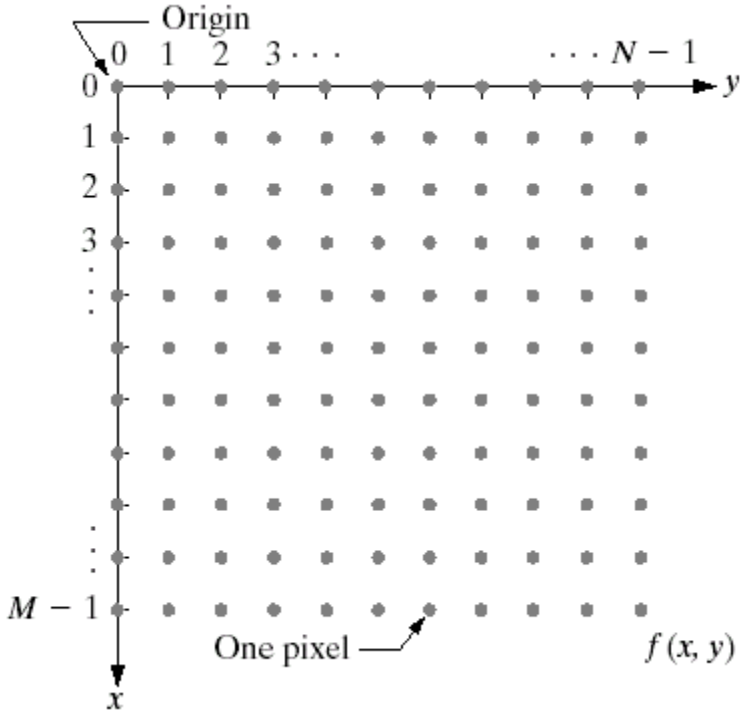
Color Image Processing



Components of an Image Processing System

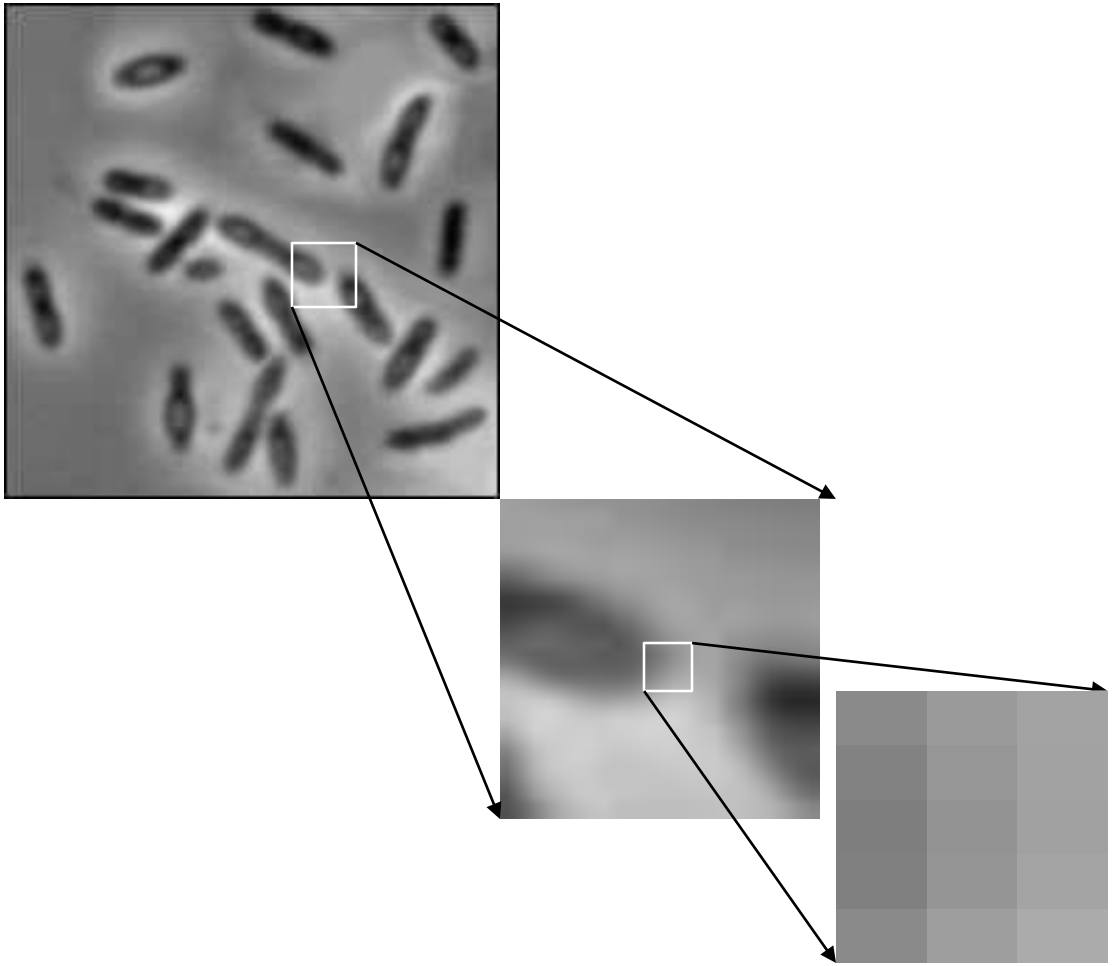
- Image sensor
- Specialized image processing hardware
  - This unit performs functions that require fast data throughputs (e.g., digitizing and averaging video images at 30 frames s) that the typical main computer cannot handle.
- Computer
- Image processing software
- Mass storage
- Image display
- Hardcopy
- Networking

Representing Digital Images



An image with size $M \times N$ and L ($=2^k$) discrete gray level for each pixel, the number of bits required to store a digitized image is $M \times N \times k$.

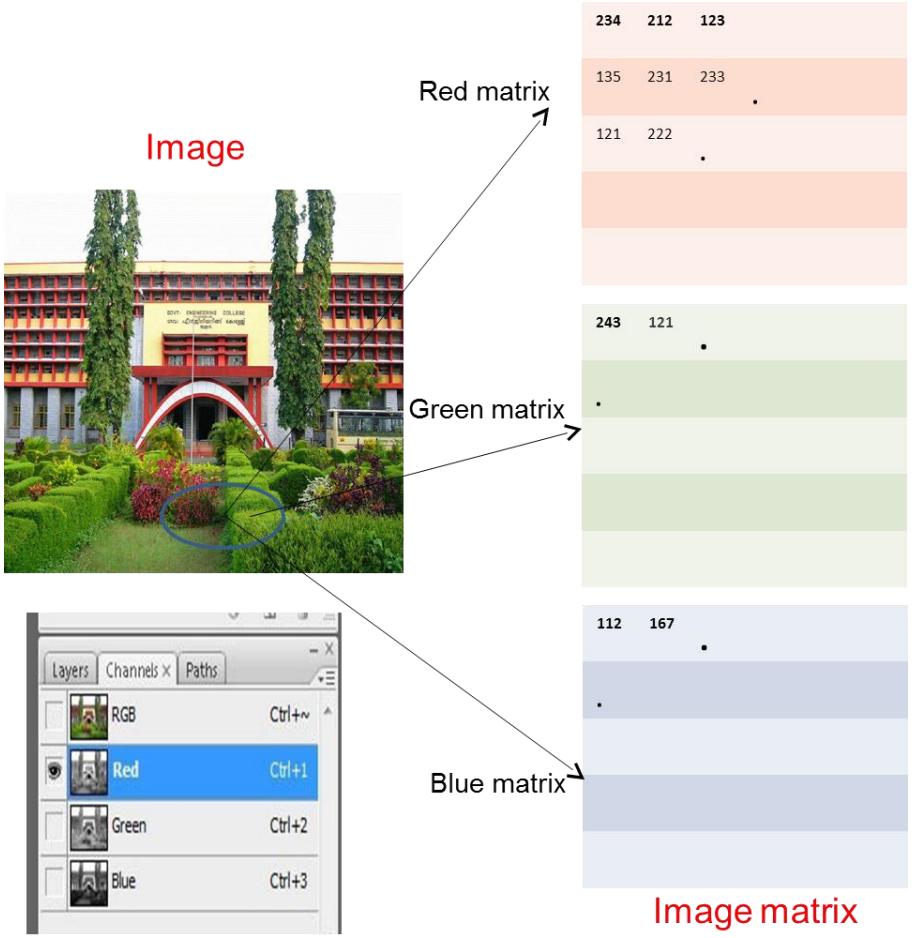
Monochromatic Image



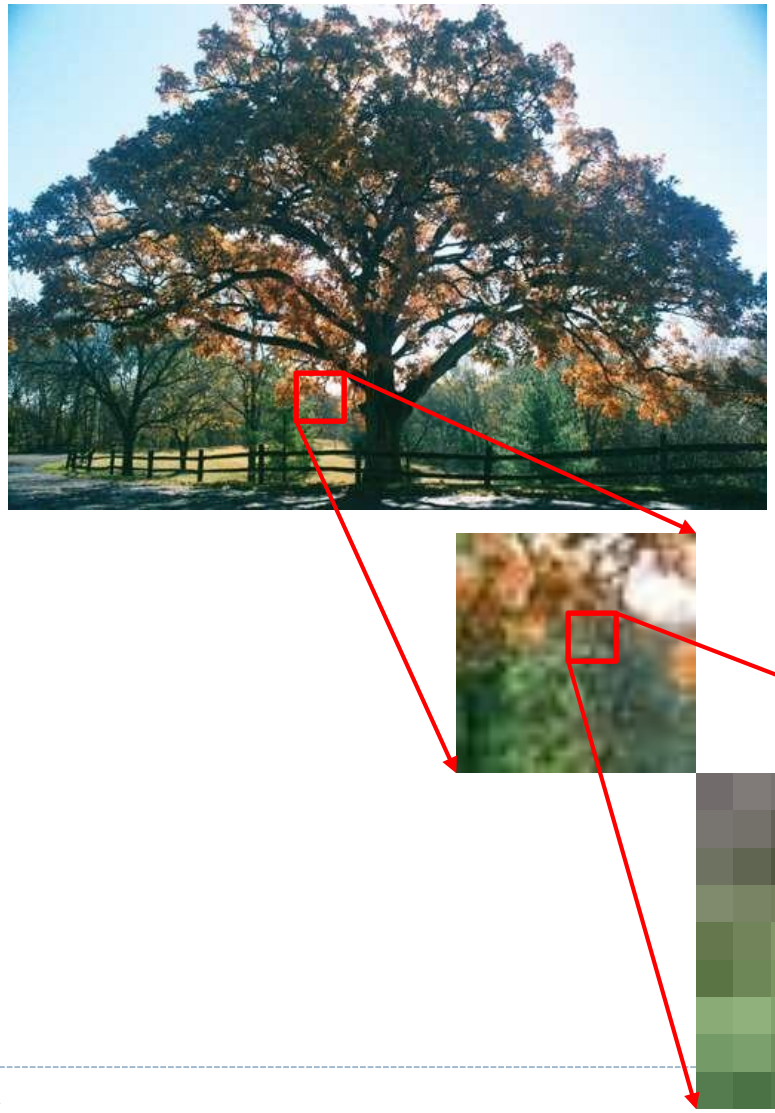
Gray scale values

$$\begin{bmatrix} 10 & 10 & 16 & 28 \\ 9 & 6 & 26 & 37 \\ 15 & 25 & 13 & 22 \\ 32 & 15 & 87 & 39 \end{bmatrix}$$

Color Image


Image
Red matrix
Green matrix
Blue matrix
234 212 123
135 231 233
121 222
243 121
112 167
Layers
Channels
Paths
RGB Ctrl+~
Red Ctrl+1
Green Ctrl+2
Blue Ctrl+3
Image matrix

Color Image

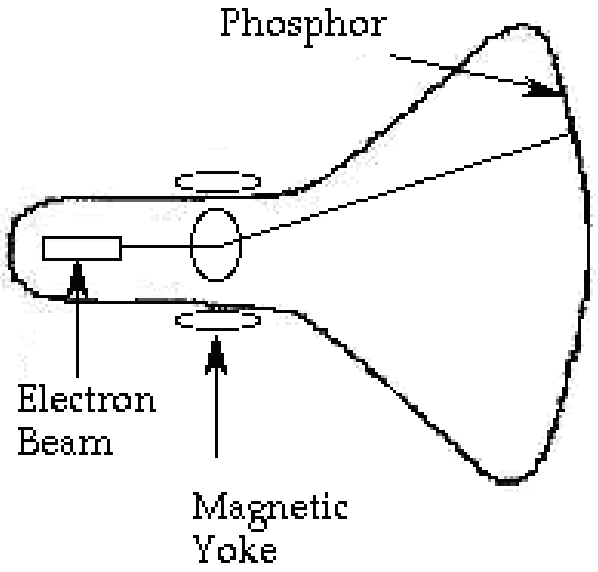


Each pixel contains a vector representing red, green and blue components

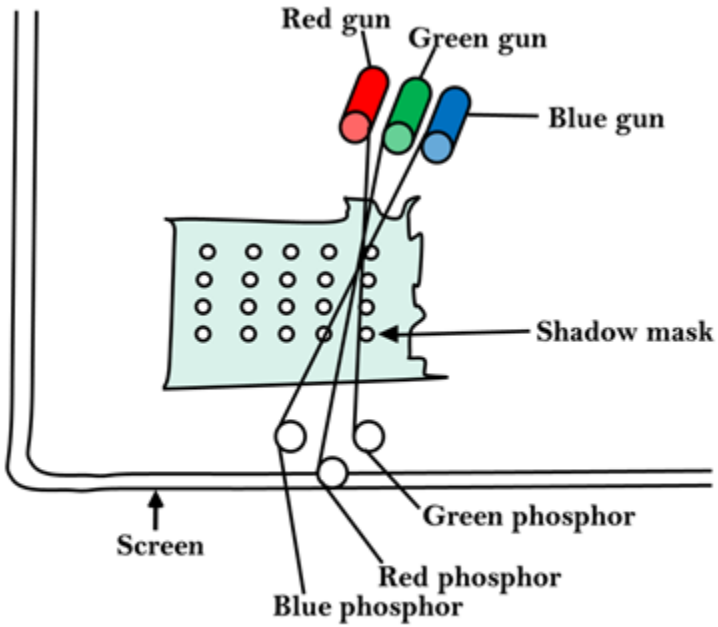
RGB components

$$\begin{bmatrix} 10 & 10 & 16 & 28 \\ 9 & \begin{bmatrix} 65 & 70 & 56 & 43 \\ 32 & \begin{bmatrix} 99 & 70 & 56 & 78 \\ 60 & 90 & 96 & 67 \\ 85 & 85 & 43 & 92 \\ 32 & 65 & 87 & 99 \end{bmatrix} \\ 21 & \\ 54 & \end{bmatrix} \\ 15 & \\ 32 & \end{bmatrix}$$

Display Device (Gray)



Display Device (Color)

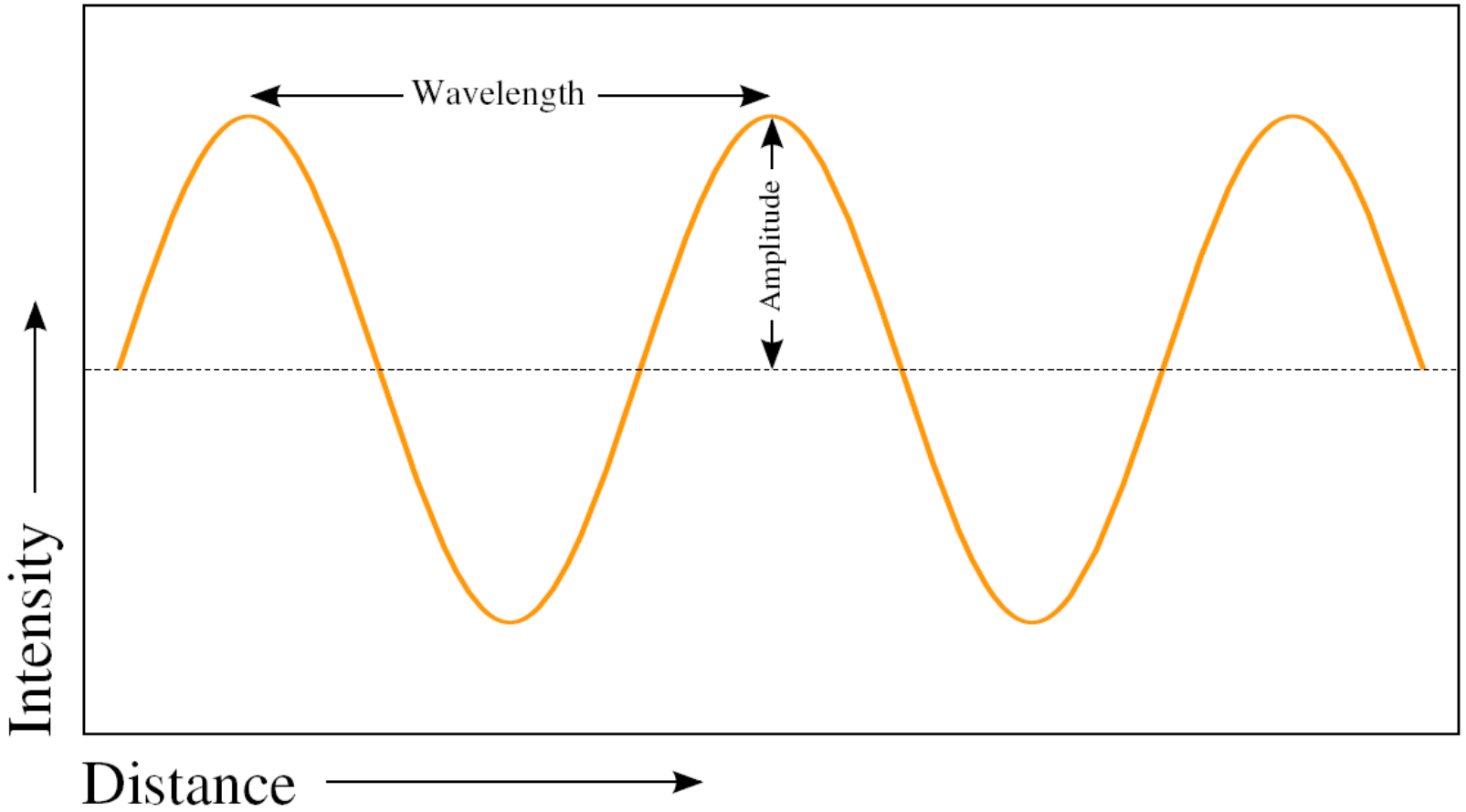


The Shadow mask CRT

2 Digital Image Fundamentals

- Electromagnetic radiation (Electromagnetic wave) can be thought of as a stream of massless particles, called photons, that carry energy in straight-line paths through space.
- They travel in a wavelike (sinusoidal) pattern and move through empty space at the speed of light.
- They travel somewhat more slowly through air, water, glass, or other media.
- Each massless particle contains a certain amount (or bundle) of energy.

Electromagnetic Wave



Electromagnetic Energy

- Electromagnetic energy is proportional to frequency ($E=h\nu$)
- The higher frequency (shorter wavelength) electromagnetic phenomena carry more energy per photon.

Electromagnetic Spectrum

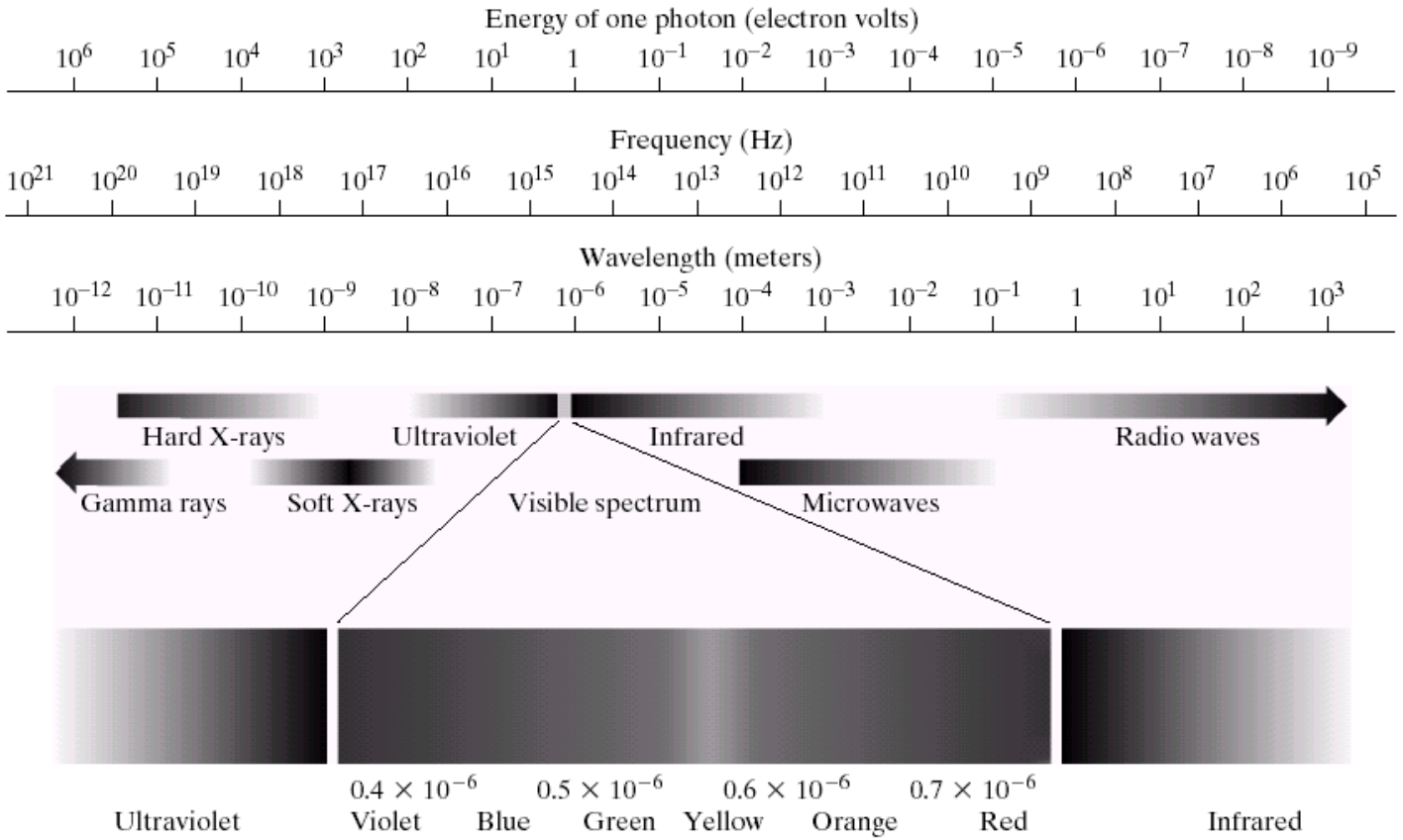


FIGURE 2.10 The electromagnetic spectrum. The visible spectrum is shown zoomed to facilitate explanation, but note that the visible spectrum is a rather narrow portion of the EM spectrum.

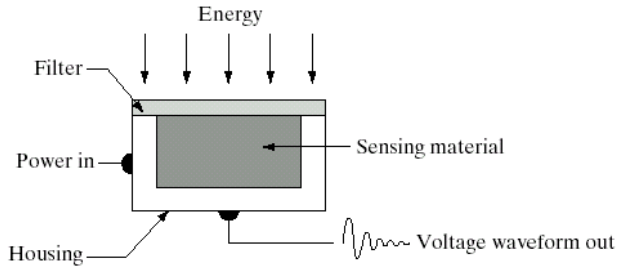
Light and Color Sensing

- Light is a particular type of electromagnetic radiation that can be seen and sensed by the human eye.
- Chromatic light spans the electromagnetic energy spectrum from approximately 0.43 to 0.79 μm.
- The colors that humans perceive in an object are determined by the nature of the light *reflected* from the object.

Quality of a Chromatic Light

- Three basic quantities are used to describe the quality of a chromatic light source:
  - Radiance
    - *Radiance* is the total amount of energy that flows from the light source, and it is usually measured in watts (W).
  - Luminance
    - *Luminance*, measured in lumens (lm), gives a measure of the amount of energy an observer *perceives* from a light source.
  - Brightness.
    - *Brightness* is a subjective descriptor of light perception that is practically impossible to measure.
    - It embodies the achromatic notion of intensity and is one of the key factors in describing color sensation.

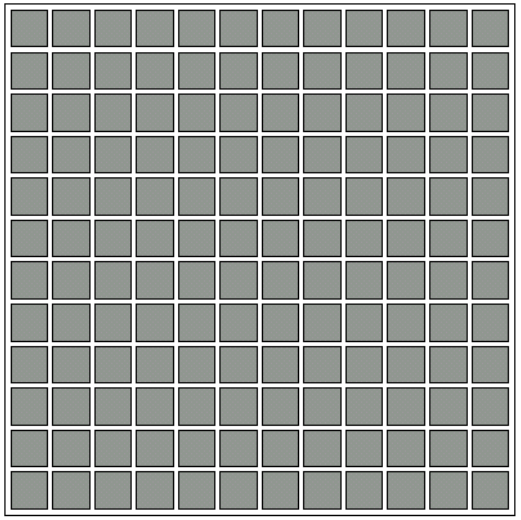
Image Sensing and Acquisition



Single Sensor



Line Sensor



Array Sensor

Image Sensing and Acquisition

- Incoming energy is transformed into a voltage by the combination of input electrical power and sensor material that is responsive to the particular type of energy being detected.
- The output voltage waveform is the response of the sensor(s), and a digital quantity is obtained from each sensor by digitizing its response.

Image Sensor

- The two main types of digital image sensors are the charge-coupled device (CCD) and the active-pixel sensor (CMOS sensor).
- Both CCD and CMOS sensors are based on MOS technology.
- MOS capacitors being the building blocks of a CCD.
- MOSFET amplifiers being the building blocks of a CMOS sensor.

Image Acquisition Using a Single Sensor

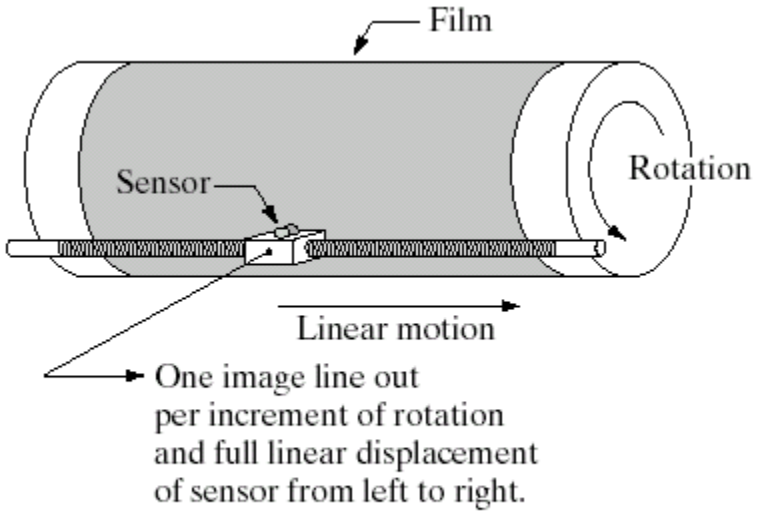
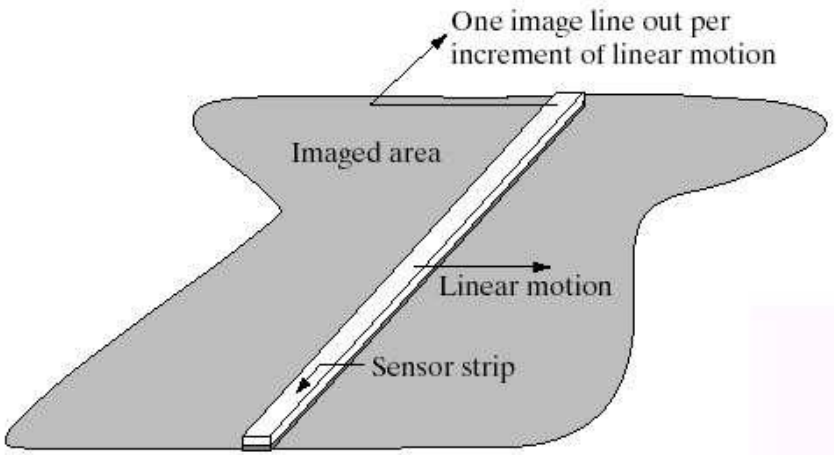


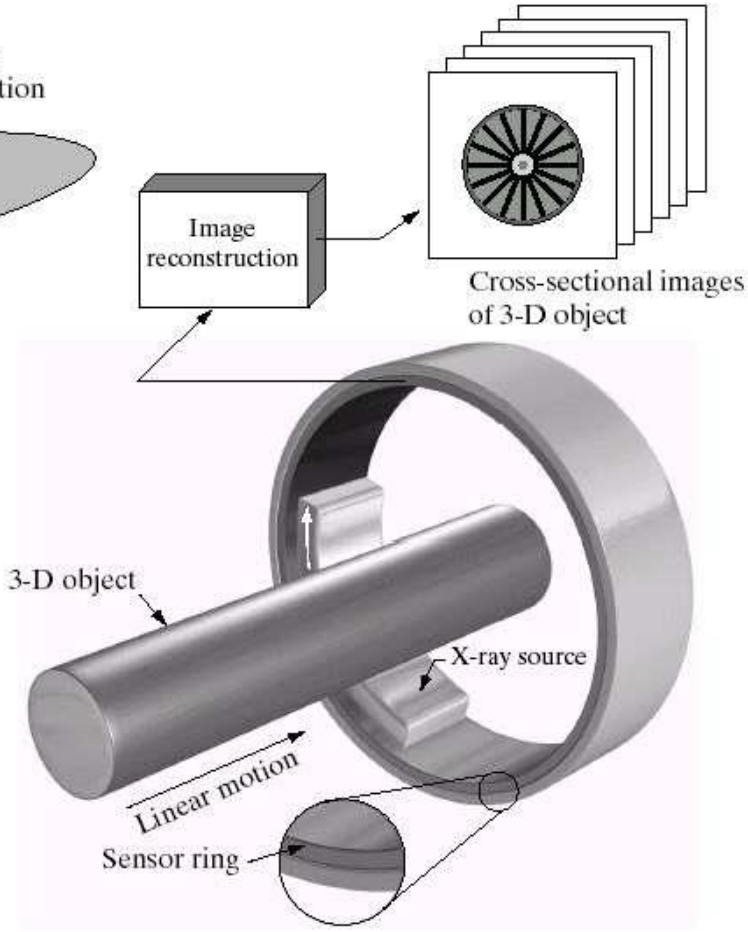
FIGURE 2.13 Combining a single sensor with motion to generate a 2-D image.

Image Acquisition Using Sensor Strips





Fingerprint sweep sensor



Computerized Axial Tomography (CAT)

Image Acquisition Using Sensor Arrays

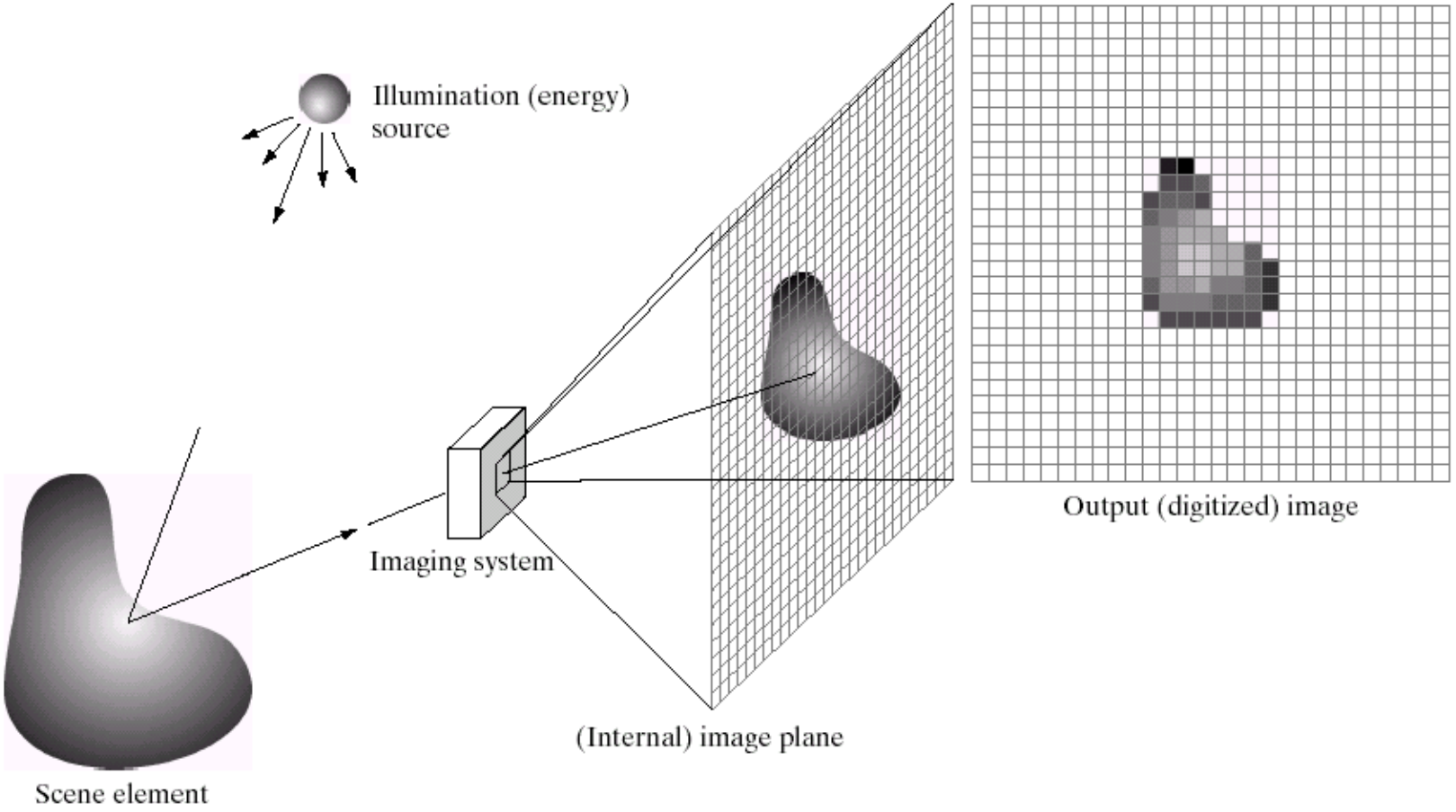


Image Acquisition Using Digital Camera

- The sunlight is used as a source of energy.
- When the sunlight falls upon the object, then the object reflects light.
- The front end of the imaging system is a lens, which projects the viewed scene onto the lens focal plane.
- The sensor array is coincident with the focal plane.
- The sensor array produces outputs (analog voltage signal) proportional to the integral of the light received at each sensor.
- This data from each sensor is converted into a digital form. This involves sampling and quantization.

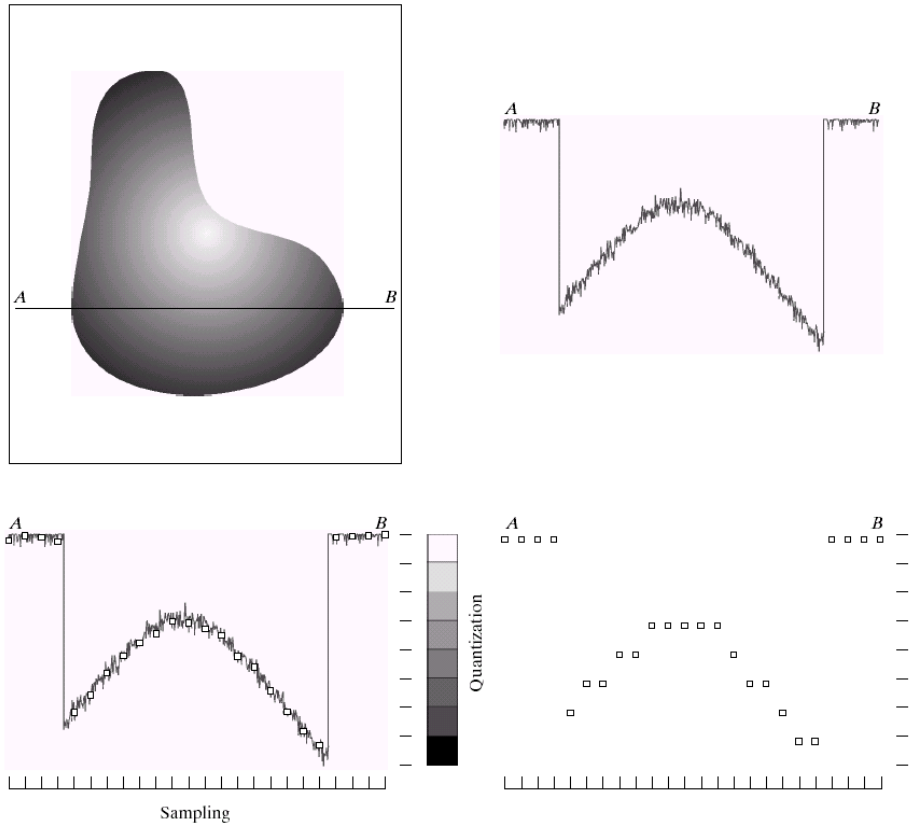
Energy Radiation

- The incoming energy is characterized by two components:
  - The amount of source illumination incident on the scene being viewed, denoted by *i*.
  - The amount of illumination reflected by the objects in the scene, denoted by *r*.
- The incoming energy is *i***r.*

Image Sampling and Quantization

- The output of most sensors is a continuous voltage waveform whose amplitude and spatial behavior are related to the physical phenomenon being sensed.
- Digitizing the coordinate values is called *sampling*.
- Digitizing the amplitude values is called *quantization*.

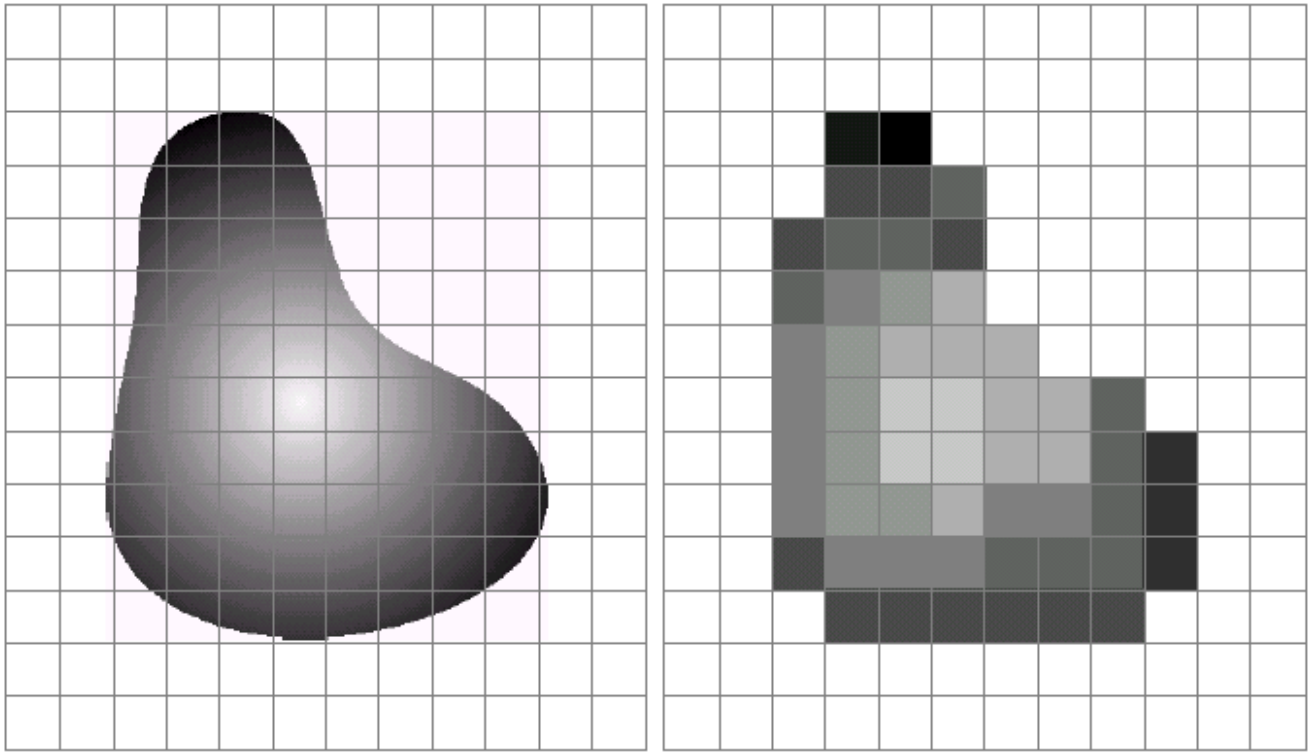
Image Sampling and Quantization



a b
c d

FIGURE 2.16 Generating a digital image. (a) Continuous image. (b) A scan line from *A* to *B* in the continuous image, used to illustrate the concepts of sampling and quantization. (c) Sampling and quantization. (d) Digital scan line.

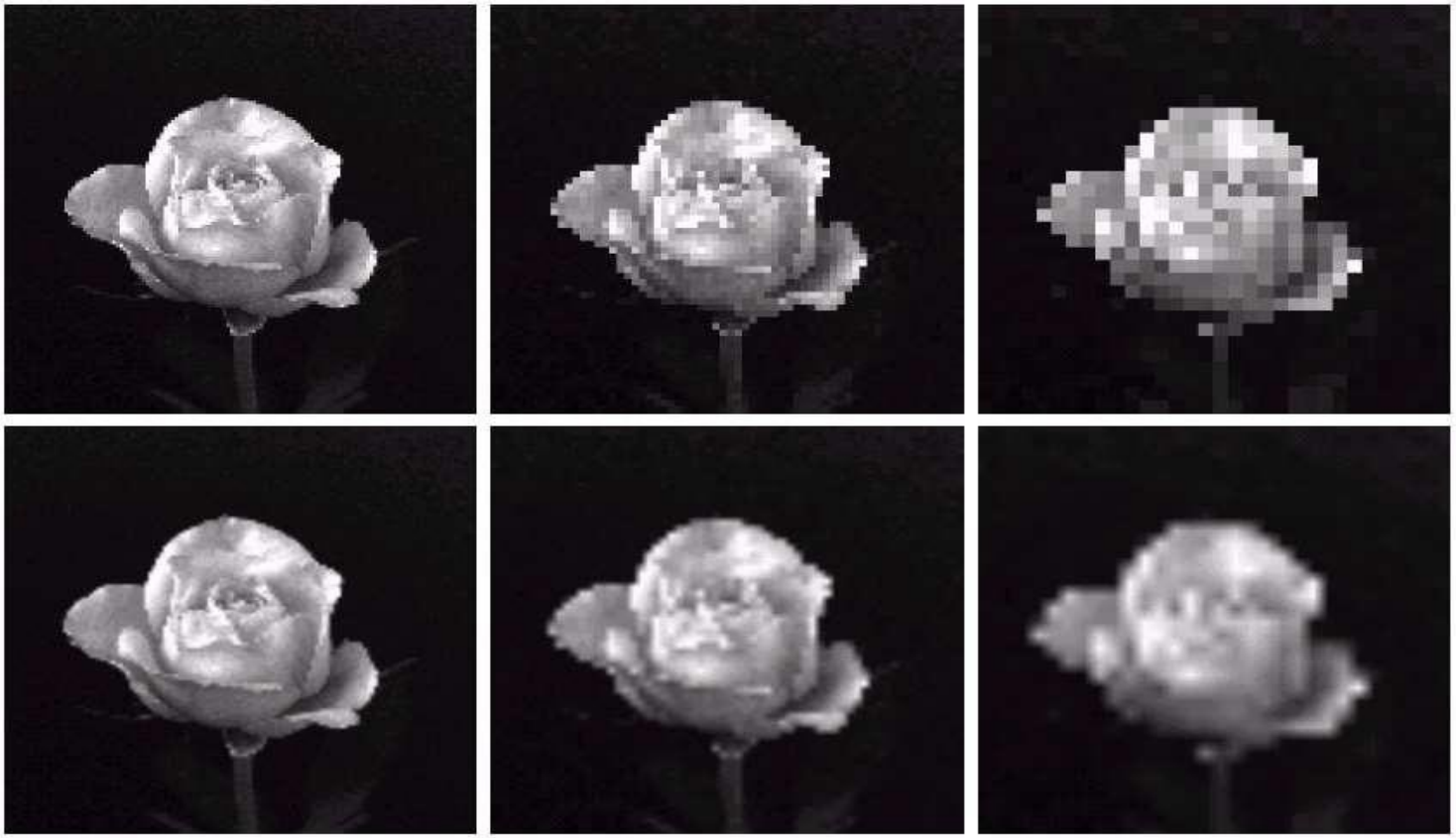
Image Sampling and Quantization



a b

FIGURE 2.17 (a) Continuos image projected onto a sensor array. (b) Result of image sampling and quantization.

Zooming



a b c
d e f

FIGURE 2.25 Top row: images zoomed from 128 × 128, 64 × 64, and 32 × 32 pixels to 1024 × 1024 pixels, using nearest neighbor gray-level interpolation. Bottom row: same sequence, but using bilinear interpolation.

Zooming

- Zooming means oversampling.
- Zooming requires two steps:
  - The creation of new pixel locations.
  - The assignment of gray levels to those new locations.
- Three zooming methods:
  - Nearest neighbor interpolation
  - Pixel replication
  - Bilinear interpolation

Zooming: Nearest Neighbor Interpolation

- Create new dimensional grid over the original image.
- The spacing in the new grid would be less than one pixel because we are fitting it over a smaller image.
- For any point in the overlay, look the closest pixel in the original image and assign its gray level to the new pixel in the grid.

Shrinking

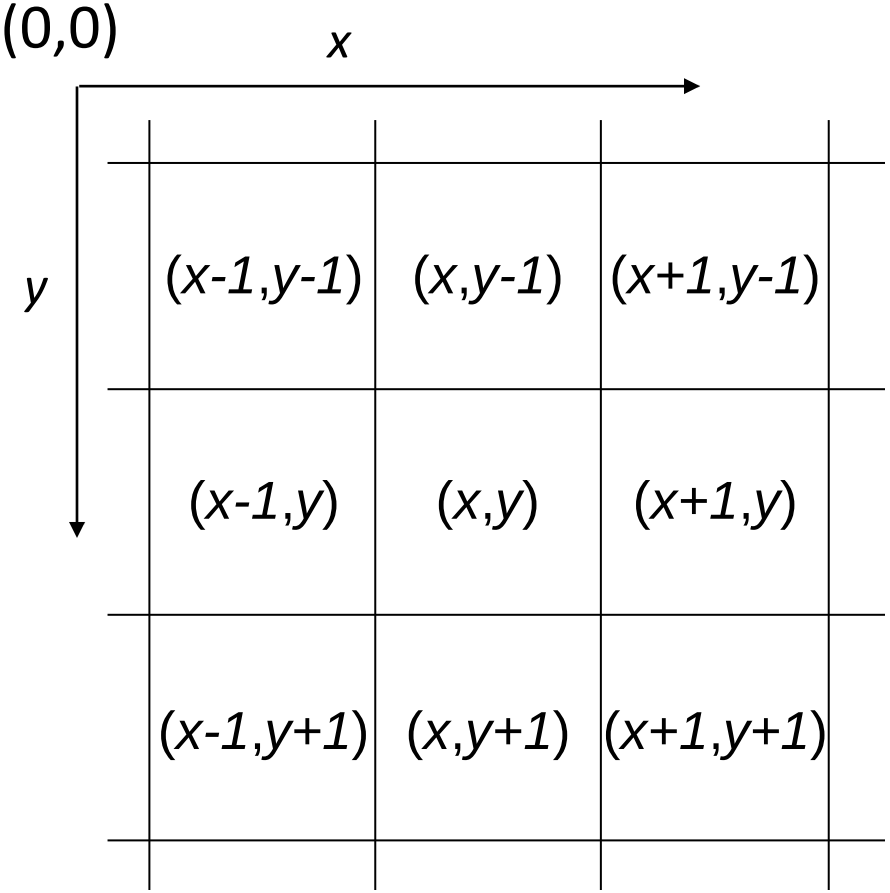
- Shrinking means undersampling.
- The shrinking is accomplished by deleting the appropriate number of rows and columns from the original image.



FIGURE 2.19 A 1024 × 1024, 8-bit image subsampled down to size 32 × 32 pixels. The number of allowable gray levels was kept at 256.

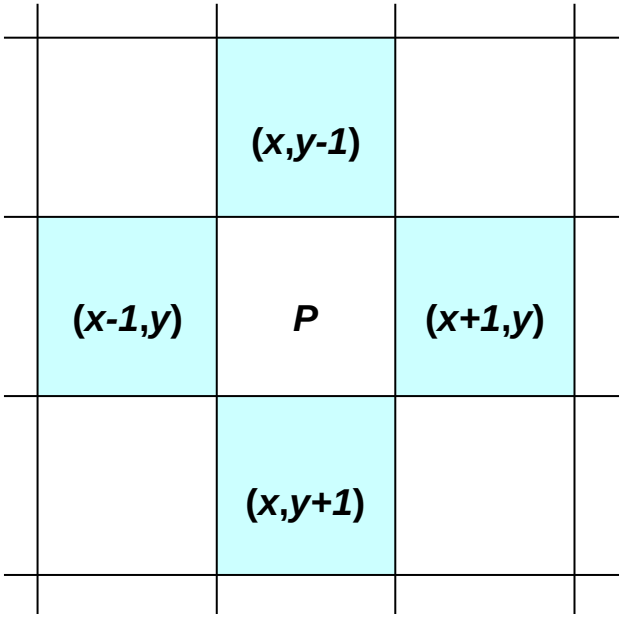
Relationships Between Pixels

- (0,0)



Neighbors of a Pixel

- Neighborhood relation is used to determine adjacent pixels. It is useful for analyzing regions.



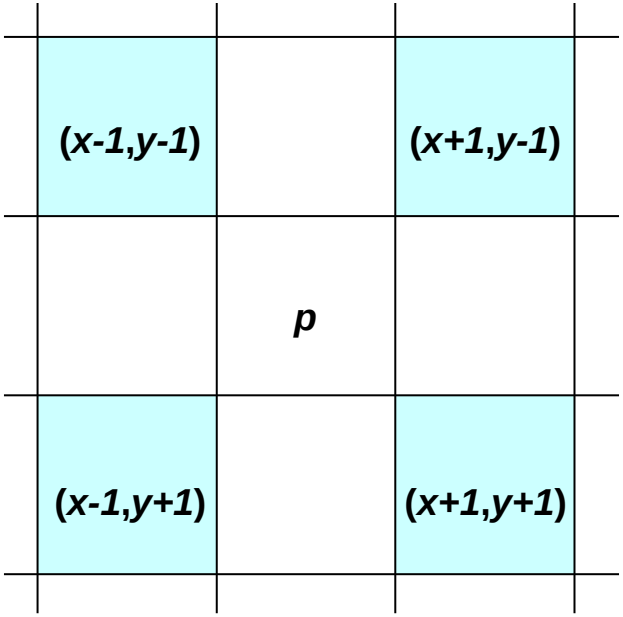
4-neighbors of *p*:

$$N_4(p) = \{ (x-1,y), (x+1,y), (x,y-1), (x,y+1) \}$$

4-neighborhood relation considers only vertical and horizontal neighbors.

Note: $q \in N_4(p)$ implies $p \in N_4(q)$

Neighbors of a Pixel

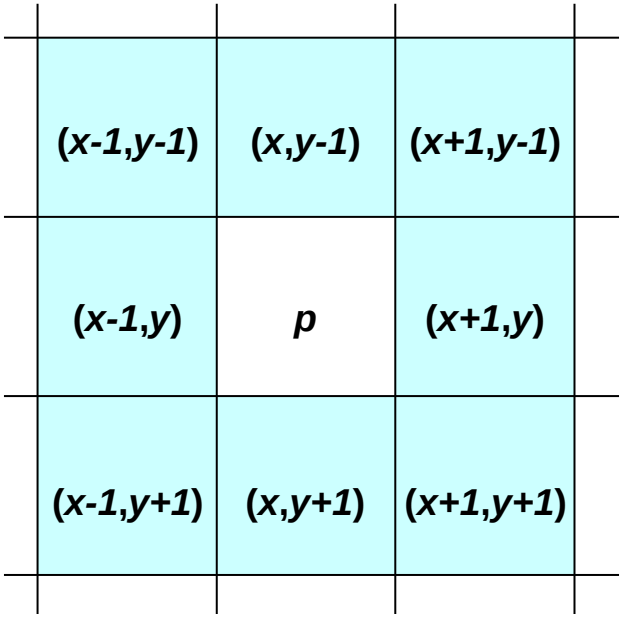


Diagonal neighbors of ***p***:

$$N_D(p) = \{ (x-1,y-1), (x+1,y-1), (x-1,y+1), (x+1,y+1) \}$$

Diagonal -neighborhood relation considers only diagonal neighbor pixels.

Neighbors of a Pixel



8-neighbors of ***p***:

$$N_8(p) = \{ (x-1,y-1), (x,y-1), (x+1,y-1), (x-1,y), (x+1,y), (x-1,y+1), (x,y+1), (x+1,y+1) \}$$

8-neighborhood relation considers all neighbor pixels, i.e., $N_8(p) = N_4(p) \cup N_D(p)$

Adjacency and Connectivity

- Connectivity between pixels determines regions and boundaries.
- Two pixels are connected if they are neighbors and if their gray levels satisfy a specified criterion of similarity (say, if their gray levels are equal).

Adjacencies

1. **4-adjacency:** Two pixels *p* and *q* with values from *V* are 4-adjacent if *q* is in the set $N_4(p)$.
2. **8-adjacency:** Two pixels *p* and *q* with values from *V* are 8-adjacent if *q* is in the set $N_8(p)$.
3. ***m*-adjacency (mixed-adjacency):**

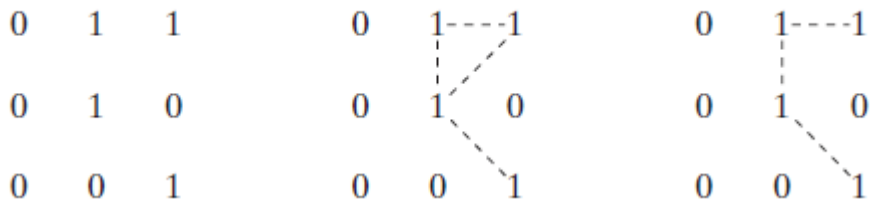
   Two pixels *p* and *q* with values from *V* are m-adjacent if :

   i. *q* is in $N_4(p)$, or

   ii. *q* is in $N_D(p)$ and the set $N_4(p) \cap N_4(q)$ has no pixel whose values are from V (i.e., $N_4(p) \cap N_4(q) = \phi$) .

m-Adjacency

- m-adjacency is a modification of 8-adjacency.
- It is introduced to eliminate the ambiguities that often arise when 8-adjacency is used.
- For example:



a b c

FIGURE 2.26 (a) Arrangement of pixels; (b) pixels that are 8-adjacent (shown dashed) to the center pixel; (c) *m*-adjacency.

Distance Measures

- For pixels p, q and z, with coordinates (x, y), (s, t) and (v, w), respectively, D is a distance function if:

  (a) $D(p, q) \geq 0$ $\quad\quad$ ($D(p, q) = 0$ iff $p = q$),

  (b) $D(p, q) = D(q, p)$, and

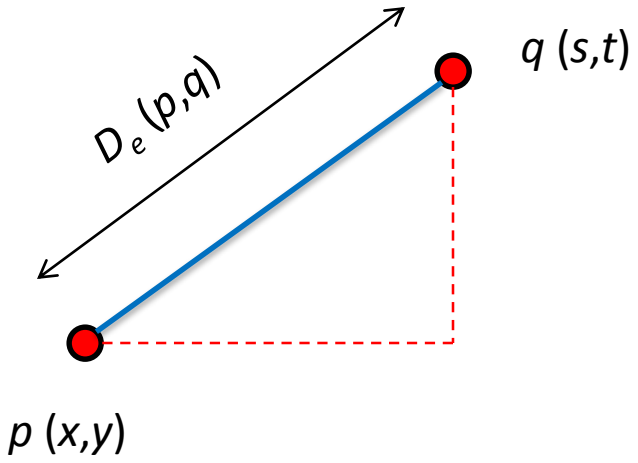
  (c) $D(p, z) \leq D(p, q) + D(q, z)$.

Euclidean Distance

- The *Euclidean Distance* between p and q is defined as:

  $D_e(p, q) = [(x - s)^2 + (y - t)^2]^{1/2}$

- Pixels having a distance less than or equal to some value r from (x, y) are the points contained in a disk of radius r centered at (x, y).

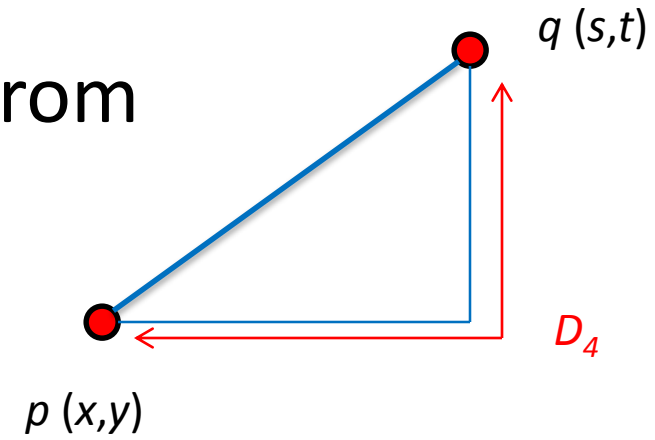


D_4 Distance Or City-Block Distance

- The D_4 *distance* (also called *city-block distance*) between p and q is defined as:

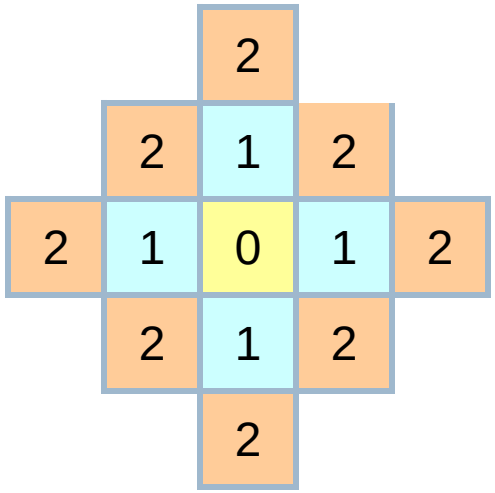
  $D_4(p, q) = |x - s| + |y - t|$

- Pixels having a D_4 distance from (x, y), less than or equal to some value r form a diamond centered at (x, y).



D_4 Distance Or City-Block Distance

- The pixels with distance $D_4 \leq 2$ from (x, y) form the following contours of constant distance.
- The pixels with $D_4 = 1$ are the 4-neighbors of (x, y).

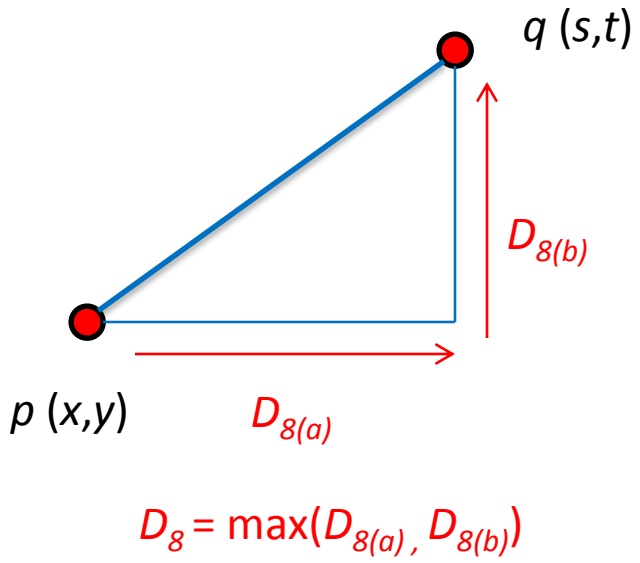


D_8 Distance Or Chessboard Distance

- The D_8 *distance* (also called c*hessboard distance*) between p and q is defined as:

  $D_8(p, q) = \max(|x - s|, |y - t|)$

- Pixels having a D_8 distance from (x, y), less than or equal to some value r form a square centered at (x, y).



D_8 Distance Or Chessboard Distance

- D_8 distance ≤ 2 from (x, y) form the following contours of constant distance.

2	2	2	2	2
2	1	1	1	2
2	1	0	1	2
2	1	1	1	2
2	2	2	2	2

3 Image Enhancement in the Spatial Domain

- Image Enhancement:
  - The objective of image enhancement is to process an image so that the result is more suitable than the original image for a specific application.
- Reasons:
  - Contrast enhancement
  - Visual improvement
  - Image understanding

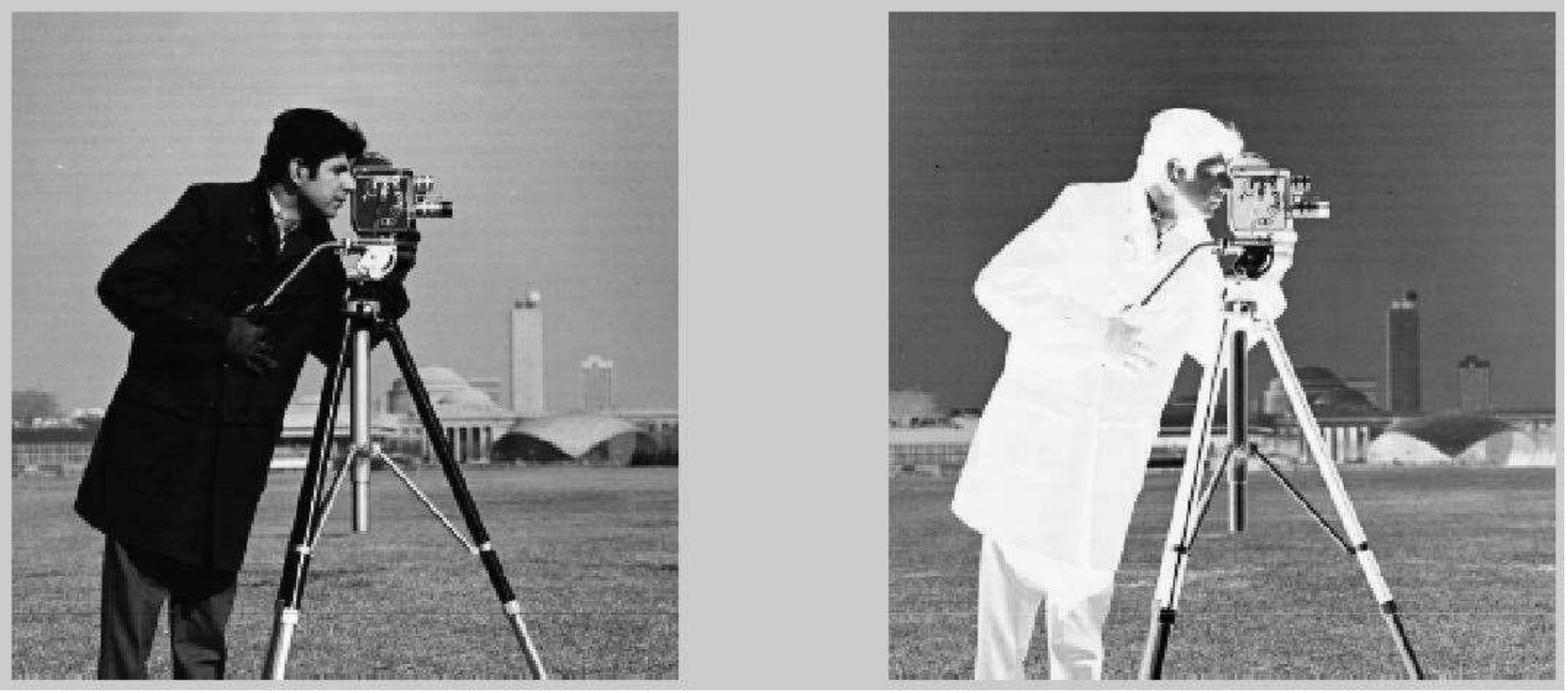
Enhancement Methods

- Spatial Domain
  - Based on direct manipulation of pixels in an image
- Frequency Domain
  - Based on modifying the Fourier transform of an image

Image Negatives

- $s = L - 1 - r$
  - s is the output intensity value
  - L is the highest intensity levels
  - r is the input intensity value
- Particularly suited for enhancing white or gray detail embedded in dark regions of an image, especially when the black areas are dominant in size.
- Produces photographic negative.
- Some details are easier to spot if we go from black and white to white and black.

Image Negative



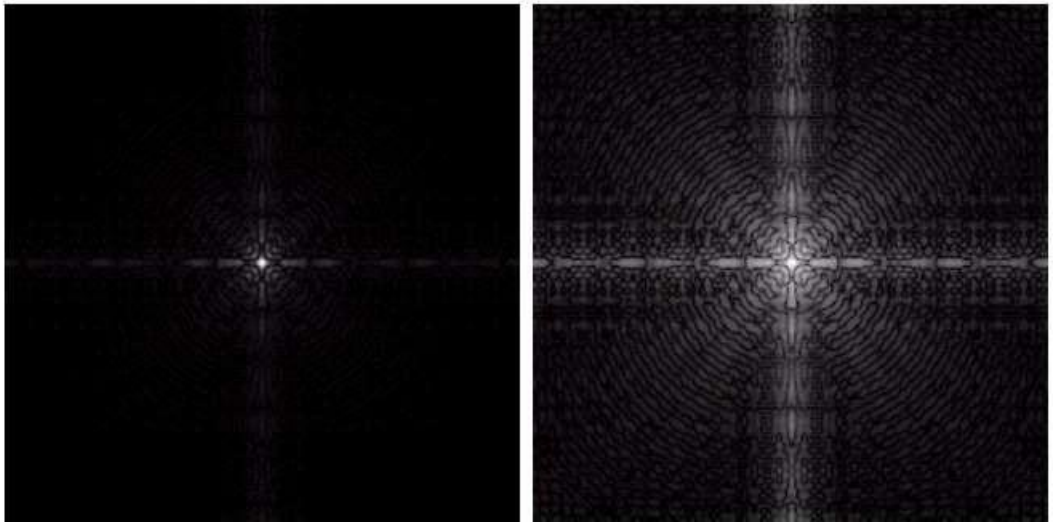
Log Transformations

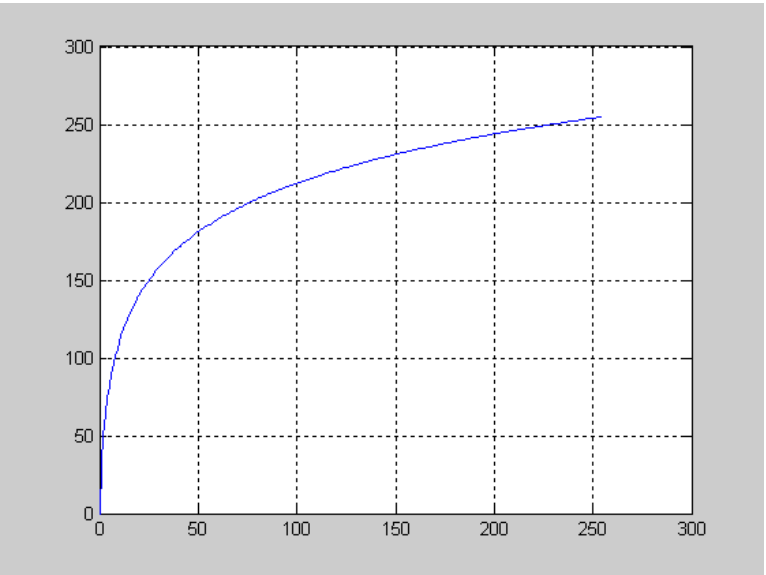
- $s = c\log(1 + r)$, c is constant.
- It maps a narrow range of low intensity values in the input into a wide range of output levels.
- The opposite is true of higher values of input levels.
- It expands the values of dark pixels in an image while compressing the higher level values.
- It compresses the dynamic range of images with large variations in pixel values.
- The opposite is true of the inverse log transformation.

Log Transformations

a b

FIGURE 3.5
(a) Fourier spectrum.
(b) Result of applying the log transformation given in Eq. (3.2-2) with $c = 1$.





$$T(r) = c\log(1+r)$$



Power Law (Gamma) Transformations

- $s = cr^{\gamma}$, c and γ are both positive constants.
- With fractional values($0<\gamma<1$) of gamma map a narrow range of dark input values into a wider range of output values, with the opposite being true for higher values ($\gamma > 1$)of input levels.
- Variety of devices used for image capture, printing, and display respond according to a power law.
- Process used to correct these power law response phenomena is called *gamma correction.*

Power Law (Gamma) Transformations

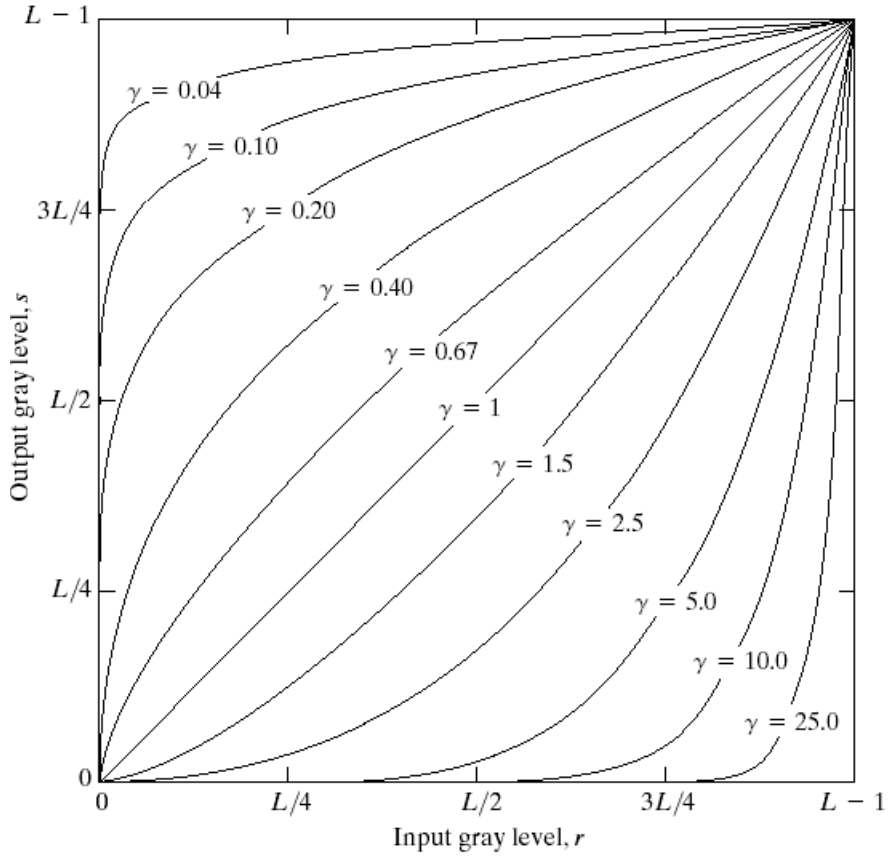
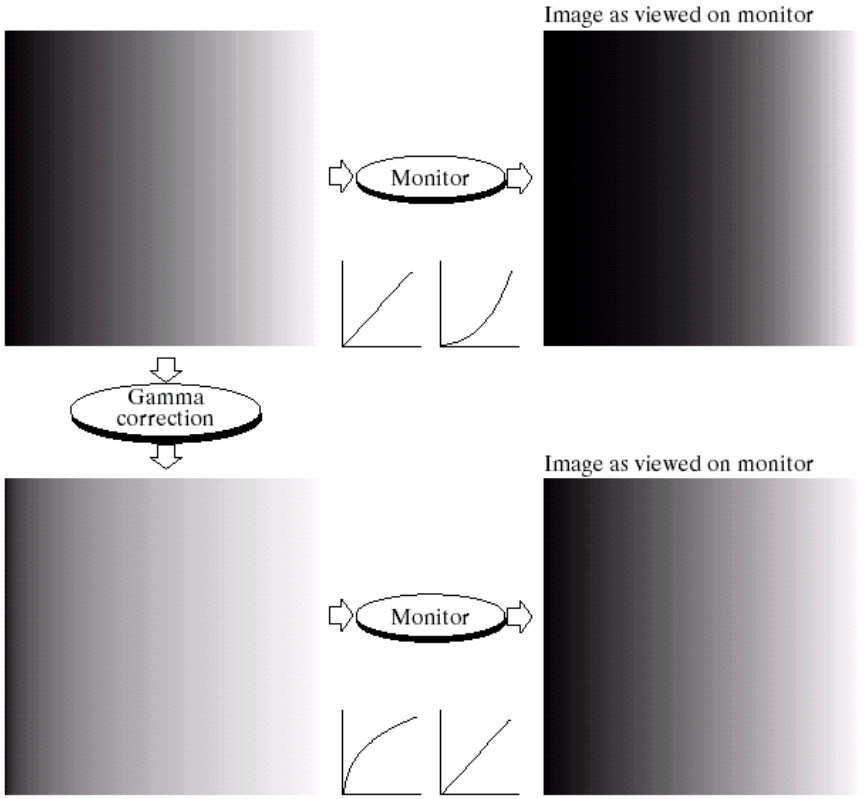


FIGURE 3.6 Plots of the equation $s = cr^{\gamma}$ for various values of γ ($c = 1$ in all cases).

Power Law (Gamma) Transformations

- CRT devices have Intensity-voltage response functions that are power functions.
- They vary in exponents from 1.8 to 2.5.



a b
c d

FIGURE 3.7
(a) Linear-wedge gray-scale image. (b) Response of monitor to linear wedge. (c) Gamma-corrected wedge. (d) Output of monitor.

$$T(r) = r^{1/2.5} = r^{0.4}$$

Histogram Processing

- Histogram of a digital image with intensity levels in the range [0,L-1] is a discrete function $h(r_k) = n_k$, where r_k is the kth intensity value and n_k is the number of pixels in the image with intensity r_k.
- Normalized histogram $p(r_k)=n_k/MN$, for k = 0,1,2..... L-1. Spatial image resolution is M×N.
- Histogram manipulation can be used for image enhancement.
- Information inherent in histogram is also quite useful in other image processing applications, such as image compression and segmentation.

Histogram



Image

0	7	3	2	3
0	0	0	6	7
7	7	2	2	0
1	1	0	4	1
0	0	7	4	1

Image matrix

Similarly

INTENSITY r	0	1	2	3	4	5	6	7
NUMBER of pixels of r h(r)	h(r0)=**8**	h(r1)=**4**	h(r2)=**3**	h(r3)=**2**	h(r4)=**2**	h(r5)=**0**	h(r6)=**1**	h(r7)=**5**

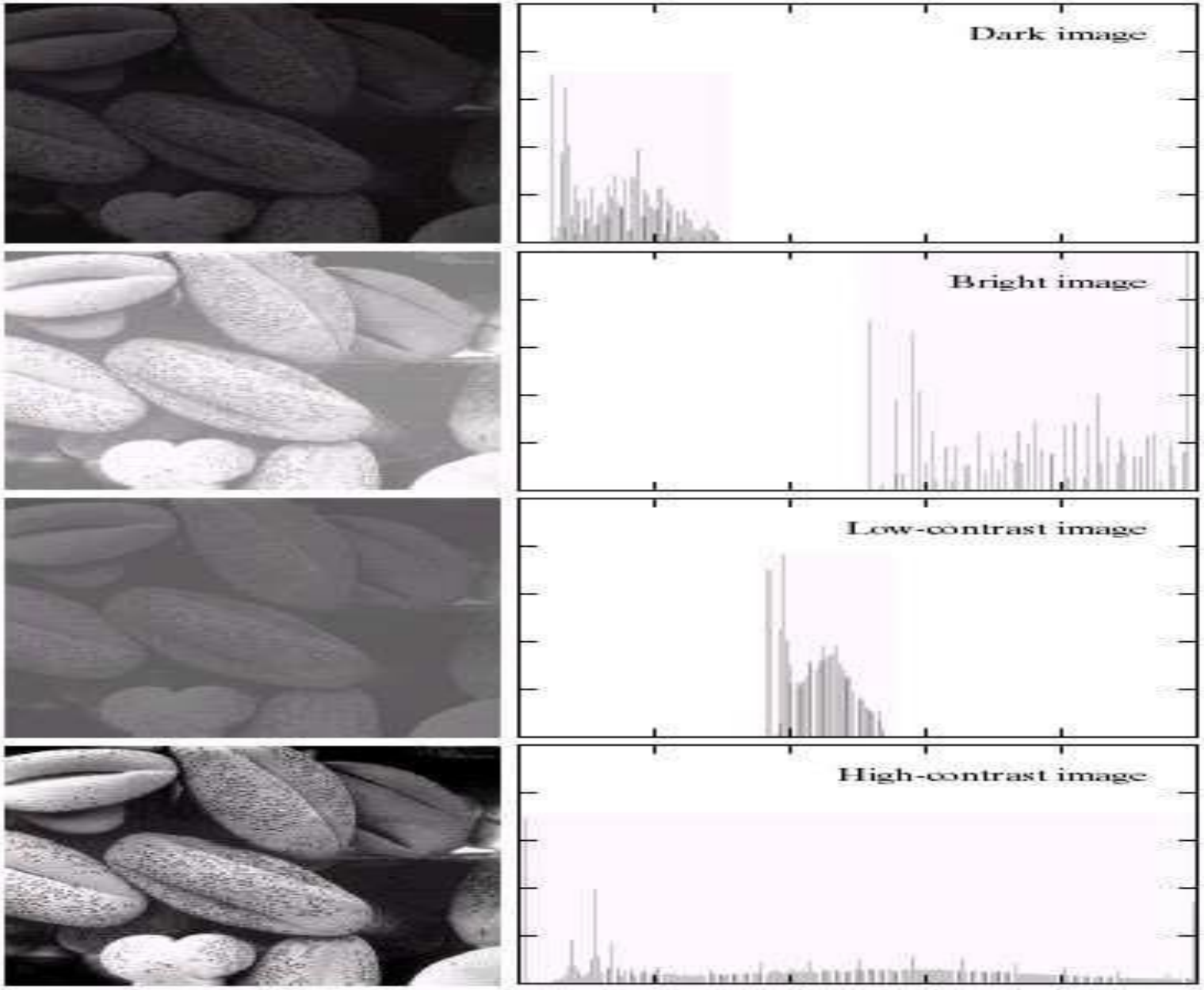
Histogram

r	0	1	2	3	4	5	6	7
h(r)	8	4	3	2	2	0	1	5
p(r) h(r)/(5*5)	8/25	4/25	3/25	2/25	2/25	0/25	1/25	5/25

HISTOGRAM - ***h(r)*** - Y axis - ***number*** of intensities

NORMALIZED *HISTOGRAM* - ***p(r)*** - Y axis - ***probability*** of intensities

Histogram Analysis



Histogram Equalization

- Intensity mapping form

$$s = T(r),\ 0 \le r \le L-1$$

Conditions:

a) T(r) is a monotonically increasing function in the interval [0, L-1] and

b) $0 \le T(r) \le L-1 \qquad for\ 0 \le r \le L-1$

In some formulations, we use the inverse in which case

$$r = T^{-1}(s),\ 0 \le s \le 1$$

(a) change to

a´) T(r) is a strictly monotonically increasing function in the interval [0, L-1]

Histogram Equalization

- Intensity levels in an image may be viewed as random variables in the interval [0,L-1]
- Fundamental descriptor of a random variable is its probability density function (PDF).
- Let $p_r(r)$ and $p_s(s)$ denote the PDFs of r and s Respectively.

$$p_s(s) = p_r(r)\left|\frac{dr}{ds}\right|$$

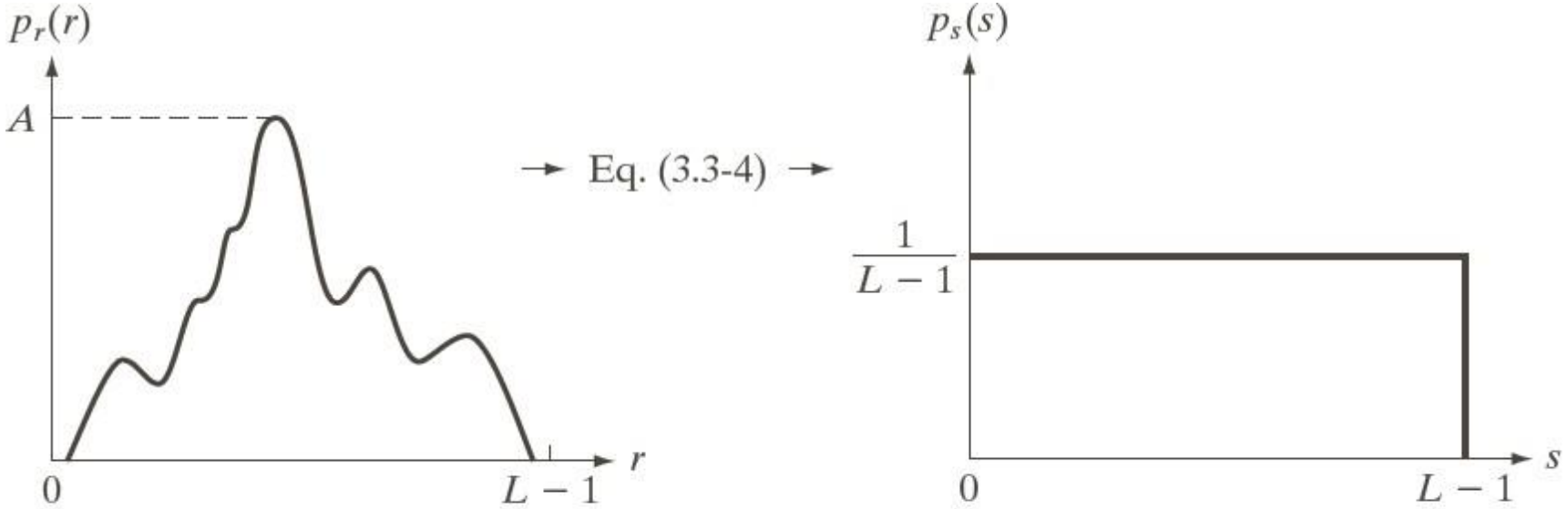
$$s = T(r) = (L-1)\int_0^r p_r(w)dw$$

$$\frac{ds}{dr} = \frac{dT(r)}{dr} = (L-1)\frac{d}{dr}\left[\int_0^r p_r(w)dw\right] = (L-1)p_r(r)$$

Histogram Equalization

$$p_s(s) = \frac{1}{L-1}$$

$$s_k = T(r_k) = (L-1)\sum_{j=0}^{k} p_r(r_j) = (L-1)\sum_{j=0}^{k} \frac{n_j}{MN}, \quad k = 0,1,2,\ldots,L-1$$



a b

FIGURE 3.18 (a) An arbitrary PDF. (b) Result of applying the transformation in Eq. (3.3-4) to all intensity levels, r. The resulting intensities, s, have a uniform PDF, independently of the form of the PDF of the r's.

Histogram Equalization

r_k	n_k	$p_r(r_k) = n_k/MN$
$r_0 = 0$	790	0.19
$r_1 = 1$	1023	0.25
$r_2 = 2$	850	0.21
$r_3 = 3$	656	0.16
$r_4 = 4$	329	0.08
$r_5 = 5$	245	0.06
$r_6 = 6$	122	0.03
$r_7 = 7$	81	0.02

TABLE 3.1 Intensity distribution and histogram values for a 3-bit, 64×64 digital image.

Histogram Equalization

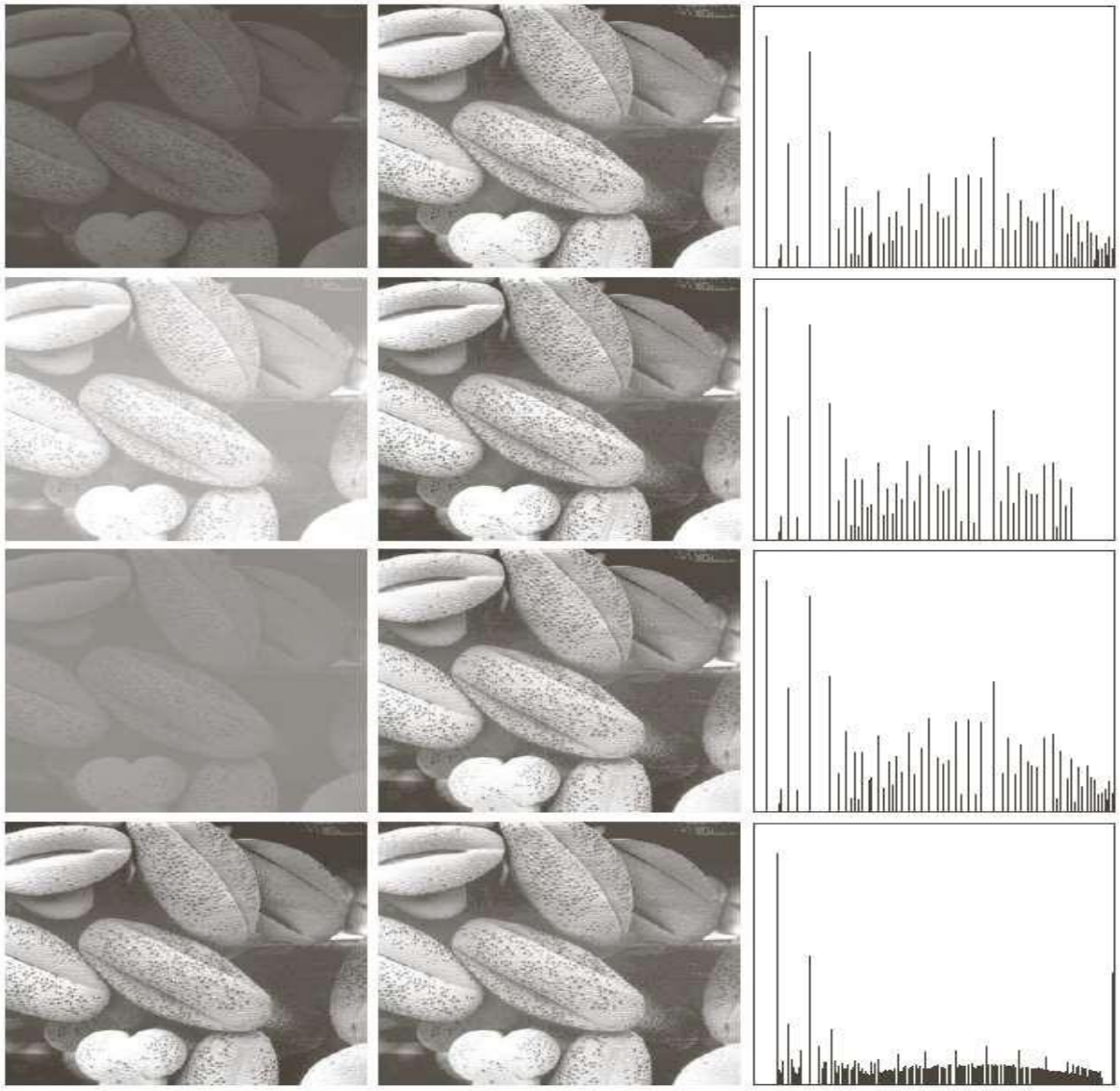


FIGURE 3.20 Left column: images from Fig. 3.16. Center column: corresponding histogram-equalized images. Right column: histograms of the images in the center column.

Mechanics of Spatial Filtering

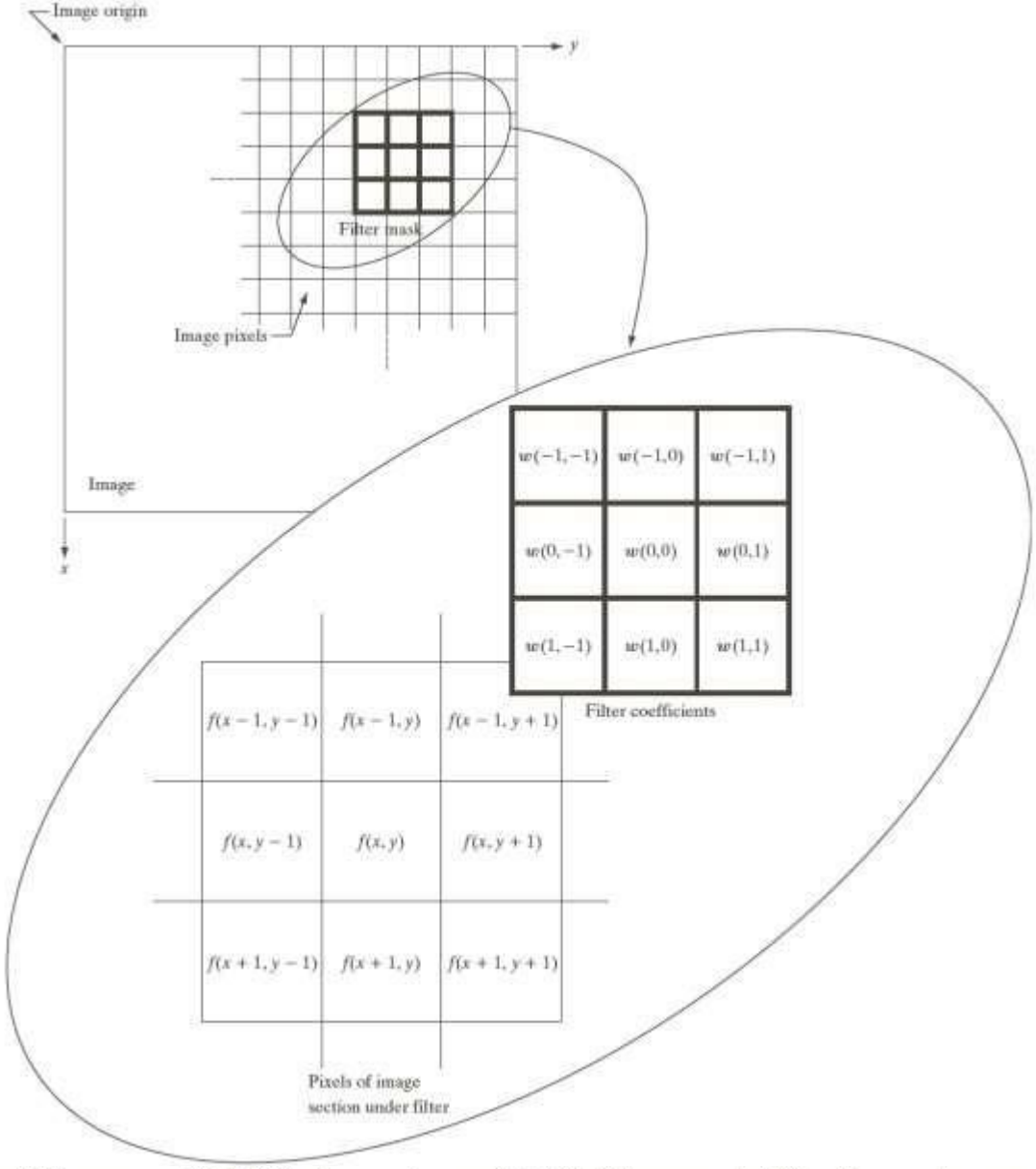


FIGURE 3.28 The mechanics of linear spatial filtering using a 3×3 filter mask. The form chosen to denote the coordinates of the filter mask coefficients simplifies writing expressions for linear filtering.

Mechanics of Spatial Filtering

- The result (or response), R, of linear filtering with the filter mask at a point (x,y) in the image is:
- $R = w(-1,-1)f(x-1,y-1) + w(-1,0)f(x-1,y) + \cdots + w(0,0)f(x,y) + \cdots + w(1,0)f(x-1,y-1) + w(1,1)f(x+1,y+1)$

Smoothing Spatial Linear Filters

- The output of a smoothing linear spatial filter is simply the average of the pixels contained in the neighborhood of the linear mask.
- By replacing the value of every pixel in an image by the average of the intensity levels in the neighborhood defined by the filter mask.
- Also called *Lowpass filter.*
- Random noise typically consists of sharp transition.
- Reduce “sharp” transition in intensities.
- If all coefficients are equal in filter then it is also called *a box filter*.

Smoothing Linear Filter

$\frac{1}{9} \times$

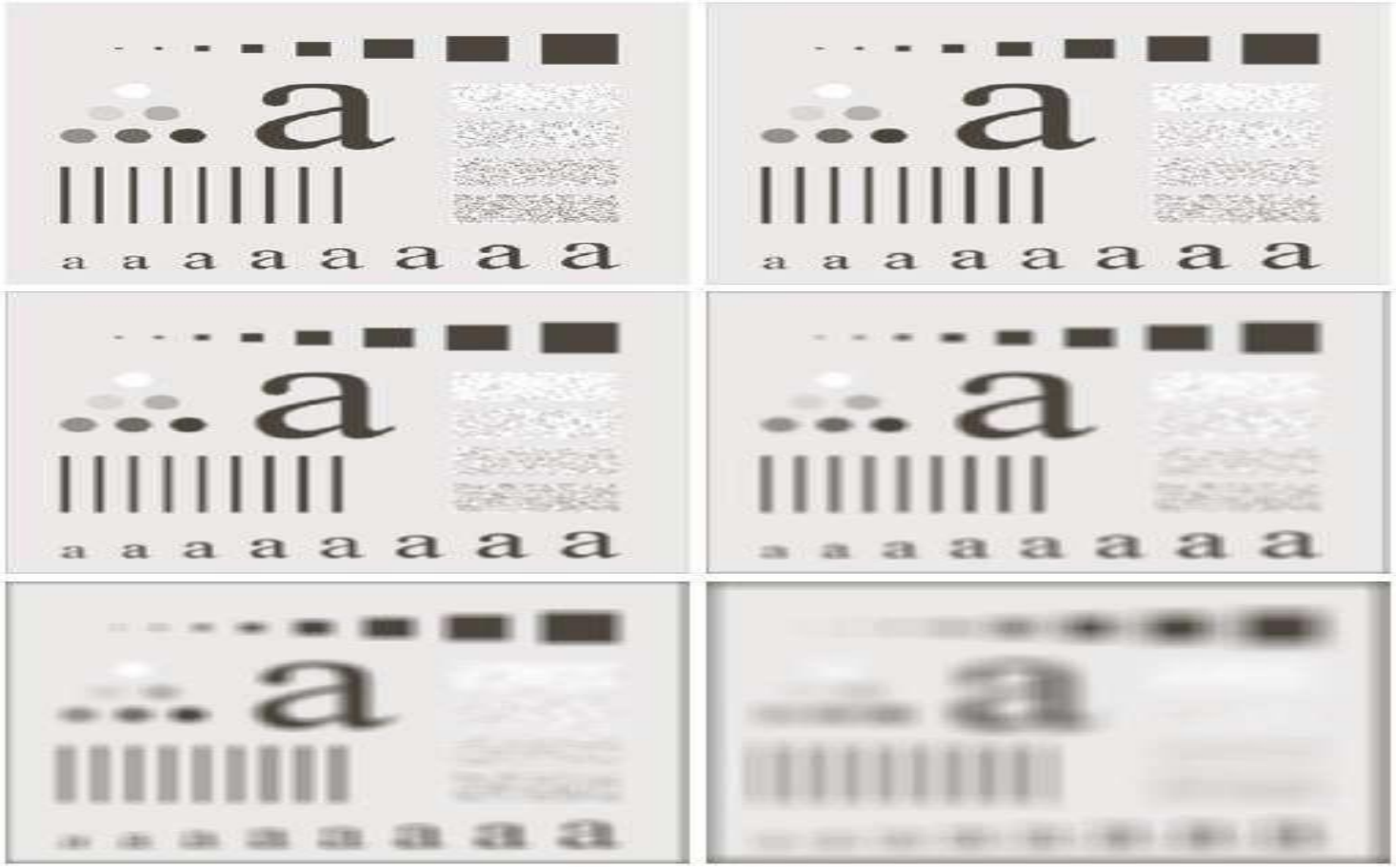
1	1	1
1	1	1
1	1	1

$\frac{1}{16} \times$

1	2	1
2	4	2
1	2	1

a b

FIGURE 3.32 Two 3×3 smoothing (averaging) filter masks. The constant multiplier in front of each mask is equal to 1 divided by the sum of the values of its coefficients, as is required to compute an average.



a b
c d
e f

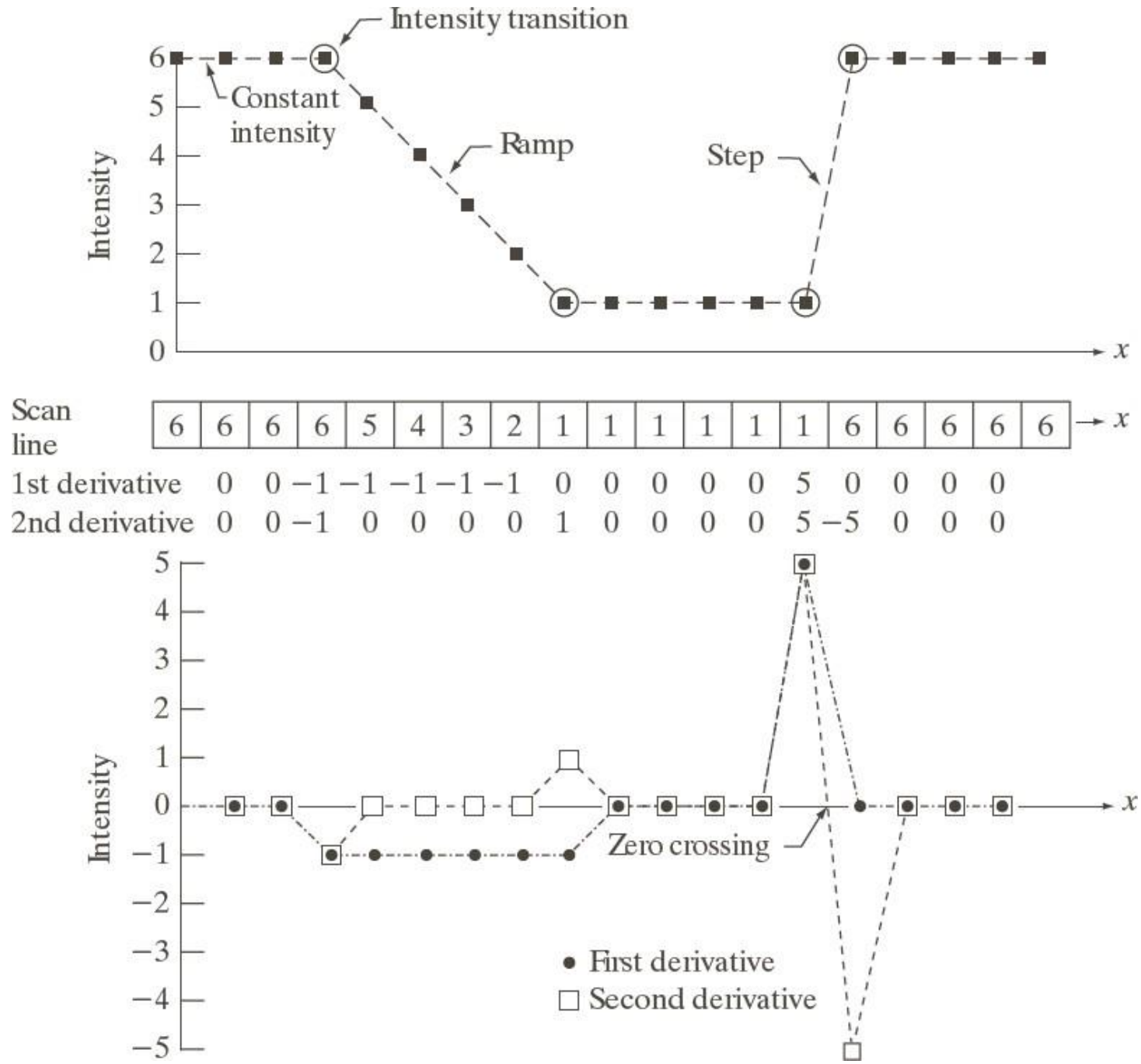
FIGURE 3.33 (a) Original image, of size 500 × 500 pixels. (b)–(f) Results of smoothing with square averaging filter masks of sizes $m = 3, 5, 9, 15$, and 35, respectively. The black squares at the top are of sizes 3, 5, 9, 15, 25, 35, 45, and 55 pixels, respectively; their borders are 25 pixels apart. The letters at the bottom range in size from 10 to 24 points, in increments of 2 points; the large letter at the top is 60 points. The vertical bars are 5 pixels wide and 100 pixels high; their separation is 20 pixels. The diameter of the circles is 25 pixels, and their borders are 15 pixels apart; their intensity levels range from 0% to 100% black in increments of 20%. The background of the image is 10% black. The noisy rectangles are of size 50 × 120 pixels.

Sharpening Spatial Filters

- Objective of sharpening is to highlight transitions in intensity.
- Uses in printing and medical imaging to industrial inspection and autonomous guidance in military systems.
- Averaging is analogous to integration, so sharpening is analogous to spatial differentiation.
- Edges also characterized by sharp intensity transitions, so averaging filters have the undesirable side effect that they blur edge.
- Thus, image differentiation enhances edges and other discontinuities (such as noise) and deemphasizes areas with slowly varying intensities.

Foundation

- Definition for a first order derivative (1) must be zero in areas of constant intensity (2) must be nonzero at the onset of an intensity step or ramp and (3) must be nonzero along ramps.
- For a second order derivatives (1) must be zero in constant areas (2) must be nonzero at the onset and (3) must be zero along ramps of constant slope.
- First order derivative of a one dimensional function f(x) is the difference of f(x+1) – f(x).
- Second order = f(x+1) + f(x-1) -2f(x)



a
b
c

FIGURE 3.36 Illustration of the first and second derivatives of a 1-D digital function representing a section of a horizontal intensity profile from an image. In (a) and (c) data points are joined by dashed lines as a visualization aid.

Second Derivatives-The Laplacian

$$\nabla^2 f = \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2}$$

$$\frac{\partial^2 f}{\partial^2 x^2} = f(x+1, y) + f(x-1, y) - 2f(x, y)$$

$$\frac{\partial^2 f}{\partial^2 y^2} = f(x, y+1) + f(x, y-1) - 2f(x, y)$$

$$\nabla^2 f = [f(x+1, y) + f(x-1, y) + f(x, y+1) + f(x, y-1)] - 4f(x, y).$$

Second Derivatives - The Laplacian

0	1	0
1	−4	1
0	1	0

1	1	1
1	−8	1
1	1	1

0	−1	0
−1	4	−1
0	−1	0

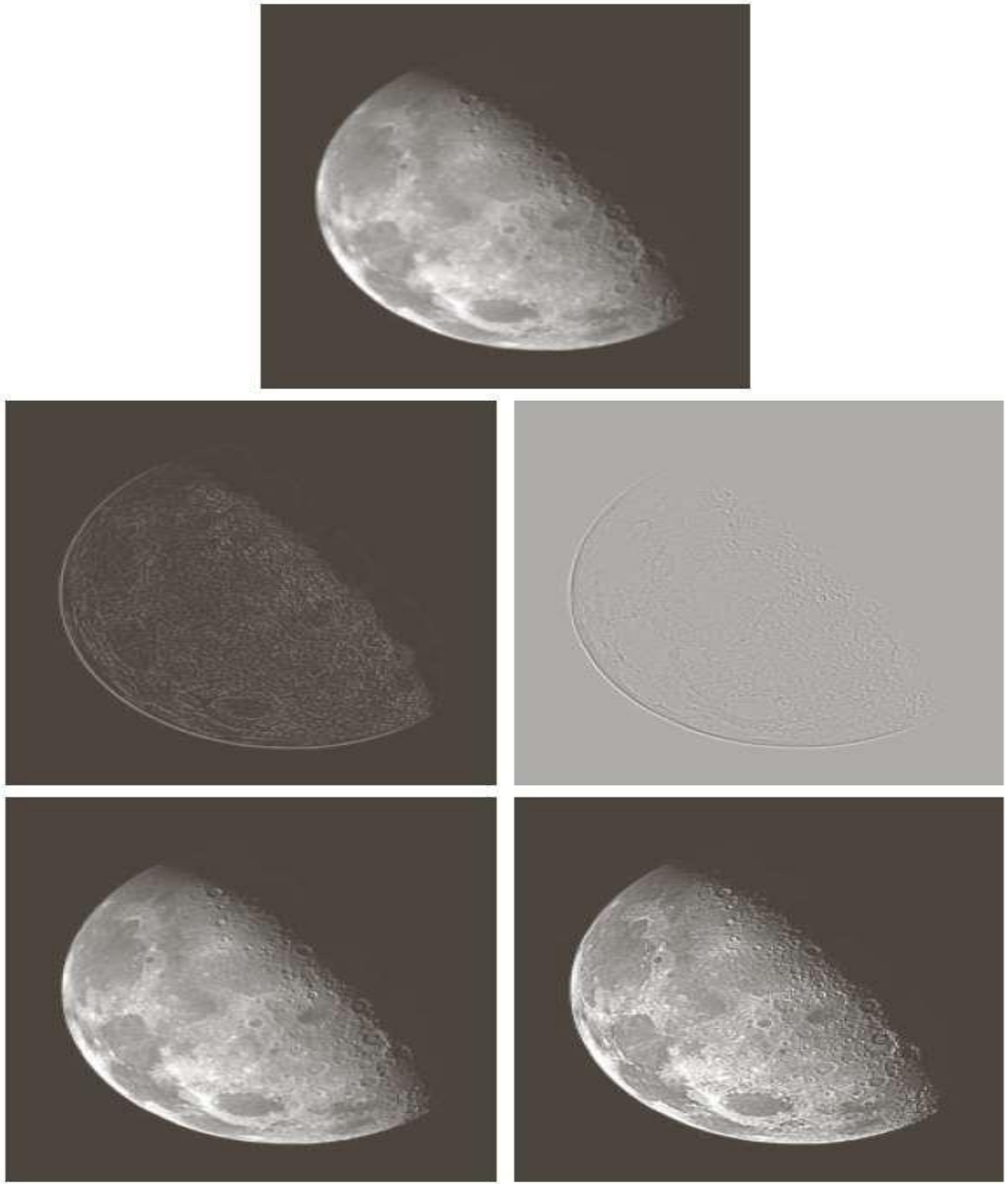
−1	−1	−1
−1	8	−1
−1	−1	−1

a b
c d

FIGURE 3.37
(a) Filter mask used to implement Eq. (3.6-6). (b) Mask used to implement an extension of this equation that includes the diagonal terms. (c) and (d) Two other implementations of the Laplacian found frequently in practice.

Second Derivatives-The Laplacian

$$g(x, y) = \begin{cases} f(x, y) - \nabla^2 f(x, y) & \text{if the center coefficient of the Laplacian mask is negative} \\ f(x, y) + \nabla^2 f(x, y) & \text{if the center coefficient of the Laplacian mask is positive.} \end{cases}$$



a
b c
d e

FIGURE 3.38
(a) Blurred image of the North Pole of the moon. (b) Laplacian without scaling. (c) Laplacian with scaling. (d) Image sharpened using the mask in Fig. 3.37(a). (e) Result of using the mask in Fig. 3.37(b). (Original image courtesy of NASA.)

Unsharp Masking and High boost Filtering

- Unsharp Masking
  - Read Original Image f(x,y)
  - Blurred original image f´(x,y)
  - Mask = f(x,y) – f´(x,y)
  - $f_s(x,y) = f(x,y) + Mask$
  - Use publishing industry to sharpen image
- High Boost Filtering
  - Read Original Image f(x,y)
  - Blurred original image f´(x,y)
  - $f_{hb}(x,y) = Af(x,y) – f´(x,y)$, where A>1
  - Use when input image is darker than desired

First Derivative – The Gradient

$$\nabla \mathbf{f} = \begin{bmatrix} G_x \\ G_y \end{bmatrix} = \begin{bmatrix} \dfrac{\partial f}{\partial x} \\ \dfrac{\partial f}{\partial y} \end{bmatrix}$$

$$\begin{aligned} \nabla f &= \operatorname{mag}(\nabla \mathbf{f}) \\ &= \left[G_x^2 + G_y^2\right]^{1/2} \\ &= \left[\left(\frac{\partial f}{\partial x}\right)^2 + \left(\frac{\partial f}{\partial y}\right)^2\right]^{1/2} \end{aligned}$$

$$\nabla f \approx |G_x| + |G_y|$$

First Derivative – The Gradient

- The center point f(x,y) denotes z_5, f(x-1,y-1) denotes z_1 and so on.
- First-order derivative approximates:
  - $G_x = (z_8 - z_5)$ and $G_y = (z_6 - z_5)$
  - $\nabla f \approx |G_x| + |G_y| = |(z_8 - z_5)| + |(z_6 - z_5)|$

Other Masks

- Robert cross-gradient operators:
  - $G_x = (z_9 - z_5)$ and $G_y = (z_8 - z_6)$
  - $\nabla f \approx |G_x| + |G_y| = |(z_9 - z_5)| + |(z_8 - z_6)|$
- Sobel operators:
  - $G_x = (z_7 + 2z_8 - z_9) - (z_1 + 2z_2 - z_3)$
  - $G_y = (z_3 + 2z_6 - z_9) - (z_1 + 2z_4 - z_7)$
  - $|G_x| + |G_y| = |(z_7 + 2z_8 + z_9) - (z_1 + 2z_2 + z_3)| + |(z_3 + 2z_6 + z_9) - (z_1 + 2z_4 + z_7)|$

z_1	z_2	z_3
z_4	z_5	z_6
z_7	z_8	z_9

−1	0
0	1

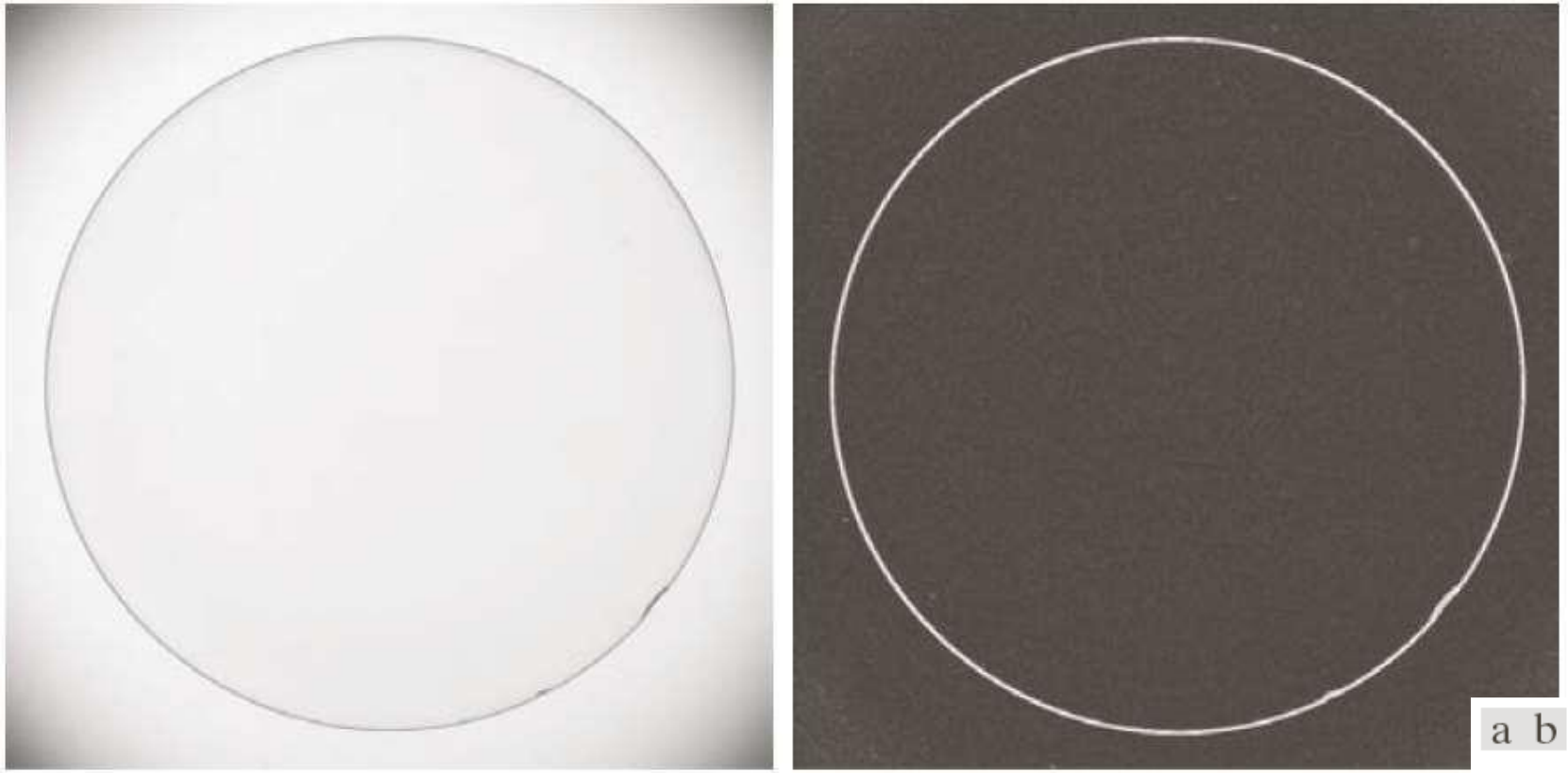
0	−1
1	0

−1	−2	−1
0	0	0
1	2	1

−1	0	1
−2	0	2
−1	0	1

a
b c
d e

FIGURE 3.41
A 3 × 3 region of an image (the zs are intensity values).
(b)–(c) Roberts cross gradient operators.
(d)–(e) Sobel operators. All the mask coefficients sum to zero, as expected of a derivative operator.



a b

FIGURE 3.42
(a) Optical image of contact lens (note defects on the boundary at 4 and 5 o'clock). (b) Sobel gradient. (Original image courtesy of Pete Sites, Perceptics Corporation.)

4 Image Enhancement in the Frequency Domain

- Frequency domain processing techniques are based on modifying the Fourier Transform of an image
- The frequency domain refers to the plane of the two dimensional discrete Fourier transform of an image.
- The purpose of the Fourier transform is to represent a signal as a linear combination of sinusoidal signals of various frequencies
- Return to the original domain without losing any information via an inverse process.

Background

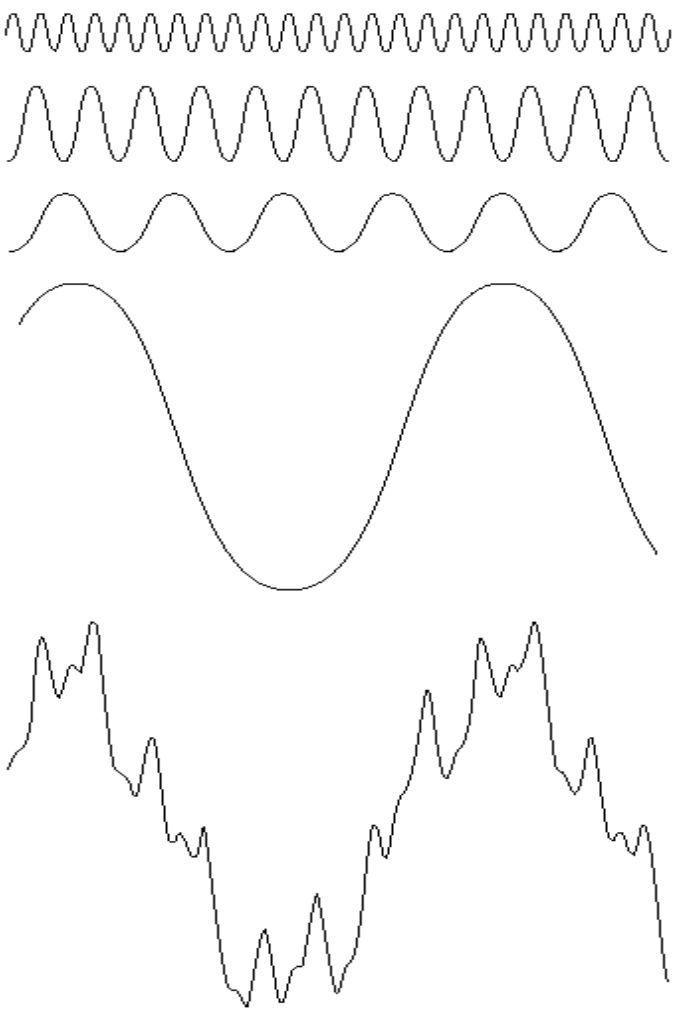


FIGURE 4.1 The function at the bottom is the sum of the four functions above it. Fourier's idea in 1807 that periodic functions could be represented as a weighted sum of sines and cosines was met with skepticism.

Fourier Transform-Discrete Case

- The two-dimensional Fourier transform and its inverse
  - Fourier transform:

$$F(u,v) = \frac{1}{MN}\sum_{x=0}^{M-1}\sum_{y=0}^{N-1} f(x,y)e^{-j2\pi(ux/M+vy/N)}$$

$$\text{for } u = 0,1,2,\ldots,M-1, v = 0,1,2,\ldots,N-1$$

$$e^{j\theta} = \cos\theta + j\sin$$

  - Inverse Fourier transform:

$$f(x,y) = \sum_{u=0}^{M-1}\sum_{v=0}^{N-1} F(u,v)e^{j2\pi(ux/M+vy/N)}$$

$$\text{for } x = 0,1,2,\ldots,M-1, y = 0,1,2,\ldots,N-1$$

    - *u*, *v* : the transform or frequency variables
    - *x*, *y* : the spatial or image variables

Basic Steps for Filtering

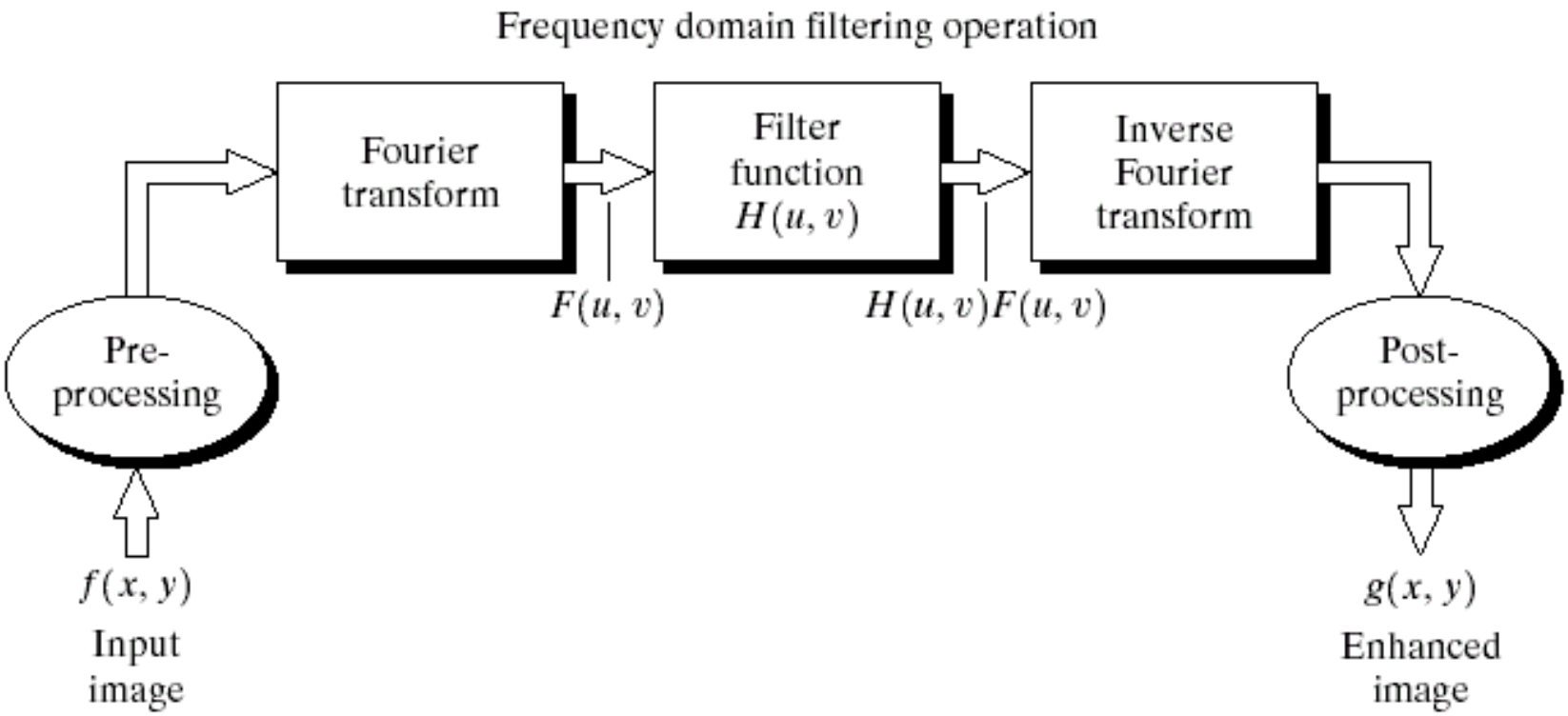


FIGURE 4.5 Basic steps for filtering in the frequency domain.

Basic Model for Filtering

- The basic model for filtering in the frequency domain

$$G(u,v) = H(u,v)F(u,v)$$

where $F(u,v)$: the Fourier transform of the image to be smoothed

$H(u,v)$: a filter transfer function

Ideal Lowpass Filter (ILPF)

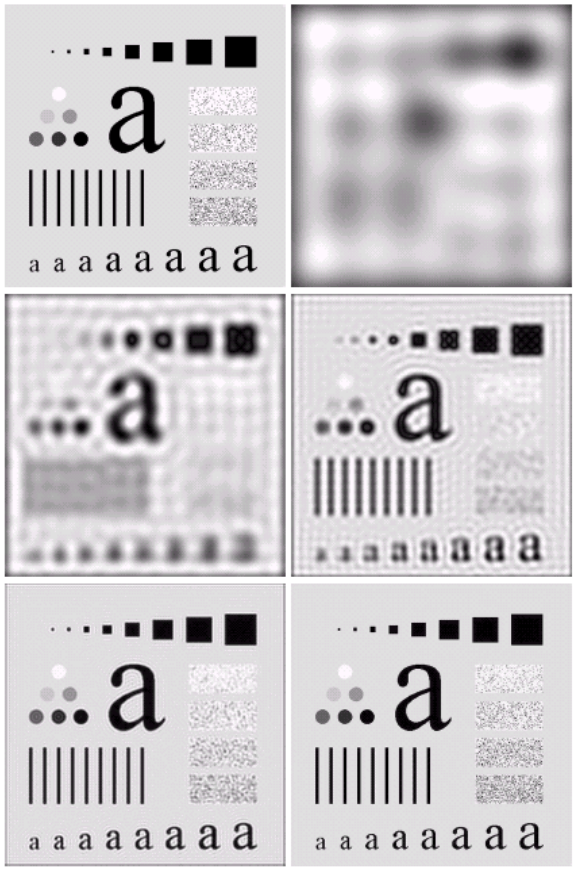
- The simplest lowpass filter is a filter that "cuts off" all high-frequency components of the Fourier transform that are at a distance greater than a specified distance D_0 from the origin of the transform.
- The transfer function of an ideal lowpass filter

$$H(u,v) = \begin{cases} 1 & \text{if } D(u,v) \le D_0 \\ 0 & \text{if } D(u,v) > D_0 \end{cases}$$

where $D(u,v)$: the distance from point (u,v) to the center of ther frequency rectangle

$$D(u,v) = \left[(u - M/2)^2 + (v - N/2)^2\right]^{\frac{1}{2}}$$

Ideal Lowpass Filter Examples

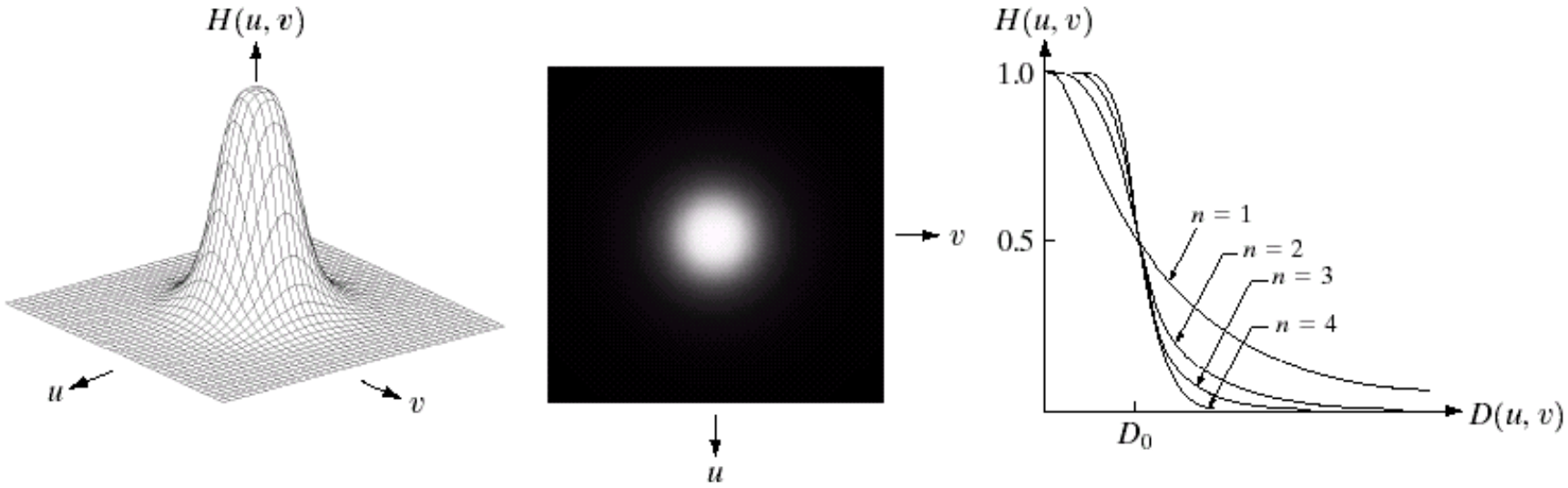


a b
c d
e f

FIGURE 4.12 (a) Original image. (b)–(f) Results of ideal lowpass filtering with cutoff frequencies set at radii values of 5, 15, 30, 80, and 230, as shown in Fig. 4.11(b). The power removed by these filters was 8, 5.4, 3.6, 2, and 0.5% of the total, respectively.

Butterworth Lowpass Filter (BLPF)

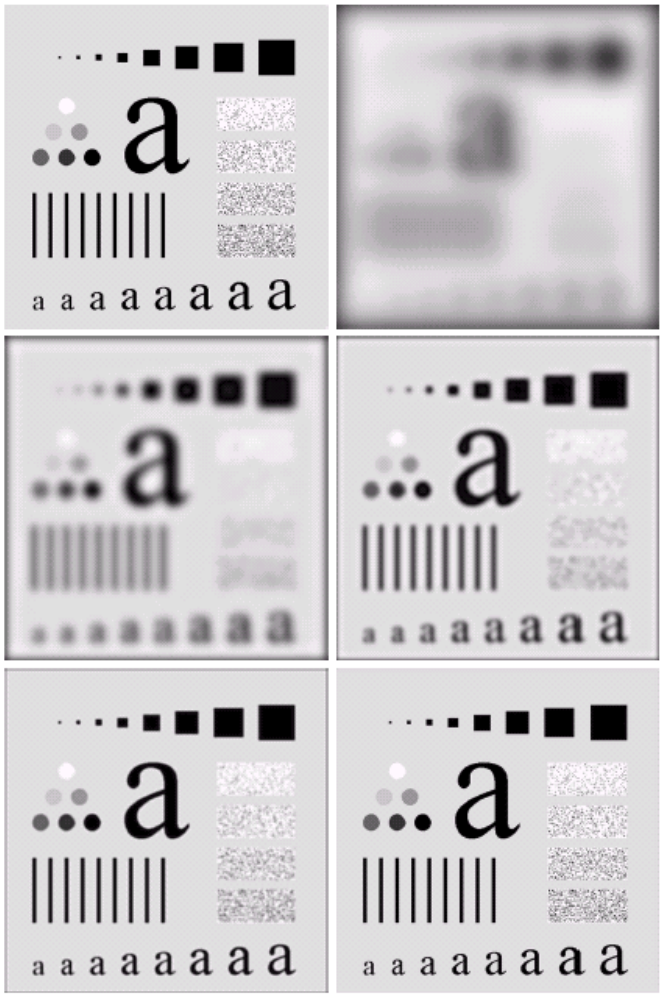
$$H(u,v) = \frac{1}{1+\left[D(u,v)/D_0\right]^{2n}}$$



a b c

FIGURE 4.14 (a) Perspective plot of a Butterworth lowpass filter transfer function. (b) Filter displayed as an image. (c) Filter radial cross sections of orders 1 through 4.

Butterworth Lowpass Filter Examples

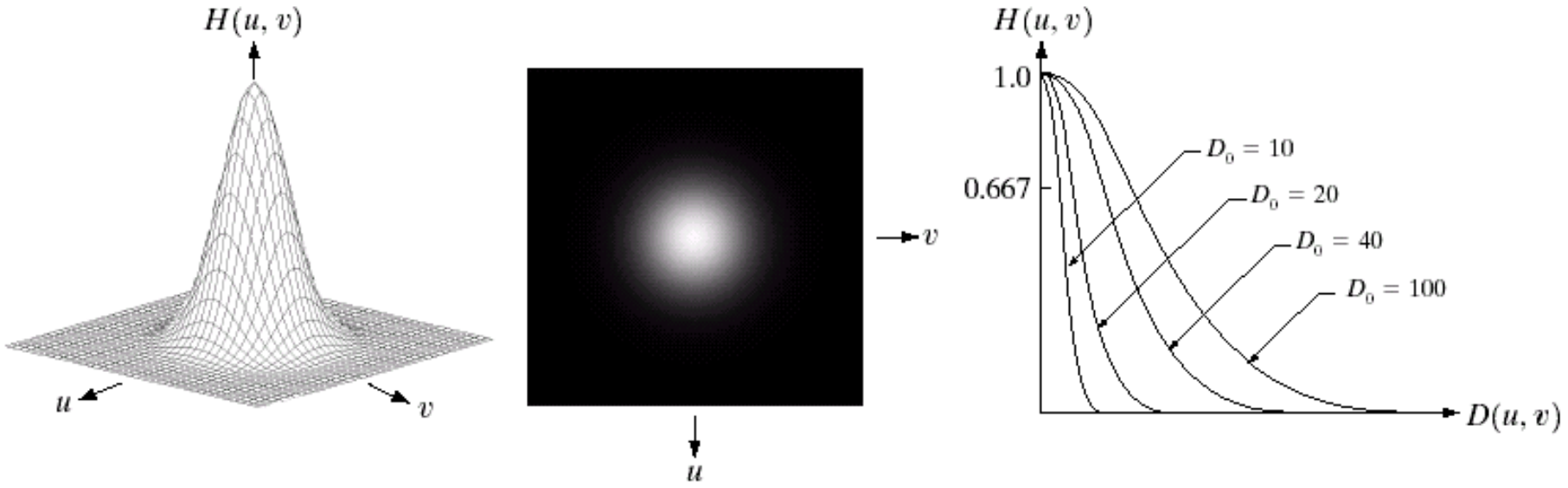


a b
c d
e f

FIGURE 4.15 (a) Original image. (b)–(f) Results of filtering with BLPFs of order 2, with cutoff frequencies at radii of 5, 15, 30, 80, and 230, as shown in Fig. 4.11(b). Compare with Fig. 4.12.

Gaussian Lowpass Filter (GLPF)

$$H(u,v) = e^{-D^2(u,v)/2D_0^2}$$



a b c

FIGURE 4.17 (a) Perspective plot of a GLPF transfer function. (b) Filter displayed as an image. (c) Filter radial cross sections for various values of D_0.

Gaussian Lowpass Filter Example

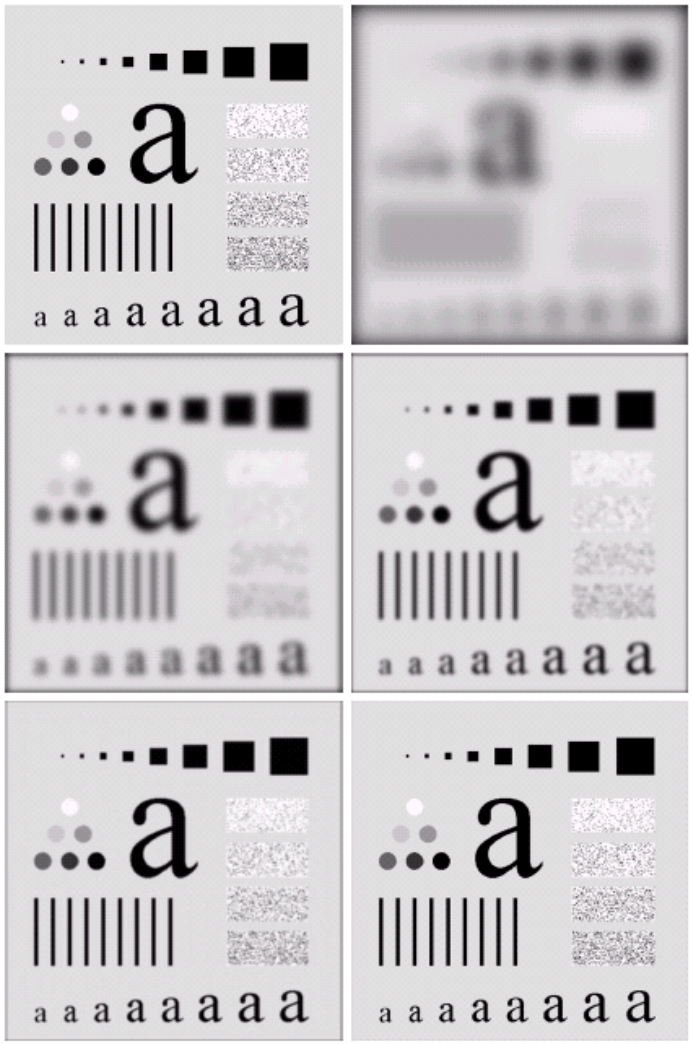


FIGURE 4.18 (a) Original image. (b)–(f) Results of filtering with Gaussian lowpass filters with cutoff frequencies set at radii values of 5, 15, 30, 80, and 230, as shown in Fig. 4.11(b). Compare with Figs. 4.12 and 4.15.

a b
c d
e f

Sharpening Frequency Domain Filter

$$H_{hp}(u,v) = H_{lp}(u,v)$$

Ideal highpass filter

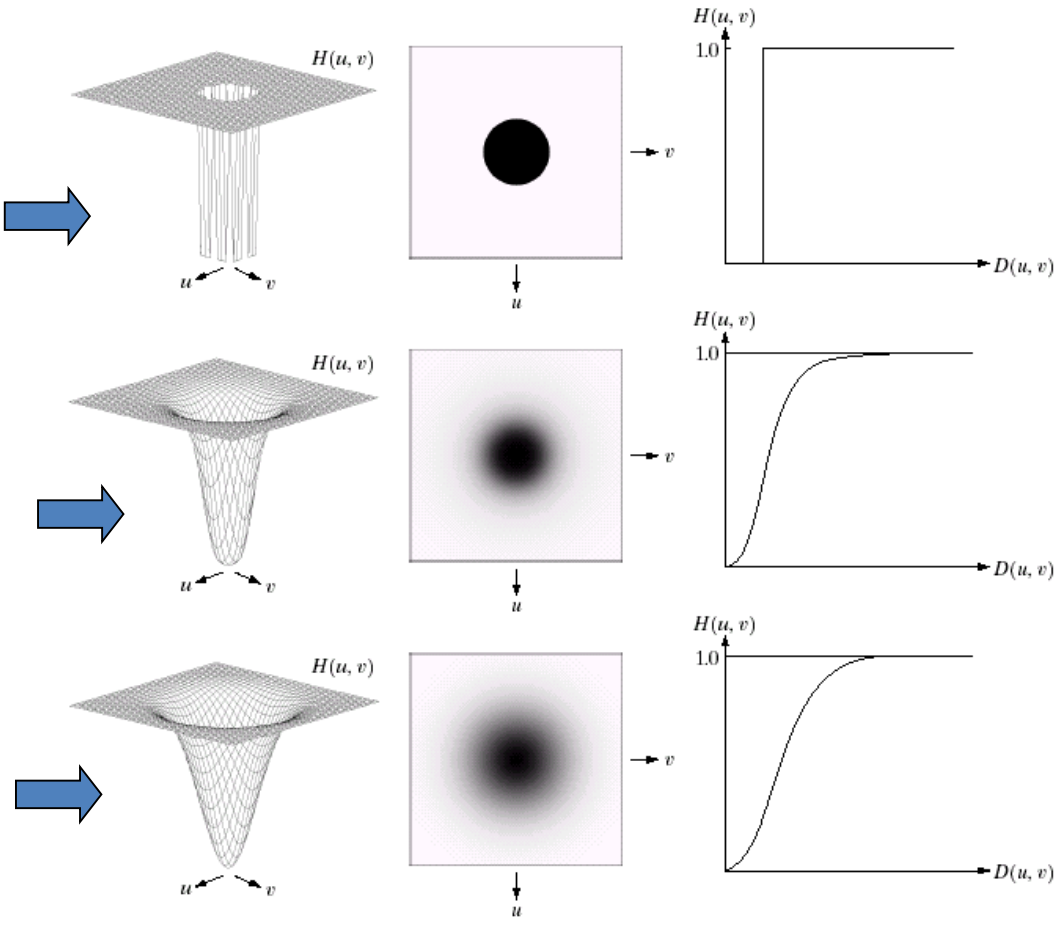
$$H(u,v) = \begin{cases} 0 & \text{if } D(u,v) \le D_0 \\ 1 & \text{if } D(u,v) > D_0 \end{cases}$$

Butterworth highpass filter

$$H(u,v) = \frac{1}{1+\left[D_0 / D(u,v)\right]^{2n}}$$

Gaussian highpass filter

$$H(u,v) = 1 - e^{-D^2(u,v)/2D_0^2}$$

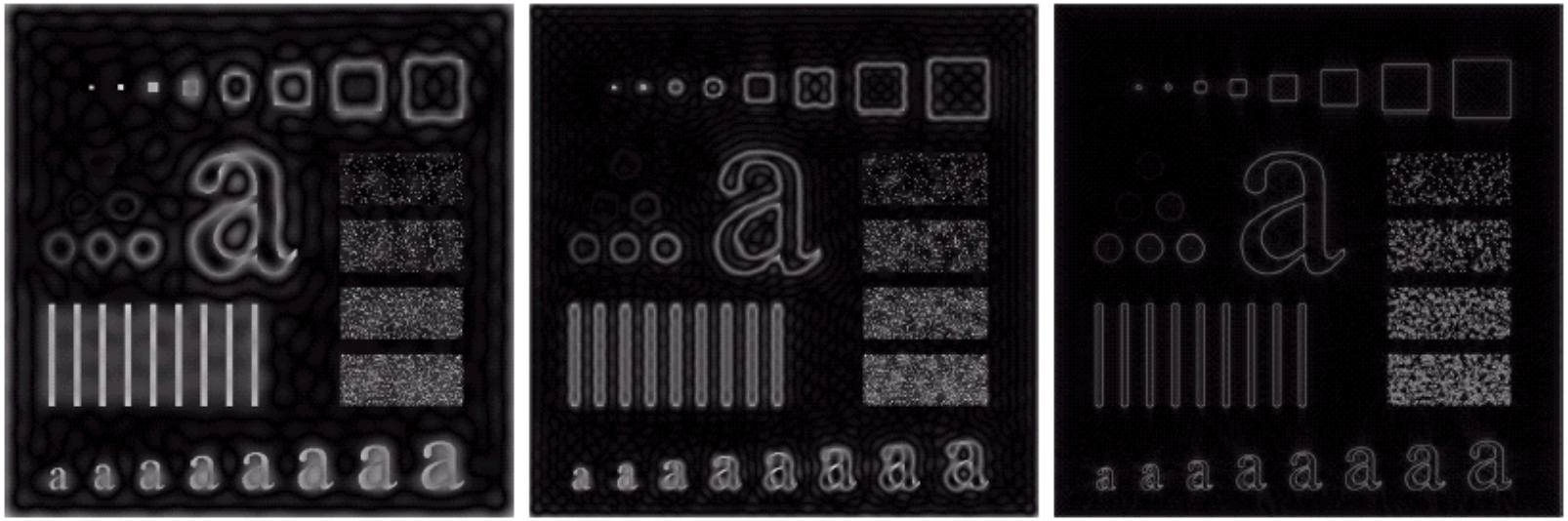


a b c
d e f
g h i

FIGURE 4.22 Top row: Perspective plot, image representation, and cross section of a typical ideal highpass filter. Middle and bottom rows: The same sequence for typical Butterworth and Gaussian highpass filters.

Ideal Highpass Filter (IHPF)

$$H(u,v) = \begin{cases} 0 & \text{if } D(u,v) \le D_0 \\ 1 & \text{if } D(u,v) > D_0 \end{cases}$$

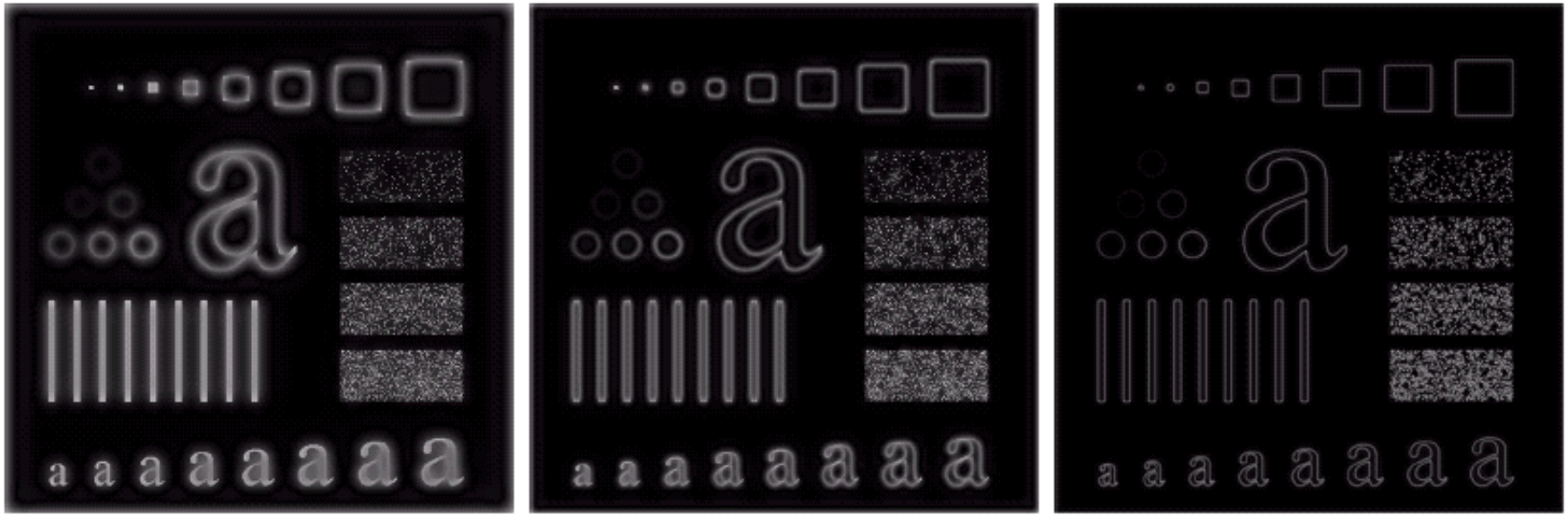


a b c

FIGURE 4.24 Results of ideal highpass filtering the image in Fig. 4.11(a) with D_0 = 15, 30, and 80, respectively. Problems with ringing are quite evident in (a) and (b).

Butterworth Highpass Filter (BHPF)

$$H(u,v) = \frac{1}{1+\left[D_0 / D(u,v)\right]^{2n}}$$



a b c

FIGURE 4.25 Results of highpass filtering the image in Fig. 4.11(a) using a BHPF of order 2 with D_0 = 15, 30, and 80, respectively. These results are much smoother than those obtained with an ILPF.

Gaussian Highpass Filter (GHPF)

$$H(u,v) = 1 - e^{-D^2(u,v)/2D_0^2}$$

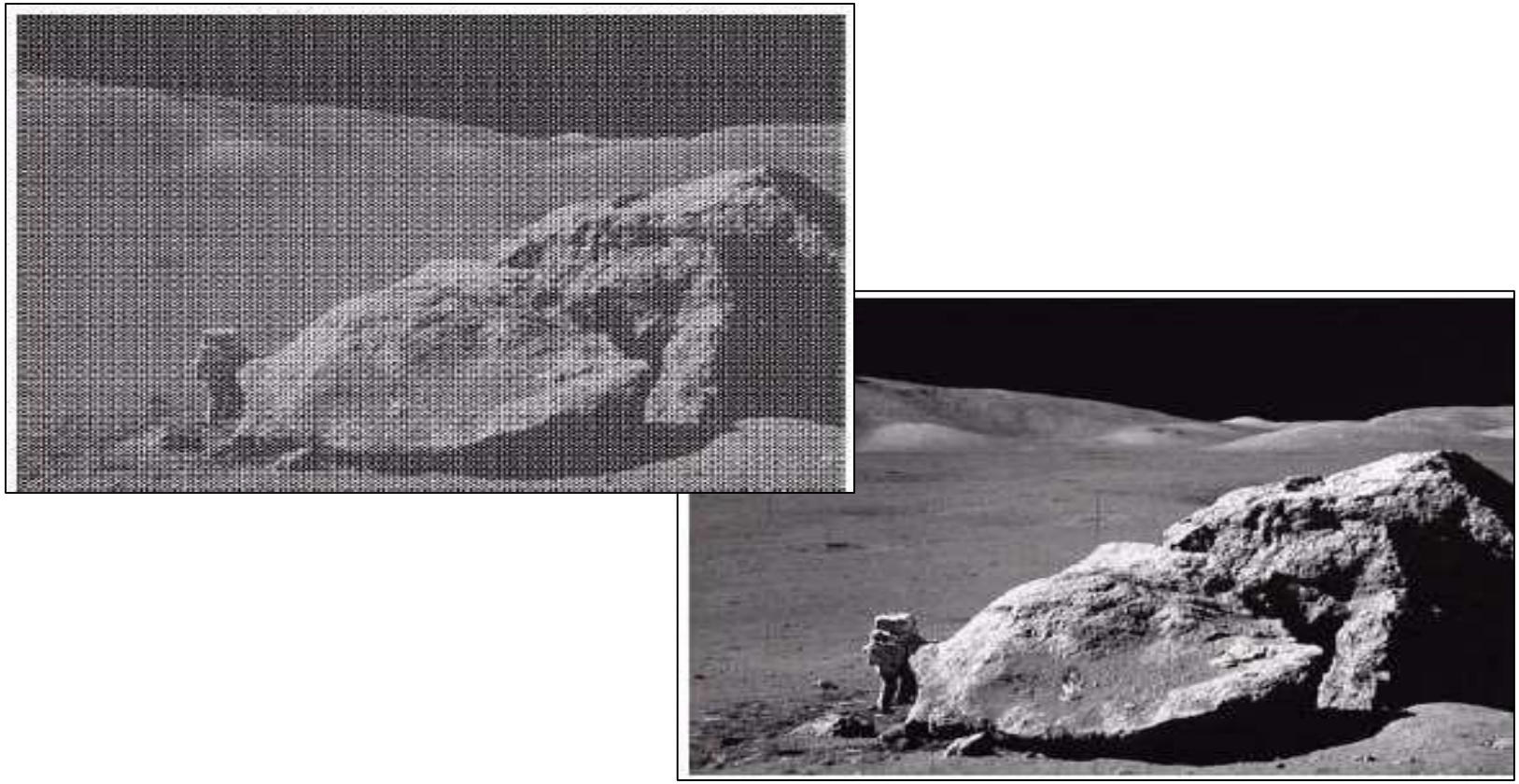


a b c

FIGURE 4.26 Results of highpass filtering the image of Fig. 4.11(a) using a GHPF of order 2 with $D_0 = 15$, 30, and 80, respectively. Compare with Figs. 4.24 and 4.25.

5 Image Restoration

- Improve an image in some predefined sense



Noise Models

- Source of Noise
  - Image acquisition (digitization)
  - Image transmission
- Gaussian Noise
  - Electronic circuit noise and sensor noise
- Impulse (Slat-and-Pepper) Noise
  - Quick transients, such as faulty switching during imaging
  - In practical, impulses are usually stronger than image signals

Spatial Filters for De-noising Additive Noise

Restoration in the Presence of Noise

1. Mean Filters

Arithmetic Mean Filter $$\hat{f}(x,y) = \frac{1}{mn} \sum_{(s,t)\epsilon S_{xy}} g(s,t)$$

Geometric Mean Filter $$\hat{f}(x,y) = \left[\prod_{(s,t)\epsilon S_{xy}} g(s,t) \right]^{\frac{1}{mn}}$$

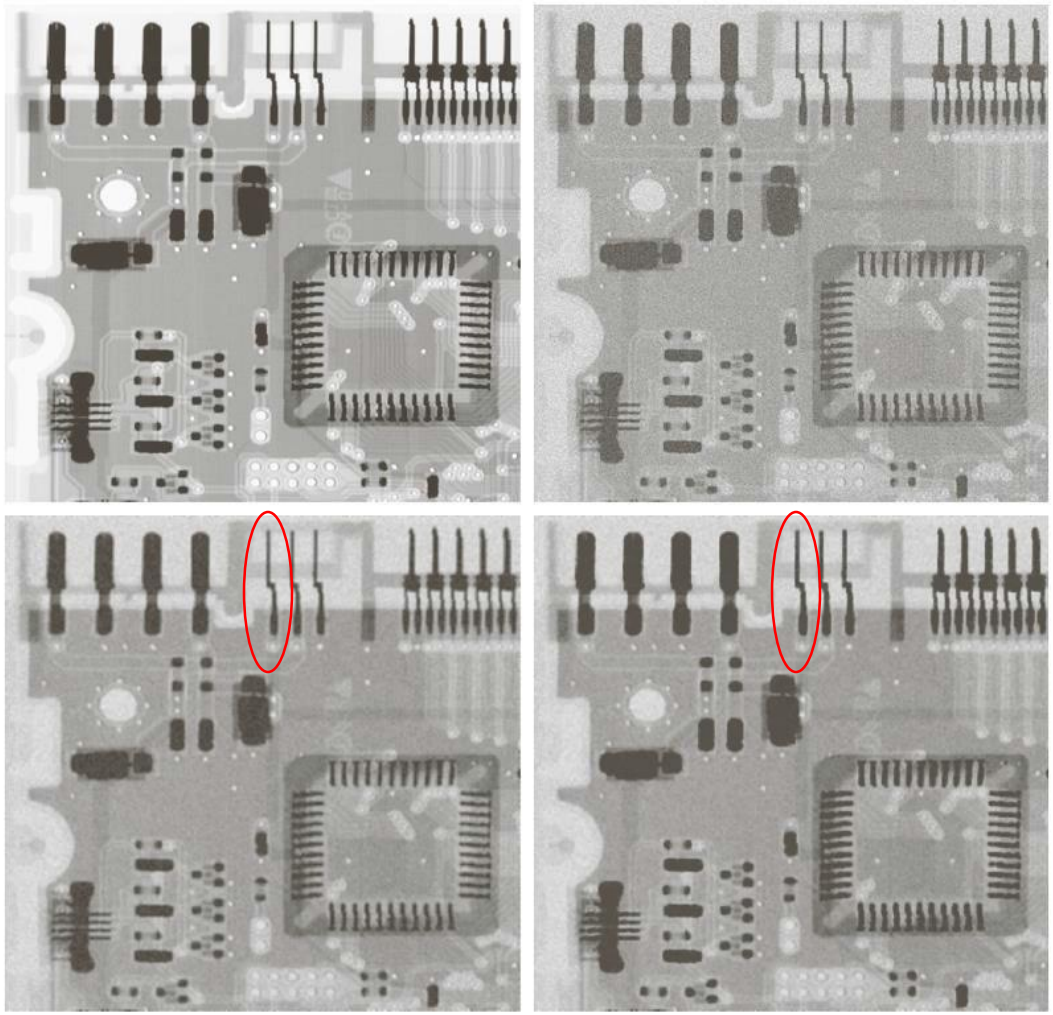
Harmonic Mean Filter $$\hat{f}(x,y) = \frac{mn}{\sum_{(s,t)\epsilon S_{xy}} \frac{1}{g(s,t}}$$

Contra-harmonic Mean Filter $$\hat{f}(x,y) = \frac{\sum_{(s,t)\epsilon S_{xy}} g(s,t)^{Q+1}}{\sum_{(s,t)\epsilon S_{xy}} g(s,t)^{Q}}$$

Mean Filters

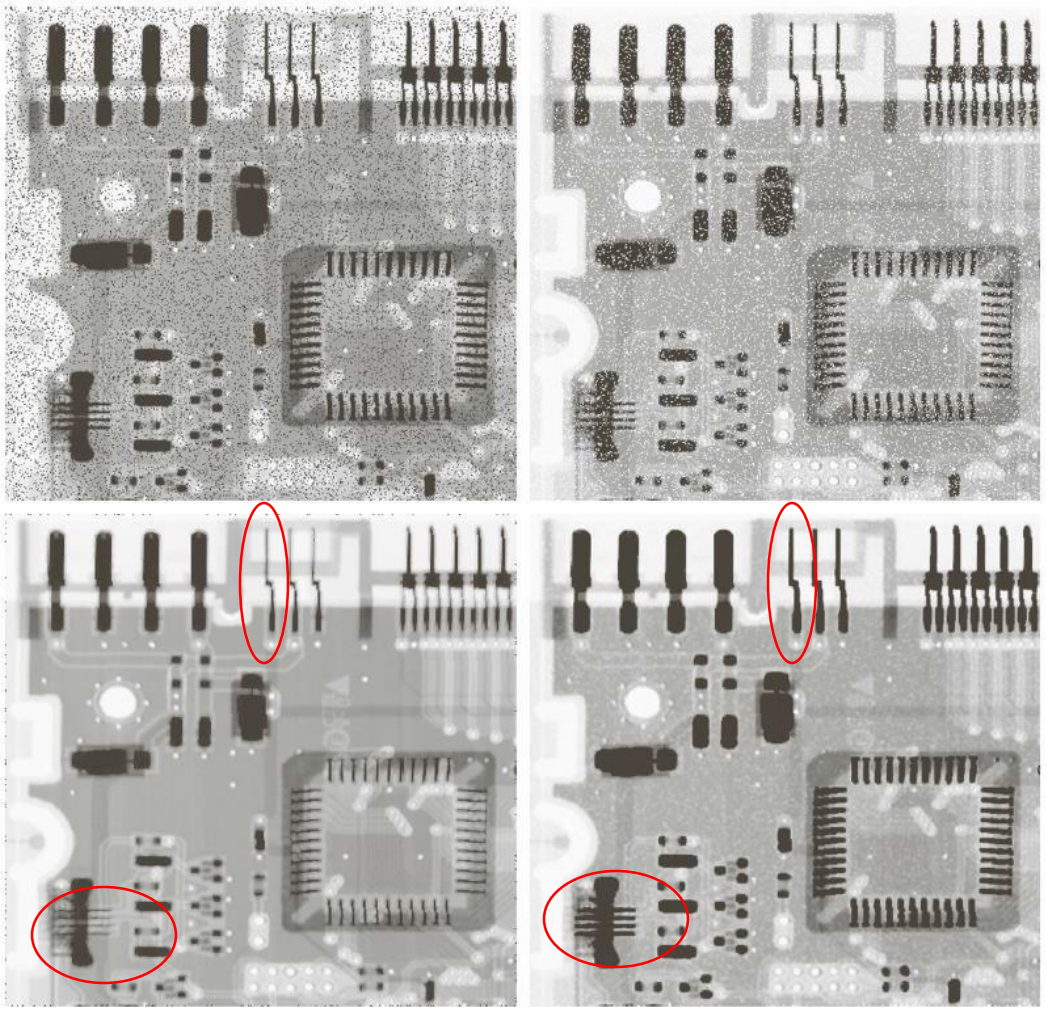
a. Original Image
b. Image corrupted by additive Gaussian Noise

c. After applying 3x3 Arithmetic mean filter.
d. After applying 3x3 Geometric mean filter.



Mean Filters

a. Image corrupted by additive pepper Noise
b. Image corrupted by additive salt Noise

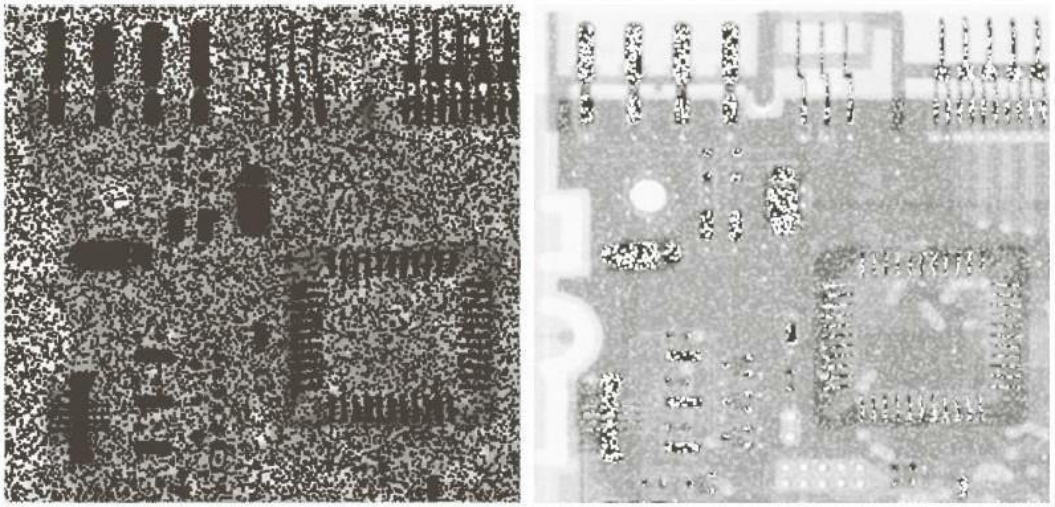


c. After applying 3x3 contra-harmonic mean filter with Q = 1.5 on a.
d. After applying 3x3 contra-harmonic mean filter with Q = -1.5 on b.

Mean filters

- Selecting the right value (and sign) of contra-harmonic mean filter is highly important.

Results of applying 3x3 contra-harmonic mean filter with Q = -1.5 on a. and Q = 1.5 on b.



Order-Statistics Filters

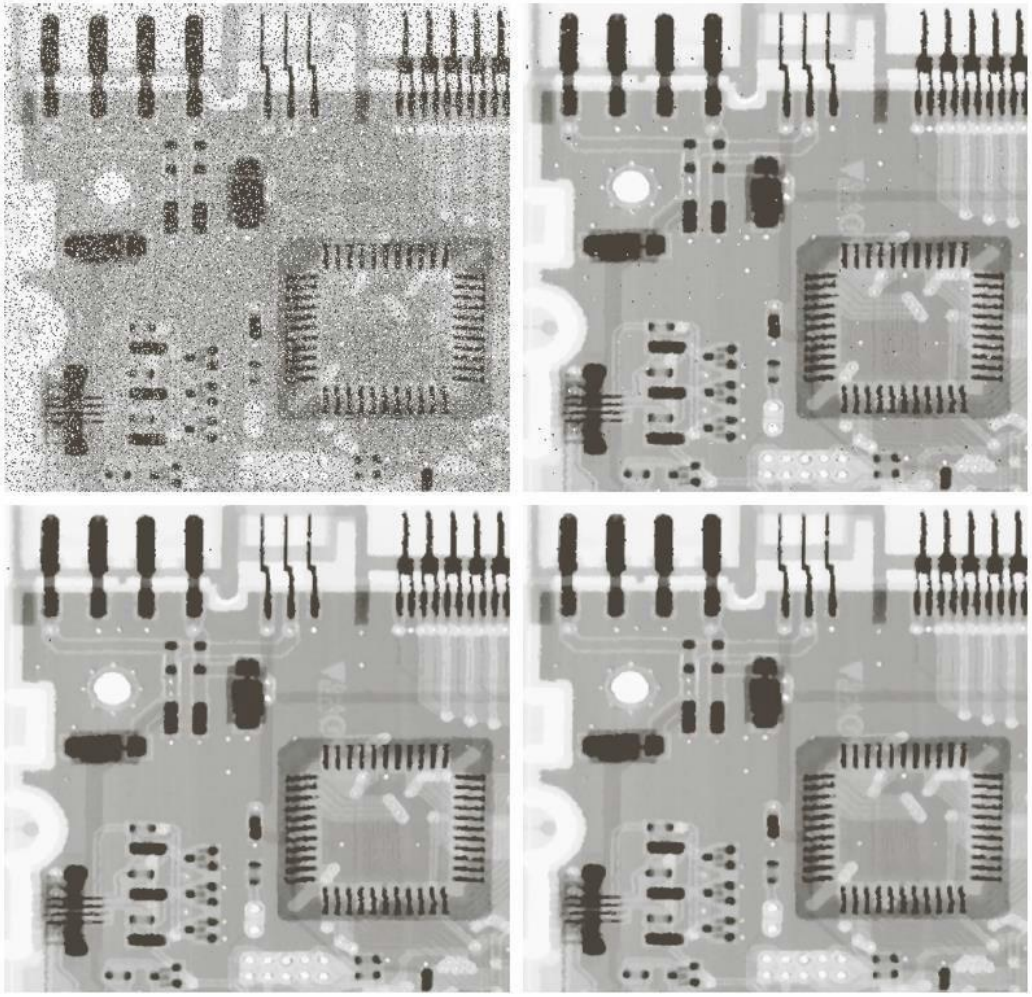
1. Median Filter $\hat{f}(x,y) = \text{median}(\text{g}(\text{s},\text{t}))$
2. Max Filter $\hat{f}(x,y) = \max(\text{g}(\text{s},\text{t}))$
3. Min Filter $\hat{f}(x,y) = \min(\text{g}(\text{s},\text{t}))$
4. Alpha-trimmed $\hat{f}(x,y) = \frac{1}{mn-d} \sum_{(s,t)\epsilon S_{xy}} g_r\,(s,t)$

   Mean Filter

   We delete the $d/2$ lowest and the $d/2$ highest gray-level values of $g(s, t)$. $g_r(s, t)$ represent the remaining mn-d.

Order-Statistics Filters

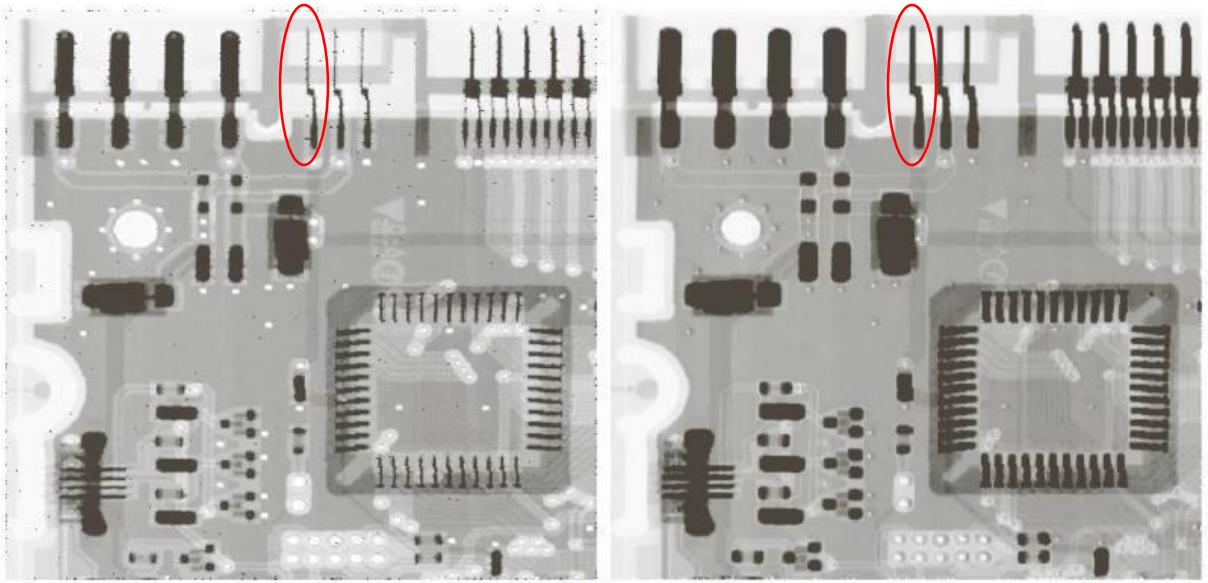
a. Image Corrupted with slat and pepper noise
b. After applying 3x3 median filter



c. After applying the same 3x3 median filter on b.
d. After applying the same 3x3 median filter on c.

Order-Statistics Filters

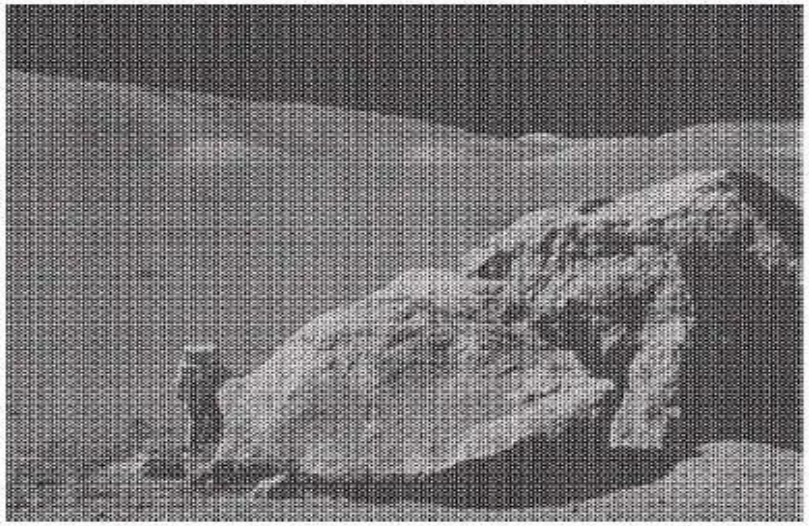
a. After applying Max filter.
b. After applying Min filter.

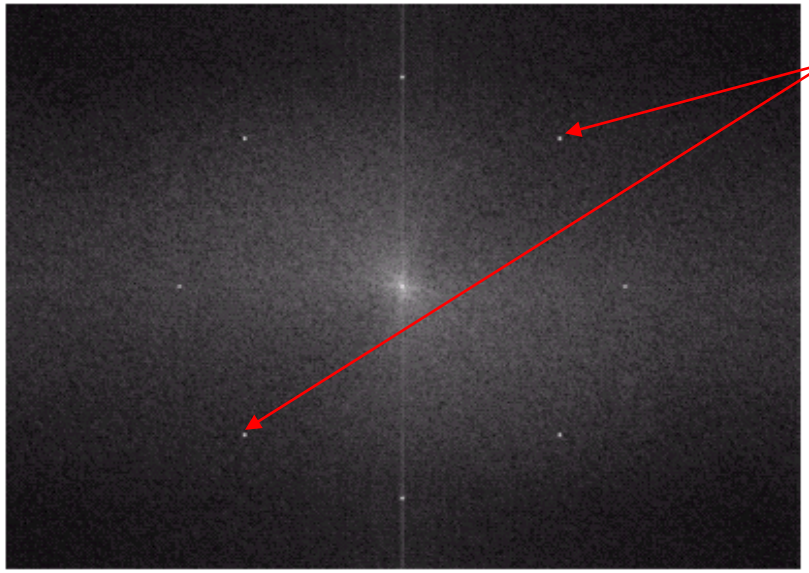


Periodic Noise

- Periodic noise arises from electrical or electromechanical interference during image acquisition
- It can be observed in the frequency domain

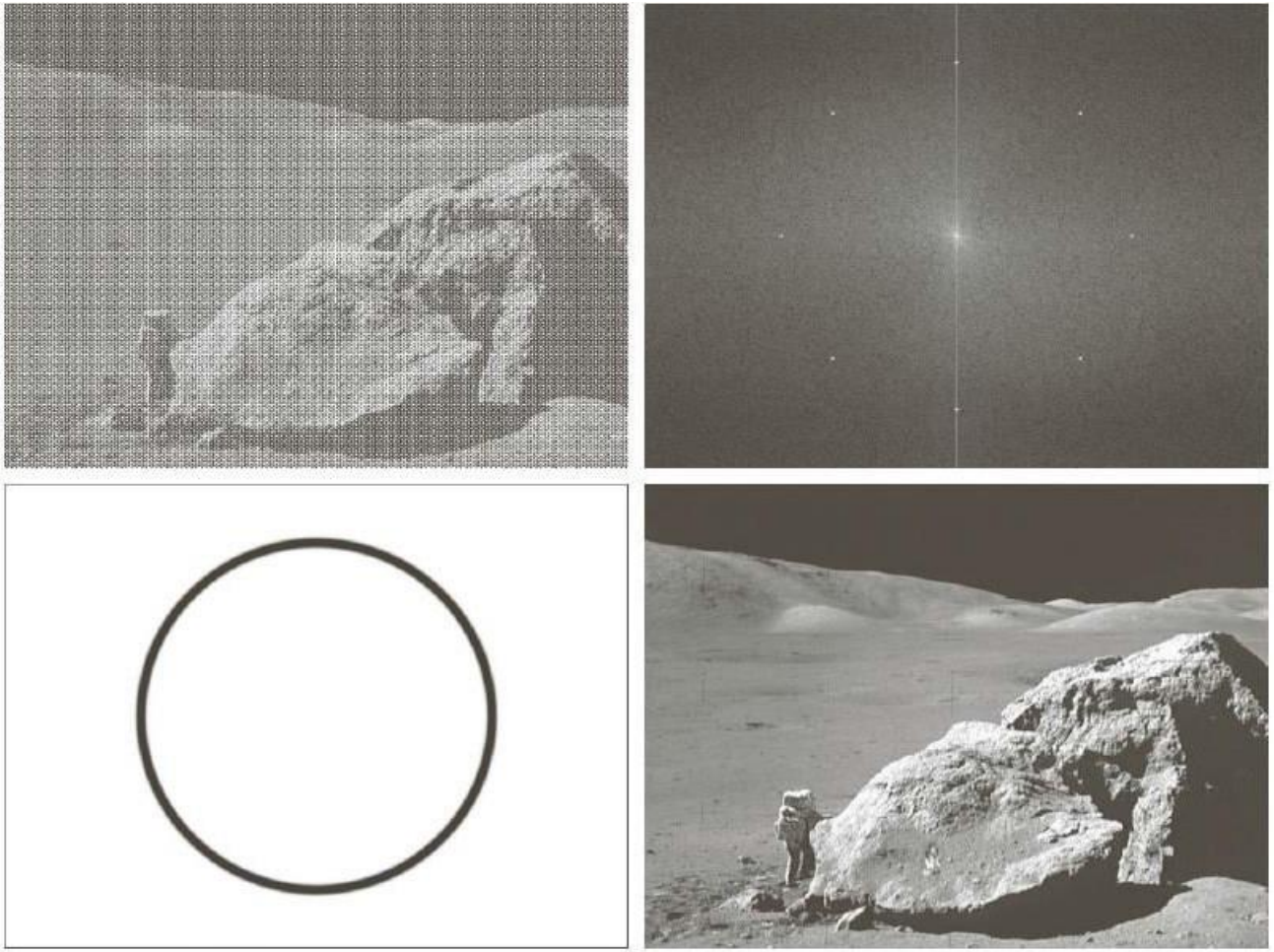
Periodic Noise Reduction





Sinusoidal noise:
Conjugate pair in
frequency domain

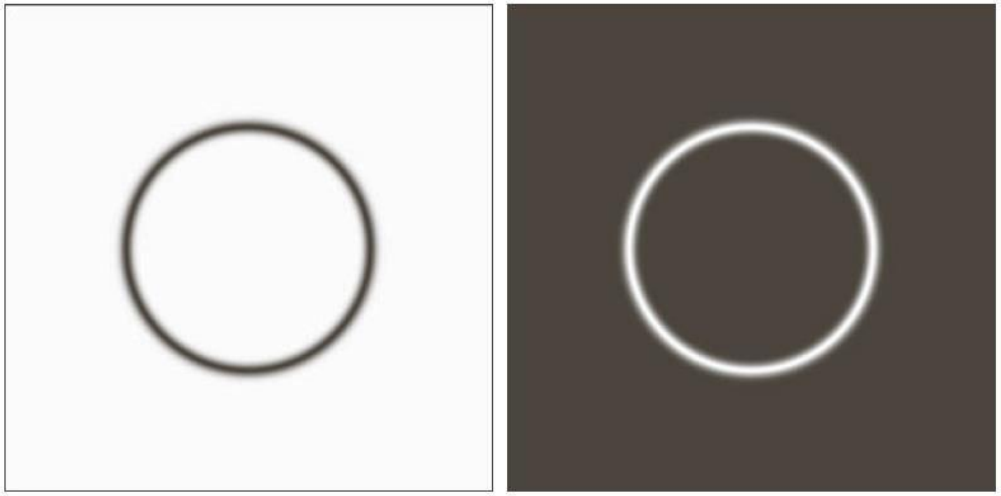
Periodic Noise Reduction



Bandpass and Band reject Filters

- Reject (pass) an isotropic frequency

$$H_{BP}(u,v) = 1 - H_{BR}(u,v)$$



6 Color Image Processing

- Color is a powerful description that often simplifies object identification and extraction from a scene
- Color Image Processing is divided into two major areas:
    1) Full-color processing
        - Images are acquired with a full-color sensor, such as a color video camera or color scanner
        - Color transformation, color complement and tone or color correction
    2) Pseudo color processing
        - Assigning a color to a particular monochrome intensity or range of intensities

RGB Color Models

- Model based on a Cartesian coordinate system
- Each color appears in its primary spectral components of red, green, and blue

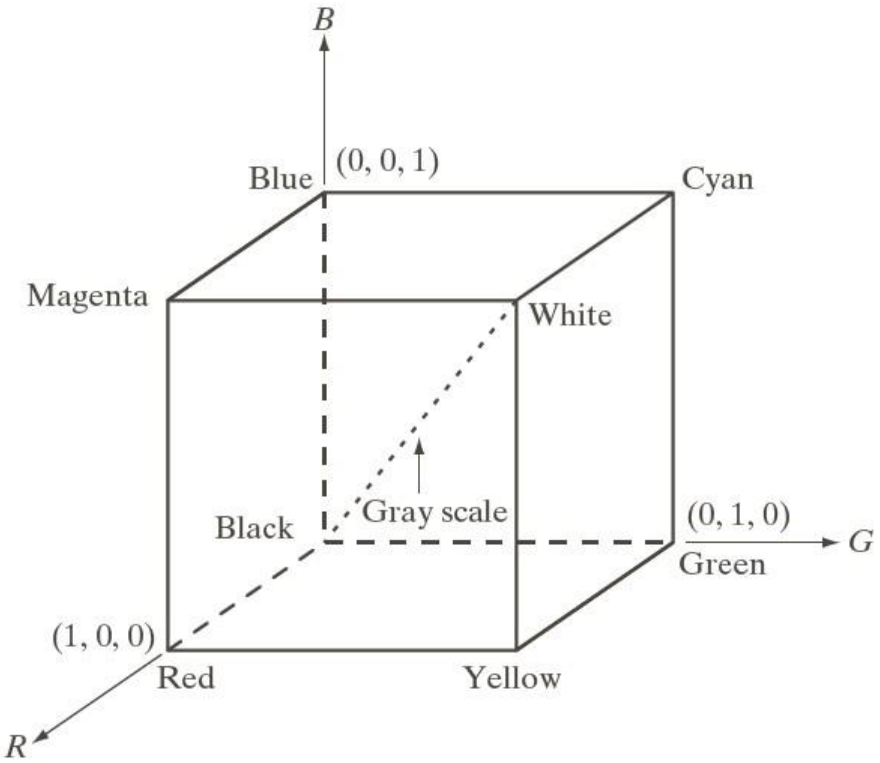


FIGURE 6.7 Schematic of the RGB color cube. Points along the main diagonal have gray values, from black at the origin to white at point (1, 1, 1).

RGB Color Models

- RGB primary values are at three corners; the secondary colors cyan, magenta, and yellow are at the other corners; black is at the origin; and white is at the corner farthest from the origin
- In this model, the gray scale (points of equal RGB values) extends from black to white along the line joining these two points
- The different colors in this model are points on or inside the cube, and are defined by vectors extending from the origin

RGB Color Models

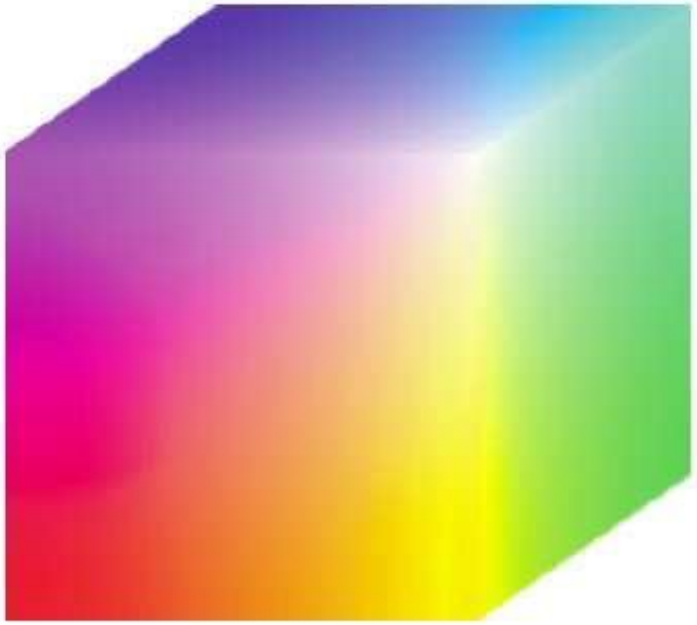


FIGURE 6.8 RGB 24-bit color cube.

- Number of bits used to represent each pixel in RGB space is called the *pixel depth*
- RGB image has 24 bit pixel depth
- True color or full color image is a 24 bit RGB image
- Total colors in 24-bit image is $(2^8)^3 = 16,777,216$

8 Image Compression

- Image Compression is the art and science of reducing amount data required to represent an image

Data Redundancies

- Types of Data Redundancies:
  1. Coding Redundancy
  2. Interpixel Redundancy
  3. Psychovisual Redundancy

Coding Redundancies

- If the gray levels of an image are coded in a way that uses more code symbols than absolutely necessary to represent each gray level, the resulting image is said to contain coding redundancies
- Variable-length coding

Interpixel Redundancy

- If the value of any given pixel can be reasonably predicted from the value of its neighbors, the image is said to contain interpixel redundancies
- Run-length coding

Psychovisual Redundancy

- The eye does not respond with equal sensitivity to all information
- Certain information simply has less relative importance than other information in normal visual processing
- It can be eliminated without significantly impairing the quality of image perception
- Lossy data compression

Error-Free Compression

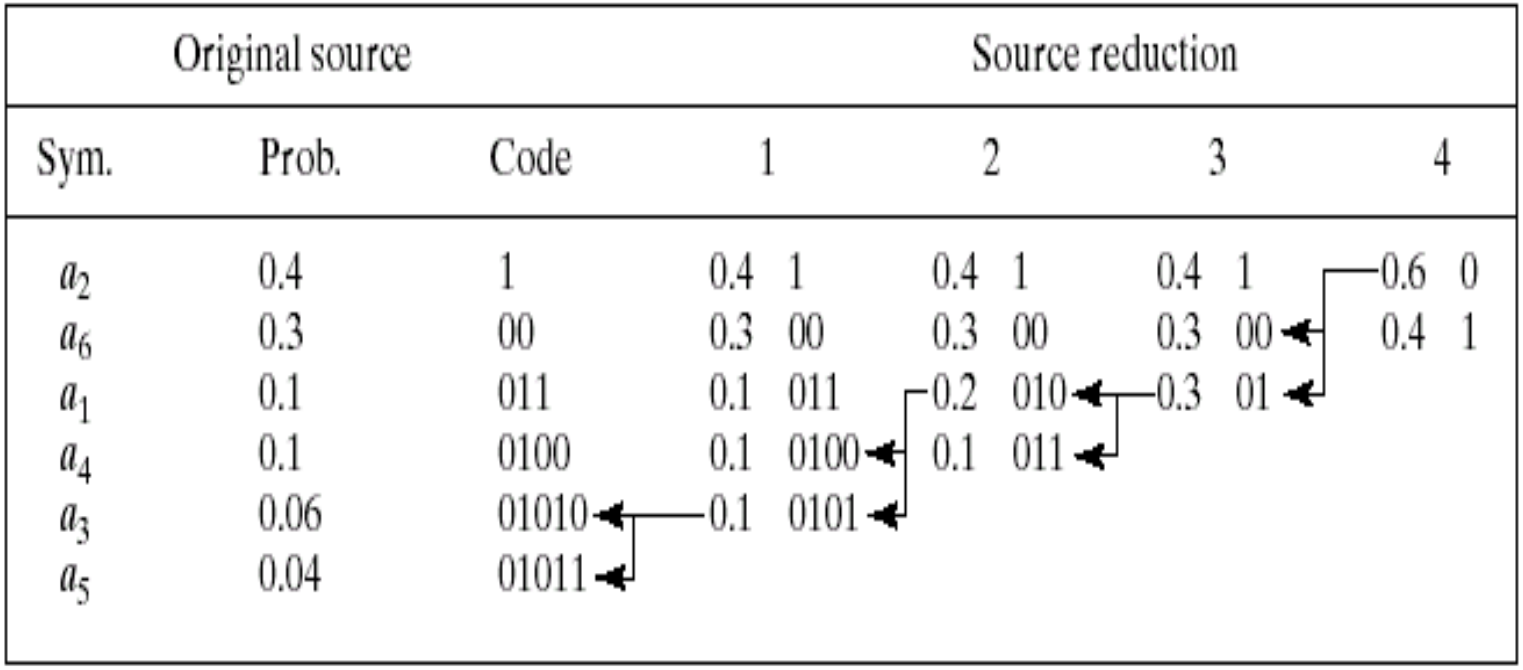
- Variable-Length Coding
  - Huffman Coding
    - Most popular for removing coding technique redundancy

Original source		Source reduction			
Symbol	Probability	1	2	3	4
a_2	0.4	0.4	0.4	0.4	0.6
a_6	0.3	0.3	0.3	0.3	0.4
a_1	0.1	0.1	0.2	0.3	
a_4	0.1	0.1	0.1		
a_3	0.06	0.1			
a_5	0.04				

FIGURE 8.11 Huffman source reductions.

Huffman Coding

FIGURE 8.12
Huffman code assignment procedure.



Original source			Source reduction									
Sym.	Prob.	Code	1		2		3		4			
a_2	0.4	1	0.4	1	0.4	1	0.4	1	0.6	0		
a_6	0.3	00	0.3	00	0.3	00	0.3	00	0.4	1		
a_1	0.1	011	0.1	011	0.2	010	0.3	01				
a_4	0.1	0100	0.1	0100	0.1	011						
a_3	0.06	01010	0.1	0101								
a_5	0.04	01011										

LZW Coding

- LZW compression has been integrated into a variety of mainstream imaging file formats, including GIF, TIFF, and PDF
- The PNG format was created to get around LZW licensing requirements

LZW Coding

39	39	126	126
39	39	126	126
39	39	126	126
39	39	126	126

Dictionary Location	Entry
0	0
1	1
⋮	⋮
255	255
256	—
⋮	⋮
511	—

EXAMPLE 8.7: LZW coding.

TABLE 8.7
LZW coding example.

Currently Recognized Sequence	Pixel Being Processed	Encoded Output	Dictionary Location (Code Word)	Dictionary Entry
	39			
39	39	39	256	39-39
39	126	39	257	39-126
126	126	126	258	126-126
126	39	126	259	126-39
39	39			
39-39	126	256	260	39-39-126
126	126			
126-126	39	258	261	126-126-39
39	39			
39-39	126			
39-39-126	126	260	262	39-39-126-126
126	39			
126-39	39	259	263	126-39-39
39	126			
39-126	126	257	264	39-126-126
126		126		

Run Length Coding

- Images with repeating intensities along their rows (columns) can often be compressed by representing runs of identical intensities as run-length pairs, where each run-length pair specifies the start of a new intensity and the number of consecutive pixels that have that intensity
- When there are few(or no) runs of identical pixels, run-length encoding results in data expansion
- BMP file format uses a form of run-length encoding

Types of File Formats

- Popular file formats are listed below:
  1) GIF (Graphics Interchange Format)
  2) JPEG (Joint Photographic Experts Group)
  3) PNG (Portable Network Group)
  4) DICOM (Digital Imaging and COMmunication)
  5) SVG (Vector Graphics file format)
  6) TIFF (Tagged Image File Format)

9 Morphological Image Processing

- The word Morphology commonly denotes a branch of biology that deals with the form and structure of animals and plants
- We use mathematical morphological as a tool for extracting image components that are useful in the representation and description of region shape, such as boundaries extraction, skeletons, convex hull, morphological filtering, thinning, pruning.

Preliminaries

- Reflections

The reflection of a set B, denoted B^{μ}, is defined as

$$B^{\mu} = \{w \mid w = -b, \text{ for } b \in B\}$$

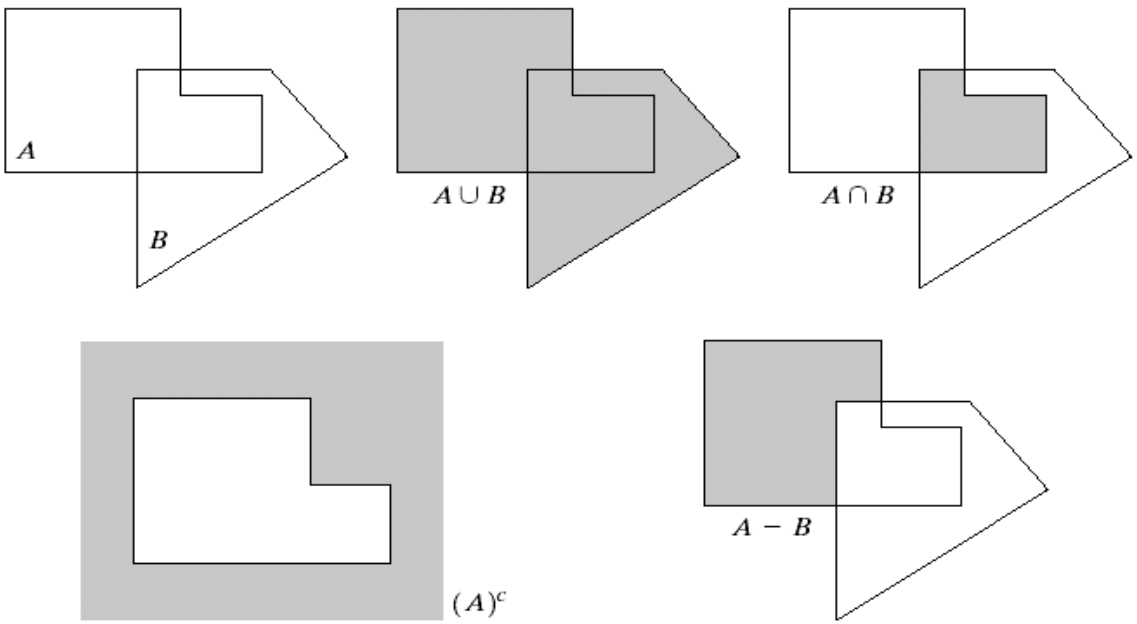
- Translations

The translation of a set B by point $z = (z_1, z_2)$, denoted $(B)_Z$, is defined as

$$(B)_Z = \{c \mid c = b + z, \text{ for } b \in B\}$$

Preliminaries

Operators by examples:



a b c
d e

FIGURE 9.1
(a) Two sets A and B. (b) The union of A and B. (c) The intersection of A and B. (d) The complement of A. (e) The difference between A and B.

Preliminaries

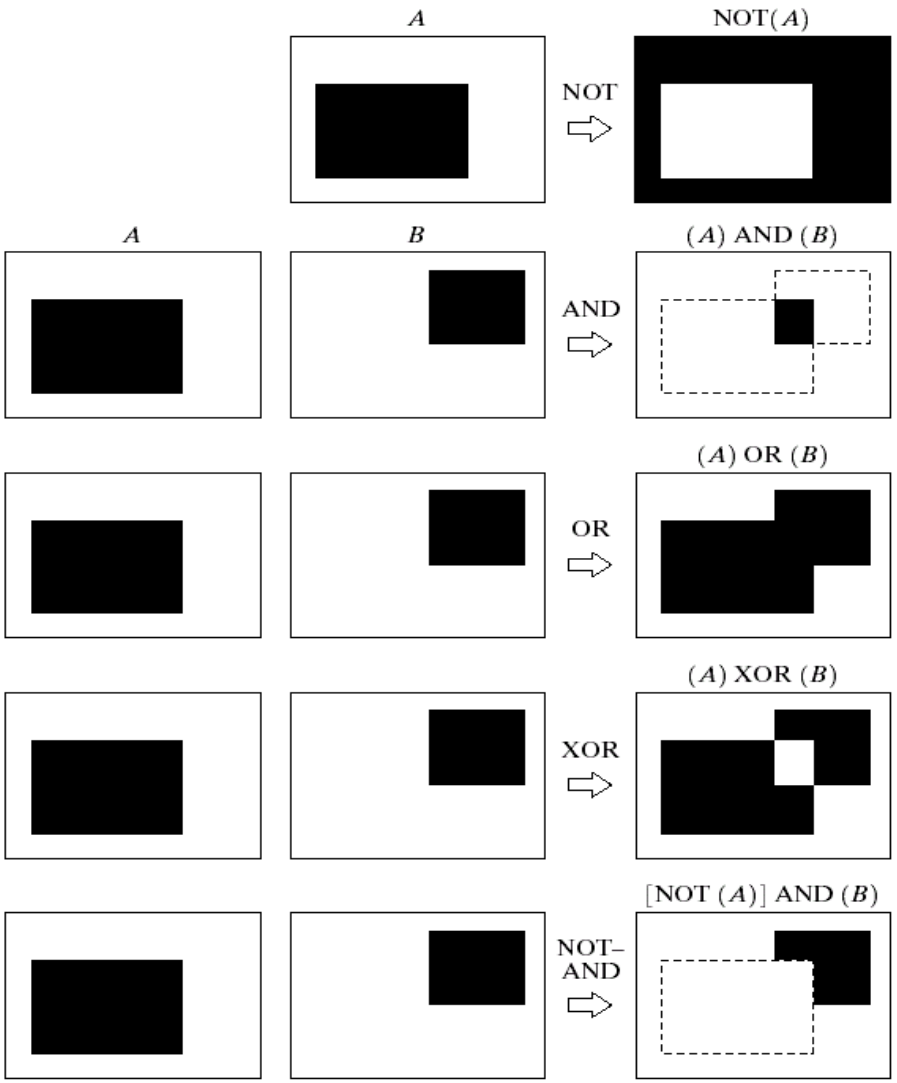
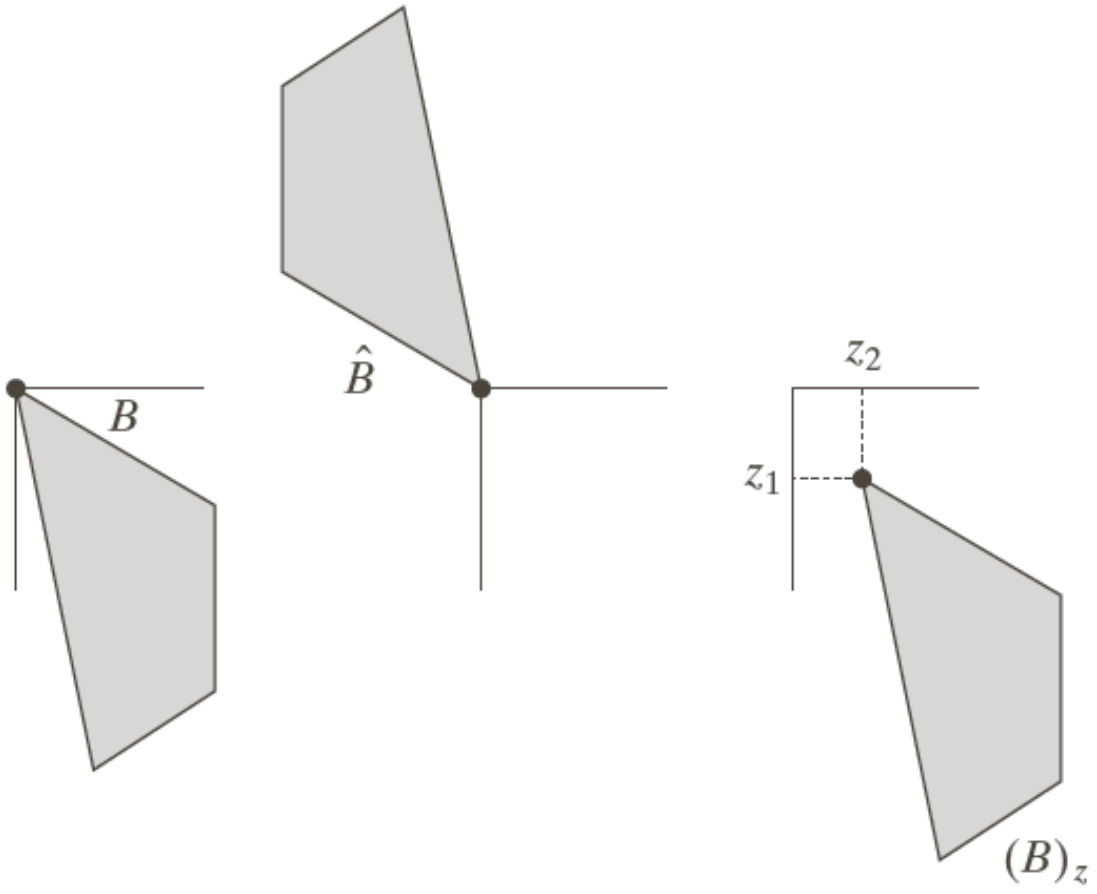


FIGURE 9.3 Some logic operations between binary images. Black represents binary 1s and white binary 0s in this example.

Examples of Reflection and Translation



a b c

FIGURE 9.1
(a) A set, (b) its reflection, and (c) its translation by z.

Structuring Elements

- Small sets or sub-images used to probe an image under study for properties of interest is called SE

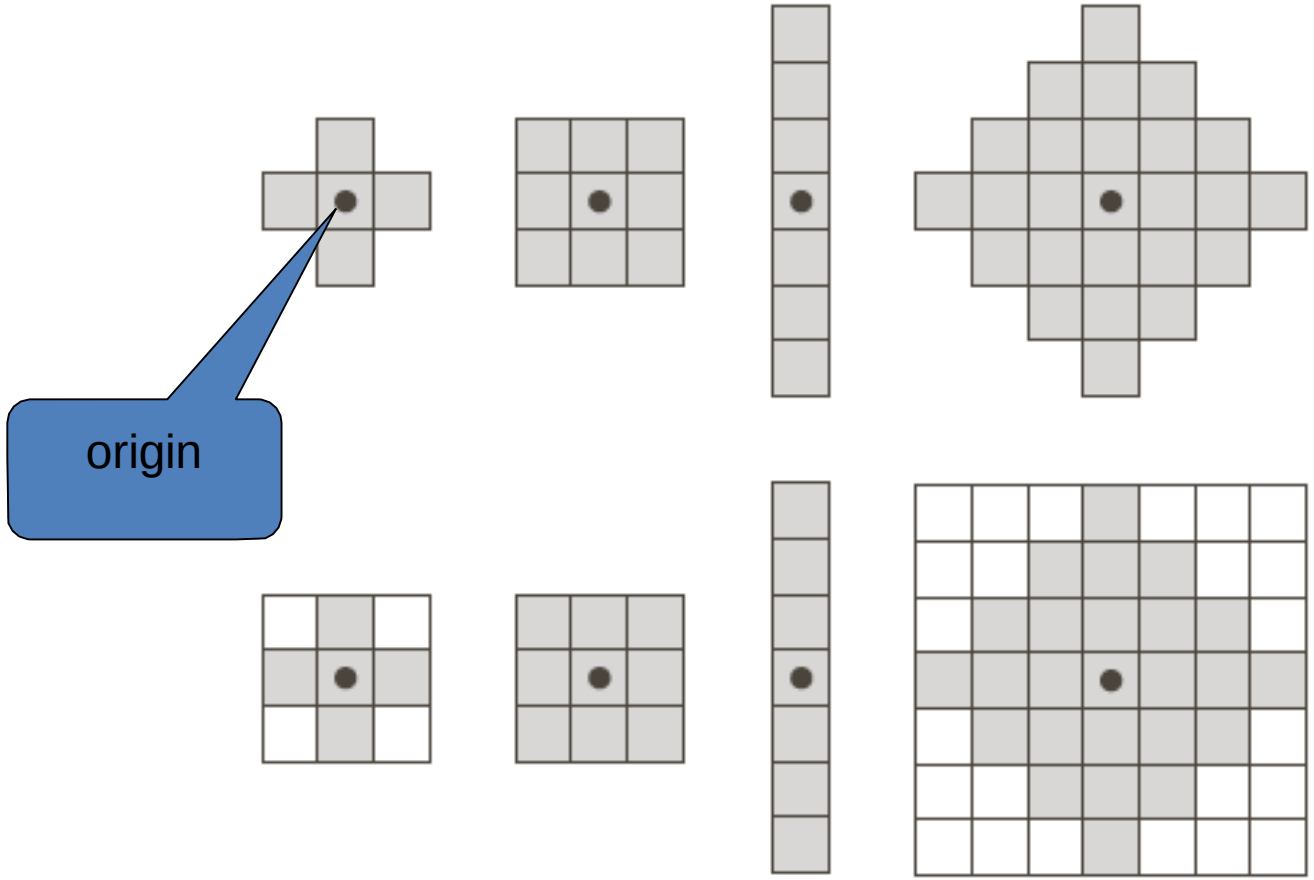
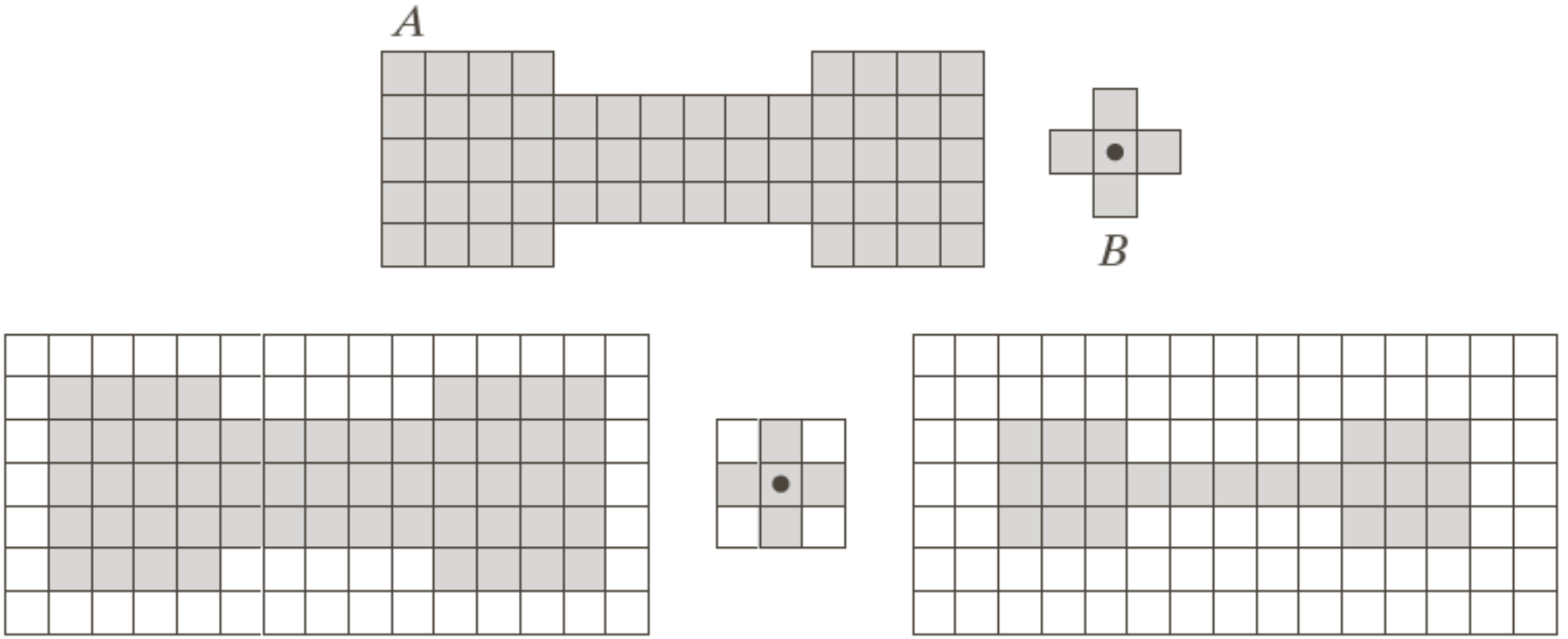


FIGURE 9.2 First row: Examples of structuring elements. Second row: Structuring elements converted to rectangular arrays. The dots denote the centers of the SEs.

Structuring Elements

- The background border is made large enough to accommodate the entire structuring element when its origin is on the border of the original set (Padding)
- SE is of size 3×3 with the origin in the center, so as a one-element border that encompasses the entire set is sufficient



Erosion

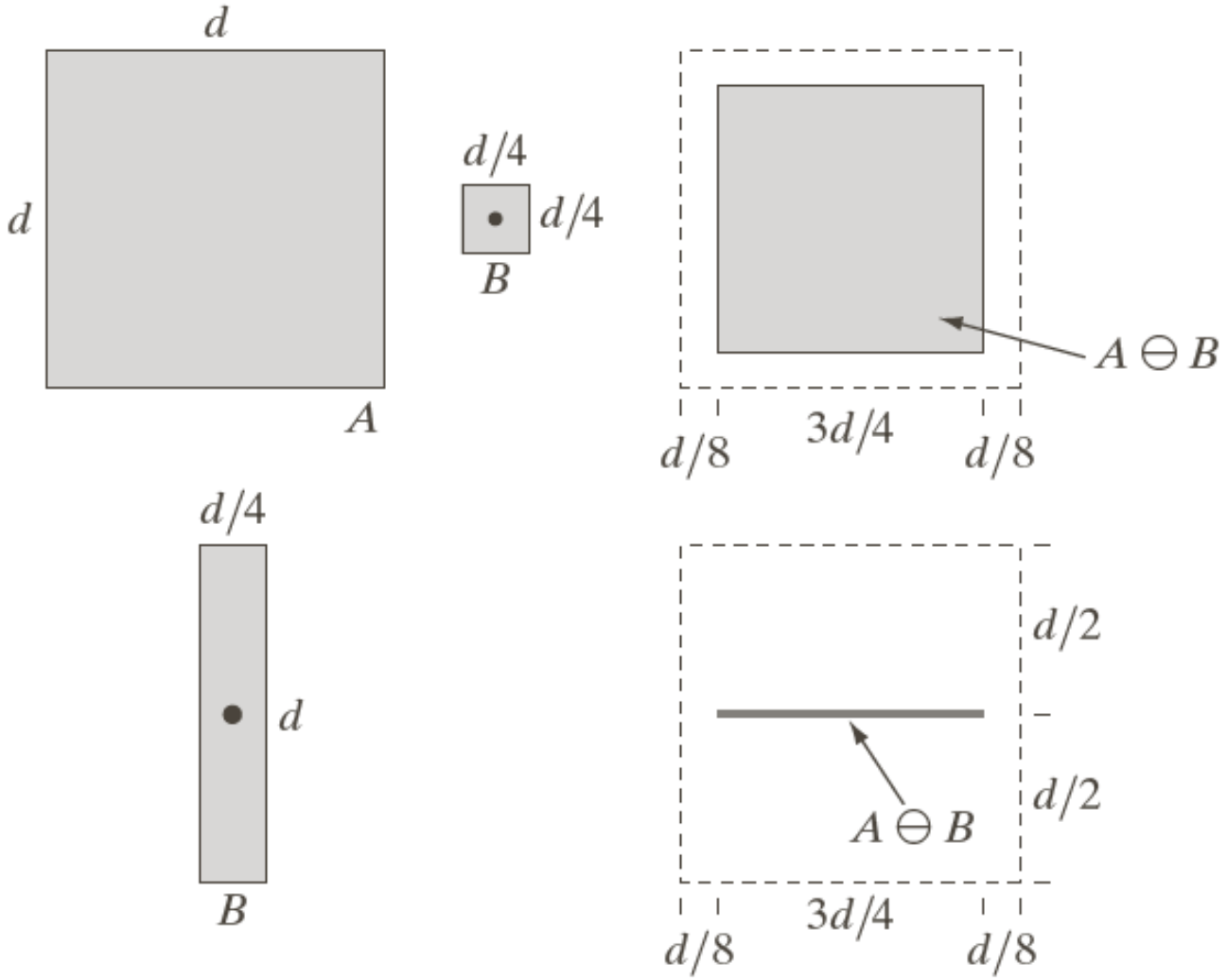
- With A and B as sets in Z^2 , the erosion of A by B is defined as

$$A \ominus B = \{z \mid (B)_Z \subseteq A\}$$

- Erosion of A by B is the set of all points z such that B, translated by z, is contained in A.

$$A \ominus B = \{z \mid (B)_Z \cap A^c = \varnothing\}$$

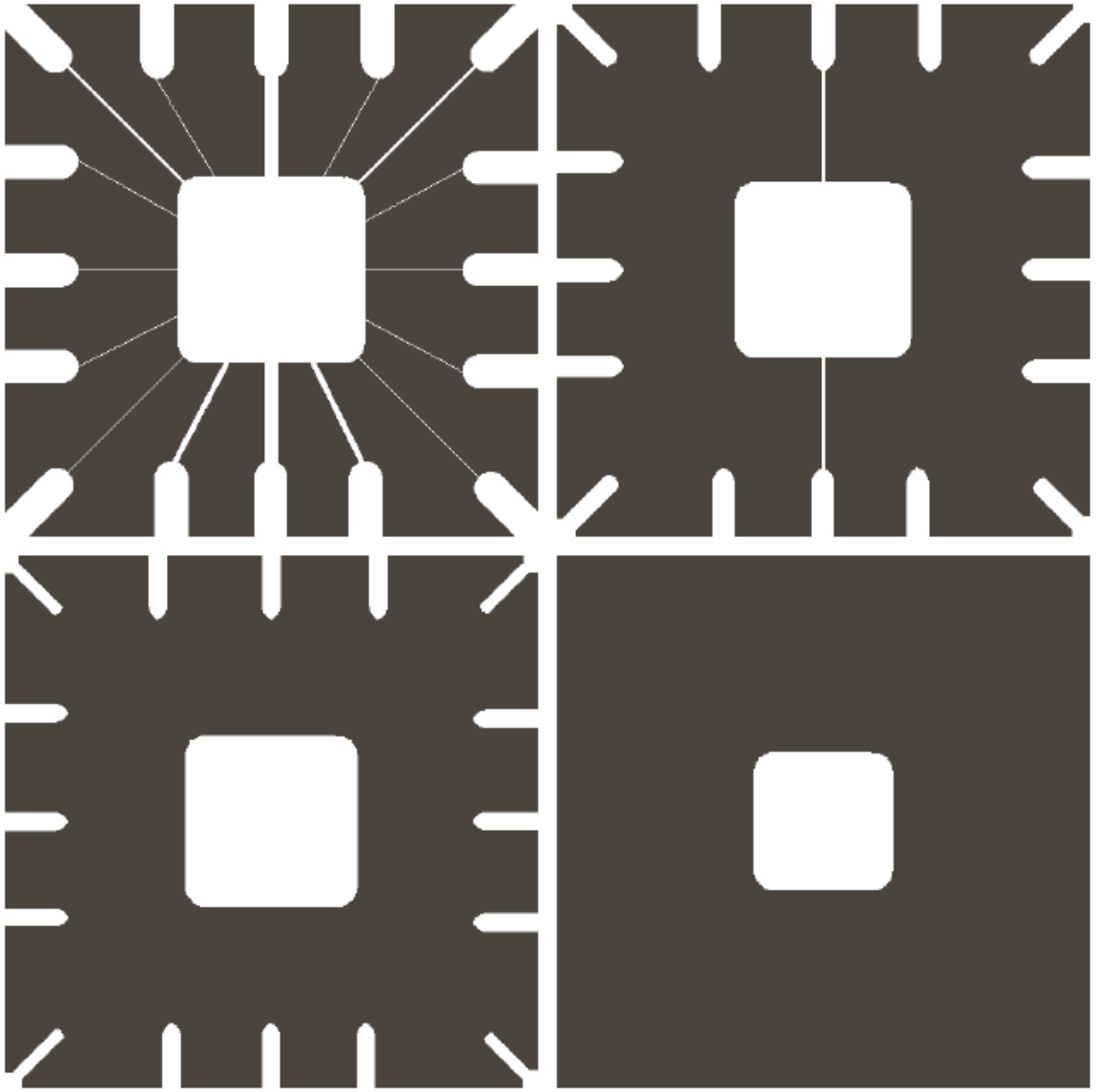
- B has to be contained in A is equivalent to B not sharing any common elements with the background



a b c
d e

FIGURE 9.4 (a) Set A. (b) Square structuring element, B. (c) Erosion of A by B, shown shaded. (d) Elongated structuring element. (e) Erosion of A by B using this element. The dotted border in (c) and (e) is the boundary of set A, shown only for reference.

Erosion



a b
c d

FIGURE 9.5 Using erosion to remove image components. (a) A 486 × 486 binary image of a wire-bond mask. (b)–(d) Image eroded using square structuring elements of sizes 11 × 11, 15 × 15, and 45 × 45, respectively. The elements of the SEs were all 1s.

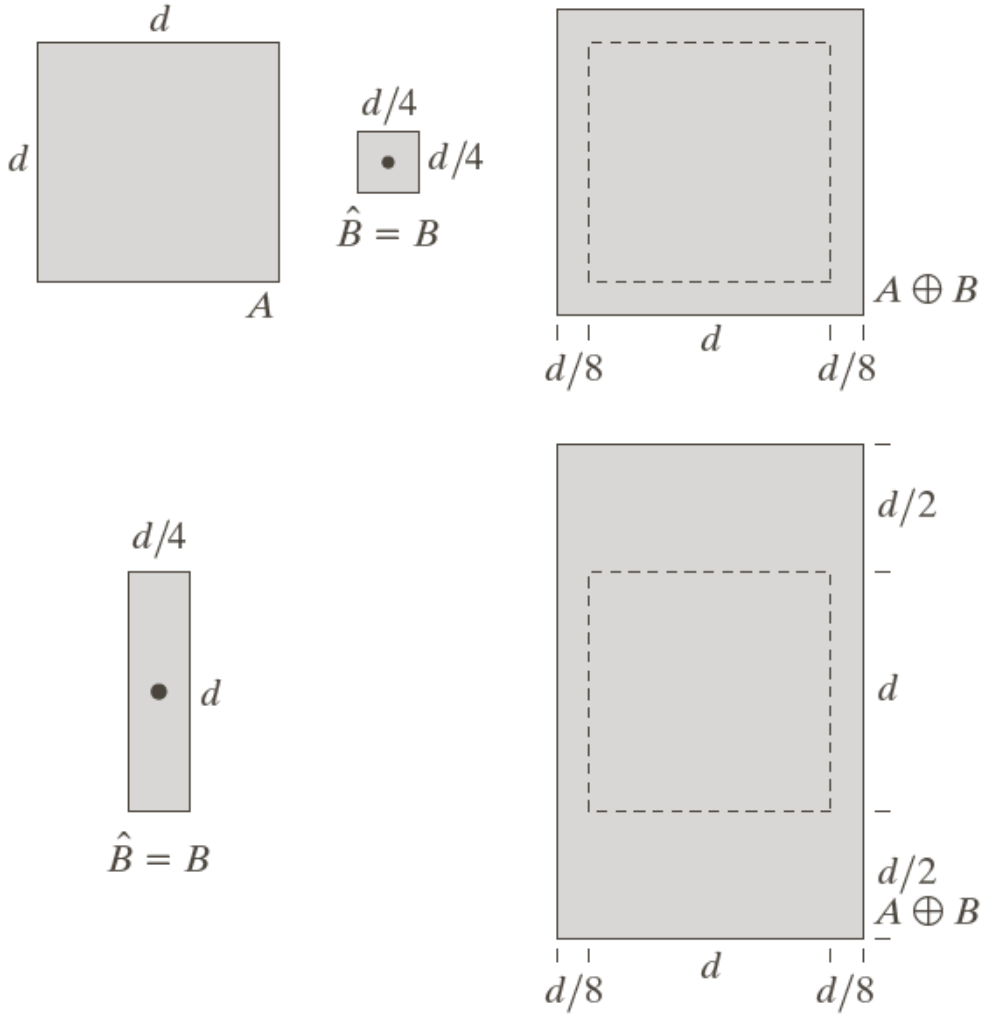
Erosion

- Erosion shrinks or thins objects in a binary image
- Erosion as a morphological filtering operation in which image details smaller than the structuring elements are filtered from the image
- Erosion performed the function of a “line filter”

Dilation

- With A and B as sets in Z^2, the dilation of A by B is defined as $A \oplus B = \left\{ z \mid \left(B^{\mu} \right)_z \cap A \neq \varnothing \right\}$
- Reflecting B about its origin, and shifting this reflection by z
- The dilation of A by B then is the set of all displacements, z, such that B^{μ} and A overlap by at least one element $A \oplus B = \left\{ z \mid \left[\left(B^{\mu} \right)_z \cap A \right] \subseteq A \right\}$

Erosion and Dilation



a b c
d e

FIGURE 9.6
(a) Set A. (b) Square structuring element (the dot denotes the origin). (c) Dilation of A by B, shown shaded. (d) Elongated structuring element. (e) Dilation of A using this element. The dotted border in (c) and (e) is the boundary of set A, shown only for reference

Dilation

- Unlike erosion, dilation “grows” or “thickens” objects in a binary image
- The specific manner and extent of this thickening is controlled by the shape of the structuring element used

Erosion and Dilation





0	1	0
1	1	1
0	1	0

a c
b

FIGURE 9.7
(a) Sample text of poor resolution with broken characters (see magnified view). (b) Structuring element. (c) Dilation of (a) by (b). Broken segments were joined.

Opening and Closing

- Opening of set A by SE B is defined

$$A \circ B = (A - B) \oplus B$$

- Thus, opening A by B is the erosion of A by B, followed by a dilation of the result by B
- Closing of set A by SE B is defined

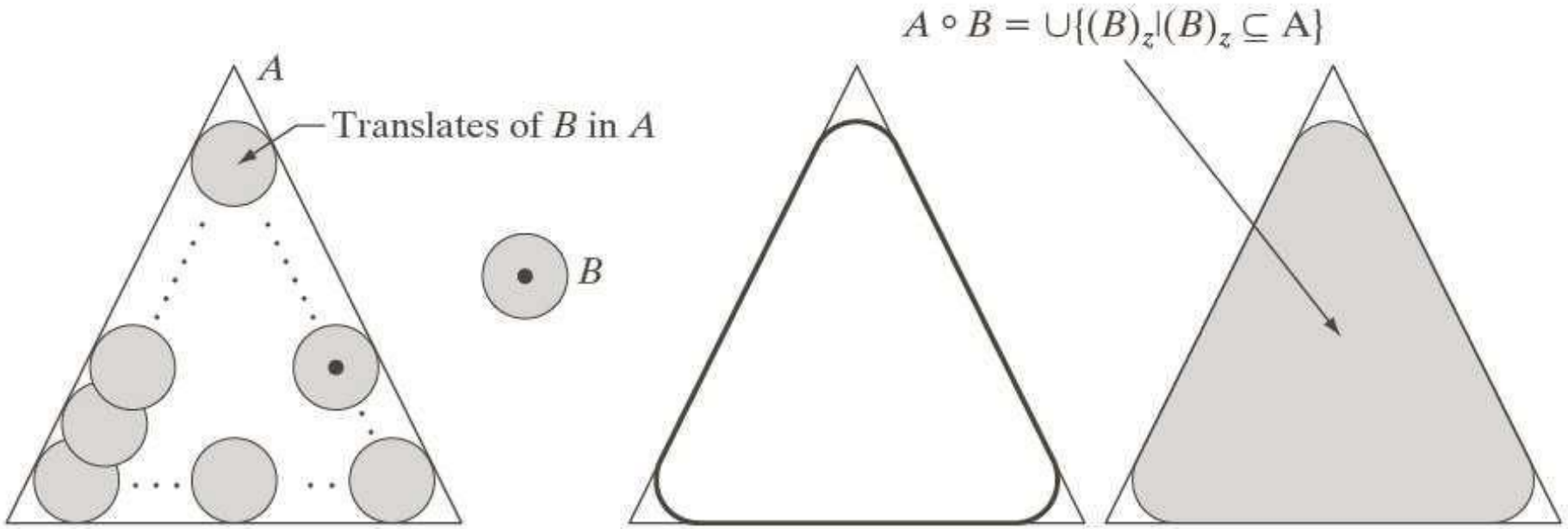
$$A \cdot B = (A \oplus B) \ominus B$$

- Closing of set A by B is simply the dilation of A by B, followed by the erosion of the result by B

Opening and Closing

- Opening smoothens the contour of an object, breaks narrow isthmuses, and eliminates thin protrusions
- Closing also tends to smooth sections of contours but, as opposed to opening, it generally fuses narrow breaks and long thin gulfs, eliminates small holes, and fills gaps in the contour

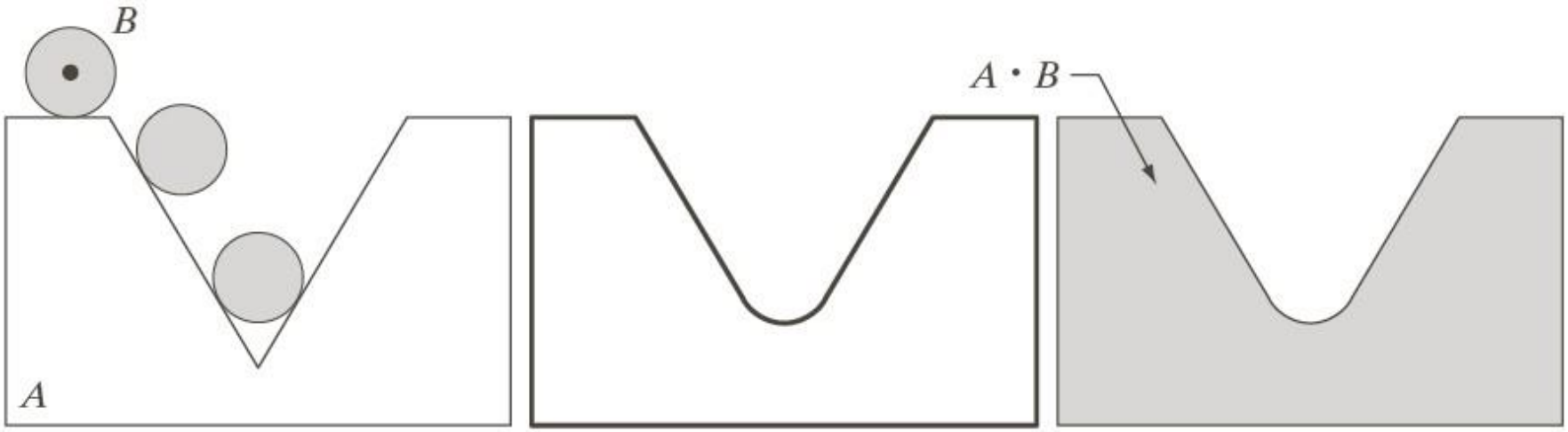
Examples: Opening



a b c d

FIGURE 9.8 (a) Structuring element B "rolling" along the inner boundary of A (the dot indicates the origin of B). (b) Structuring element. (c) The heavy line is the outer boundary of the opening. (d) Complete opening (shaded). We did not shade A in (a) for clarity.

Examples: Closing



a b c

FIGURE 9.9 (a) Structuring element B "rolling" on the outer boundary of set A. (b) The heavy line is the outer boundary of the closing. (c) Complete closing (shaded). We did not shade A in (a) for clarity.

Opening and Closing



a b
d c
e f

FIGURE 9.11
(a) Noisy image.
(b) Structuring element.
(c) Eroded image.
(d) Opening of A.
(e) Dilation of the opening.
(f) Closing of the opening.
(Original image courtesy of the National Institute of Standards and Technology.)

10 Image Segmentation

- Image segmentation divides an image into regions that are connected and have some similarity within the region and some difference between adjacent regions.
- The goal is usually to find individual objects in an image.
- There are fundamentally two kinds of approaches to segmentation: discontinuity and similarity.
  - Partition an image based on abrupt changes in intensity, color or texture but especially sudden changes in intensity along a boundary line, which is called an edge.
  - Partition an image into regions that are similar according to set of predefined criteria

Region-Based Segmentation

- Region-based segmentation is based on the connectivity of similar pixels in a region
  - Each region must be uniform
  - Connectivity of the pixels within the region is very important
- There are two main approaches to region-based segmentation: region growing and region splitting

Basic Formulation

- Let R represent the entire image region.
- Region based Segmentation is a process that partitions R into subregions, $R_1, R_2, \ldots, R_n$, such that

(a) $\bigcup_{i=1}^{n} R_i = R$

(b) R_i is a connected region, $i = 1,2,\ldots, n$

(c) $R_i \cap R_j = \phi$ for all i and j, $i \neq j$

(d) $P(R_i)$ = TRUE for $i = 1,2,\ldots, n$

(e) $P(R_i \cup R_j) =$ FALSE for any adjacent regions R_i and R_j

where $P(R_k)$: a logical predicate defined over the points in set R_k

For example: $P(R_k)$=TRUE if all pixels in R_k have the same gray level

Region Growing Algorithm

- It is a process of grouping the pixels or subregions to get a bigger region present in an image
- **Selection of the initial seed:** Initial seed that represent the ROI should be given typically by the user. Can be chosen automatically. The seeds can be either single or multiple
- **Seed growing criteria:** Similarity criterion denotes the minimum difference in the grey levels or the average of the set of pixels. Thus, the initial seed 'grows' by adding the neighbors if they share the same properties as the initial seed
- **Terminate process:** If further growing is not possible then terminate region growing process

Region Splitting

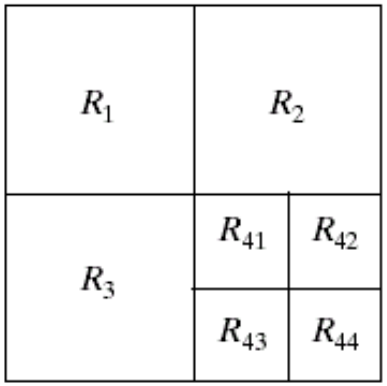
- Region splitting is the opposite of region growing
- Algorithm:
  - First there is a large region (possible the entire image)
  - Then a predicate (measurement) is used to determine if the region is uniform
  - If not, then the method requires that the region be split into two regions
  - Then each of these two regions is independently tested by the predicate (measurement)
  - This procedure continues until all resulting regions are uniform

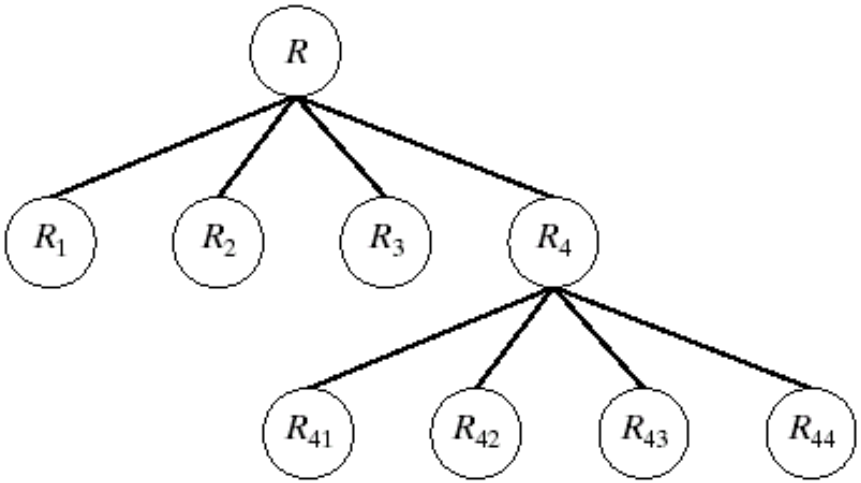
Region Splitting

- The main problem with region splitting is determining where to split a region
- One method to divide a region is to use a quadtree structure
- Quadtree: a tree in which nodes have exactly four descendants

a b

FIGURE 10.42
(a) Partitioned image.
(b) Corresponding quadtree.





Region Merging

- The merge procedure:
  - Merge any adjacent regions R_j and R_k for which $P(R_j \cup R_k)$ = TRUE (the quadtree structure may not be preserved)
  - Stop when no further merging or splitting is possible

Digital Image Processing Lab Assignment 1

Objective

- To understand how images are represented using arrays.
- To learn grayscale and RGB image construction.
- To analyze pixel intensity differences.

Task 1: Grayscale Image from Array

Instructions:

1. Create a 2D array (matrix) with values ranging from 0 to 255.
2. Convert this array into a grayscale image.
3. Display the image.
4. Observe how different values in the array affect the brightness (intensity) of the image.
5. Convert the generated grayscale image back into an array.

Task 2: RGB Image Construction (Half Green, Half Red)

Instructions:

1. Create a 3D array representing an RGB image.

   Shape: (height × width × 3)

2. Divide the image into two equal halves:

   Left half → Red color (R=255, G=0, B=0)

   Right half → Green color (R=0, G=255, B=0)

3. Convert the array into an image and display it.

Submission Requirements

- Python code for both tasks.
- Output images.

Tools/Libraries Suggested

- NumPy
- OpenCV or PILLOW

Digital Image Processing Lab Assignment 2

Assignment: RGB to Grayscale and Negative Image Conversion

Input Image

Each member of the team must use a separate RGB image.

Tasks

Task 1: RGB to Grayscale Conversion

Convert the RGB image into a grayscale image.
You may use the standard grayscale conversion formula:
Gray=0.299R+0.587G+0.114B
Save the output as:
gray_image.jpg

Task 2: Grayscale to Negative Image

Using the grayscale image obtained in Task 1, generate its negative image using the following formula:
Negative=255−Gray
Save the output as:
Negative_image.jpg

Example Transformation :



Task 3: (You don't have to write code for task 3, but you do have to answer this in the lab.)

Answer the following question:
Can the final negative image be converted back into the original RGB image? Why or why not?

Expected Output

- Source code
- Original RGB image
- Grayscale image
- Negative image

DIP Lab Assignment 03

Objective

The objective of this experiment is to understand and implement histogram equalization using the Probability Density Function (PDF) and Cumulative Distribution Function (CDF) of a grayscale image.

Task

You are given a digital image of your choice.
Perform the following operations on the image:

Step 1: Read the Image
convert it into a grayscale image.
Do not use a completely uniform or completely black/white image.

Step 2: Calculate the Histogram
For the grayscale image, calculate the frequency of each intensity level.
Plot the histogram of the grayscale image.

Step 3: Calculate the PDF
Calculate the Probability Density Function from the histogram.
Display or print the PDF values.

Step 4: Calculate the CDF
Calculate the Cumulative Distribution Function from the PDF.
Plot the CDF.

Step 5: Perform Histogram Equalization
Use the CDF to calculate the new intensity value for every pixel.
Use the resulting mapping to create the histogram-equalized image.

Step 6: Calculate the Equalized Histogram
Calculate the histogram of the equalized image.

Display:

1. Original grayscale image
2. Histogram of original image
3. PDF of original image
4. CDF of original image
5. Histogram-equalized image
6. Histogram of equalized image

Output	Requirement
Original Image	Display
Grayscale Image	Display
Original Histogram	Plot
PDF	Calculate and display
CDF	Calculate and plot
Equalized Image	Display
Equalized Histogram	Plot

You must implement the PDF, CDF, and histogram equalization calculations yourselves.

Do not directly use a built-in function such as:

- equalizeHist()
- or any other function that directly performs histogram equalization.

You may use libraries for:

- Reading images
- Converting RGB images to grayscale
- Displaying images
- Plotting graphs
- Basic numerical operations

Submit your complete source code.

Lab Assignment 04

Unsharp Masking and High-Boost Filtering

Write a program that takes a grayscale image as user input and performs:

1. **Unsharp masking**
2. **High-boost filtering**

The program should allow the user to select the required parameters.

Required Steps

First, apply a smoothing filter to the input image:

Blurred Image = Low-pass Filter(Image)

Then calculate the mask:

Mask = Original Image − Blurred Image

01. Unsharp Masking

Generate the sharpened image using:
Sharpened Image = Original Image + Mask
or equivalently:

$$g(x,y) = f(x,y) + [f(x,y) - \bar{f}(x,y)]$$

The program should display:

1. Original image
2. Blurred image
3. Mask
4. Unsharp-masked image

02. High-Boost Filtering

Implement high-boost filtering using:

$$g(x,y) = A\, f(x,y) - \bar{f}(x,y)$$

Where: $A > 1$

The value of must be provided by the user.
Test the algorithm using at least three different values, such as:

$A = 1.5$
$A = 2.0$
$A = 3.0$

Display the results and compare the effect of different values.